



Internal Use Only

MOBILE PHONE SERVICE MANUAL

CAUTION

BEFORE SERVICING THE UNIT, READ THE "SAFETY PRECAUTIONS" IN THIS MANUAL

MODEL : LG-H502f

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1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

This material is prohibited to share and release to unauthorized person, in accordance with the regulations, LG Electronics, Civil / criminal responsibility in accordance with the relevant provisions violate.

B. Precautions for repair

- In case of Disassembly or Assembly to repair product, be careful of a product failure caused by RF signals and Static electricity.
- When using Magnetic tool for the Phone's SVC repair, you should check affect the Electric parts according to effect of Magnet.
- When fastening the screw, be careful not to damage the head of screw and even product.

C. Attention

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign.

Following information is ESD handling:

- Service personal should ground themselves by using a wrist, strap when exchange system board.
- When repair are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system board or parts like EEPROM to the Factory, use the protective package as described.

2.1 Band Specification

Support Band	TX Freq (MHz)	RX Freq (MHz)
WCDMA(FDD1)	1920 – 1980	2110 – 2170
WCDMA(FDD5)	824 – 849	869 – 894
WCDMA(FDD8)	880 – 915	925 – 960
EGSM	880 – 915	925 – 960
GSM850	824 – 849	869 – 894
DCS1800	1710 – 1785	1805 – 1880
PCS1900	1850 – 1910	1930 – 1990

2.2 HW Features

List	Type / Spec.	
1. Phone Type	DOP Type	
2. Size	139.7mm x 69.8mm x 10.2mm	
3. Weight	135g / 137g (with Do-cad / LGC Battery)	
4. Battery	2,100mAh(Min.) , 2.540mAh(Typ) (Li-Ion)	
5. Chipset	MT6582 1.3GHz Quad core	
6. Memory	8GB(EMMC) + 8GB(LPDDR3) External Memory(SD Card) : Up to 32GB	
7. LCD	Size	5 inch
	Display Type	Transmissive Type
	Color	16.7M colors
	Resolution	1280(H) X 720(V), 294ppi
8. Touch	Type	5 inch Capacitive type
9. Main Camera (5M)	Type	CMOS image sensor
	Resolution	3264(H) X 2448(V) pixels.
	Image Scaling Down	8M(3264x2448), W6M(3264x1840), 5M(2448x2448), 3M(12048x1536)
	Format	Image : JPG, Video : MP4

2.2 HW Features

10. VT Camera	Type	CMOS image sensor
	Resolution	2592(H) X 1944(V) pixels.
	Image Scaling Down	5M
	Format	Image : JPG, Video : MP4
11. Audio	Receiver	13 X 07 X 2.7T Receiver
	Speaker	15 X 11 X 3T Speaker
	Format	MP3, AAC, MIDI, EAAC+, OGG, AMR
12. Bluetooth	Standard	Bluetooth 4.0
	Effective Distance	10M
	Distance	0 m ~ 10 m (depend on environment)
13. WLAN	Standard	IEEE 802.11 b/g/n
	Throughput	Max 40Mbps (SDIO Driver performance)
	Depend on environment	0 ~ 50m (depend on environment)
14. GPS	type	A-GPS
15. FM	type	FM Radio, 3.5pi Ear-jack

2.3 RSSI Display

Antenna BAR	Specification		Unit
	WCDMA(RSCP)	GSM(RSSI)	
5→4	-86 ±3dBm	-93 ±3dBm	dBm
4→3	-90 ±3dBm	-97 ±3dBm	
3→2	-96 ±3dBm	-101 ±3dBm	
2→1	-106 ±3dBm	-105 ±3dBm	
1→0	-113 ±3dBm(idle) -115±3dBm(call)	-110 ±3dBm(idle) -116±3dBm(call)	

2.4 Current consumption

Item	Specification	
	WCDMA	GSM
1. Sleep Mode	Under 7mA	Under 7mA
2. Sleep : connector Ear jack	Under 7mA	Under 7mA
3. Current(Sleep & Idle AVG)	Under 7mA @ DRX 7	Under 7mA @ P.P 5

2.5 Battery bar

Battery Bar	Specification	
BAR 20 (Full)	98% over	remain%
BAR 20 --> 19	98% ◇ 97%	
BAR 19 --> 18	93% ◇ 92%	
BAR 18 --> 17	88% ◇ 87%	
BAR 17 --> 16	83% ◇ 82%	
BAR 16 --> 15	78% ◇ 77%	
BAR 15 --> 14	73% ◇ 72%	
BAR 14 --> 13	68% ◇ 67%	
BAR 13 --> 12	63% ◇ 62%	
BAR 12 --> 11	58% ◇ 57%	
BAR 11 --> 10	53% ◇ 52%	
BAR 10 --> 9	48% ◇ 47%	
BAR 9 --> 8	43% ◇ 42%	
BAR 8 --> 7	38% ◇ 37%	
BAR 7 --> 6	33% ◇ 32%	
BAR 6 --> 5	28% ◇ 27%	
BAR 5 --> 4	23% ◇ 22%	
BAR 4 --> 3	16% ◇ 15%	
BAR 3 --> 2	13% ◇ 12%	
BAR 2 --> 1	8% ◇ 7%	
BAR 1 --> 0	3% ◇ 2%	
Low Battery Pop-up	4% ~ 15% : One Time popup (No call)	

2.6 SW Specification

Item	Feature	Comment
RSSI	0 ~ 5 Levels	
Battery Charging	0 ~ 20 Levels	
Key Volume	0 ~ 7 Level	
Audio Volume	0 ~ 15 Level	
Time / Date Display	Yes	
Multi-Language	Yes	depending on build language
Quick Access Mode	Phone / Messaging / Browser/ Applications	Phone / Contact / Messaging / Applications
PC Sync	Yes	
Speed Dial	Yes	Voice mail center -> 1 key
Profile	Yes	not same with feature phone setting
CLIP / CLIR	Yes	
Phone Book	Name / Number / Email / Groups / Postal addresses / Organizations / IM / Note / Nickname / Website / Event /	There is no limitation on the number of items. It depends on available memory amount.
Last Dial Number	Yes	
Last Received Number	Yes	
Last Missed Number	Yes	
Search by Number/Name	Yes	
Group	Yes	There is no limitation on the number of items. It depends on available memory amount.
Fixed Dial Number	Yes	
Own Number	Yes	My Profile (add/edit/delete are supported)

2.6 SW Specification

Voice Memo	Yes	Support voice recorder
Network Selection	Automatic	
Mute	Yes	
Call Divert	Yes	
Call Barring	Yes	
Call Charge (AoC)	No	
Call Duration	Yes	
SMS (EMS)	There is no limitation on the number of items It depends on available memory amount.	EMS does not support.
SMS Over GPRS	Yes	
EMS Melody / Picture Send / Receive / Save	No	
MMS MPEG4 Send / Receive / Save	Yes	➤ Send / Receive : Yes ➤ Save : depends on content type Support video content type list 1. video/mp4 2. video/h263 3. video/3gpp2 - video/3gpp
Long Message	MAX 2000 characters	The standard of Open vender
Cell Broadcast	Yes	
Download	Over the Web	
Calendar	Yes	
Memo	Yes	There is no limitation on the number of items. It depends on available memory amount.
World Clock	Yes	

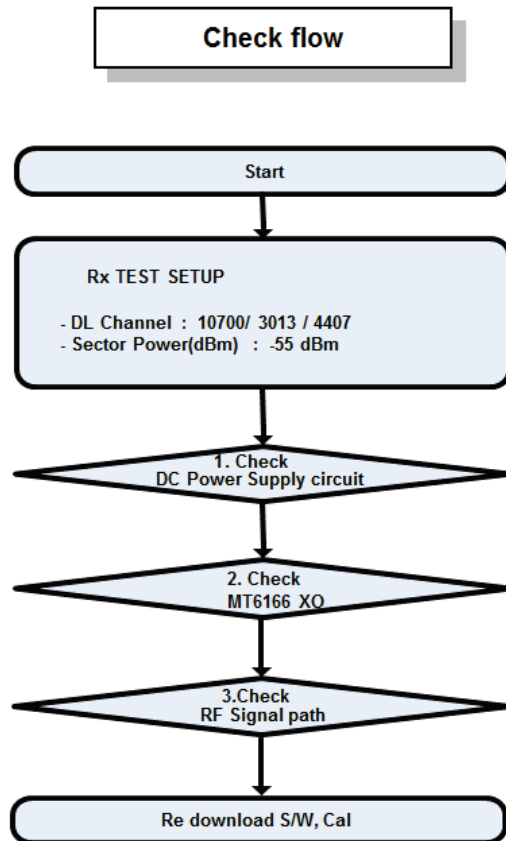
2.6 SW Specification

Stop Watch	Yes	
Wall Paper	Yes	
Download Melody / Wallpaper	Yes	Over web browser
SIM Toolkit	Yes	
MMS	Yes	Google MMS Client
EONS	Yes	
CPHS	Yes	V4.2
Camera	Yes	8M AF / Digital Zoom : x4
Bluetooth	Yes	Ver. 4.0
FM radio	Yes	
GPRS	Yes	Class 12
EDGE	Yes	Class 12(Rx only)
Hold / Retrieve	Yes	
Conference Call	Yes	Max. 6
DTMF	Yes	
AMR	Yes	
IM	Yes	Google Hangout
Email	Yes	

3.1 WCDMA Rx Part

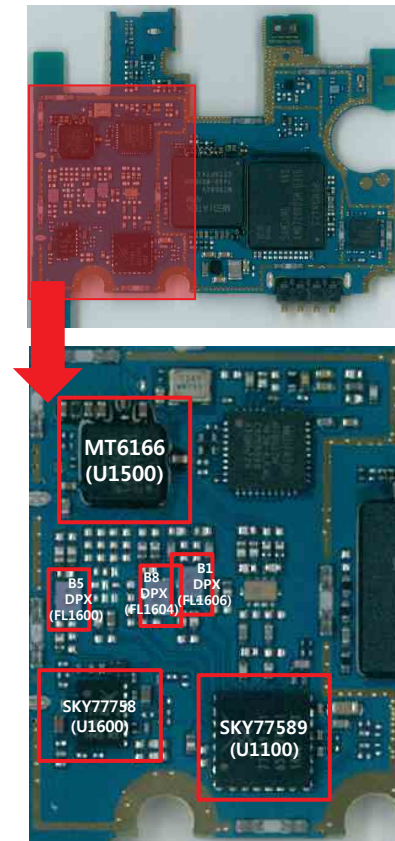
WCDMA B1/B5/B8 Rx

Checking Flow



Image

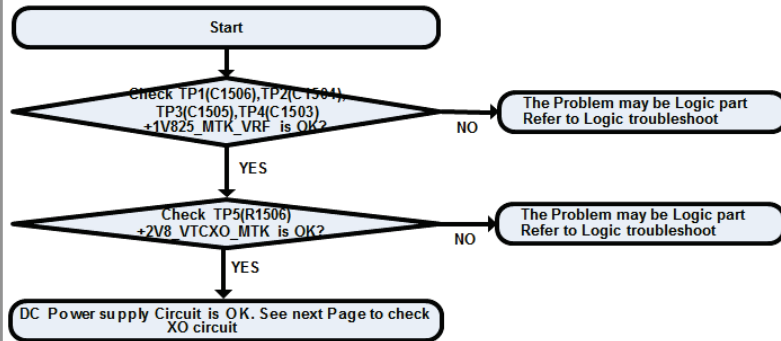
Main
Top



3.1 WCDMA Rx Part

Checking DC Power supply circuit

Checking Flow



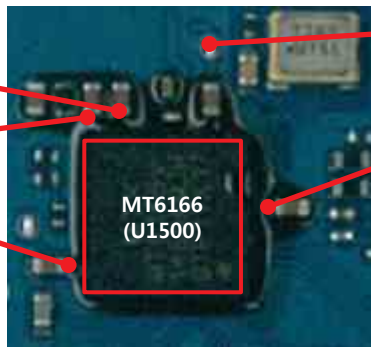
Image

Main Top

TP2: C1505
(+1V825MMTK_VRF)

TP4: C1503
(+1V825MMTK_VRF)

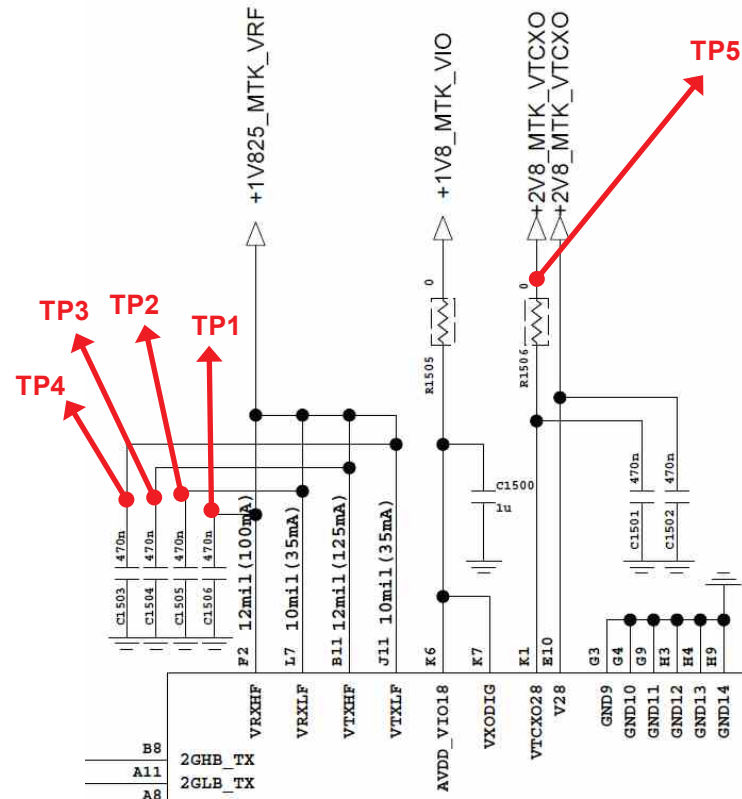
TP3: C1504
(+1V825MMTK_VRF)



TP5: R1506
(+2V8_MTK_VTCXO)

TP1: C1506
(+1V825MMTK_VRF)

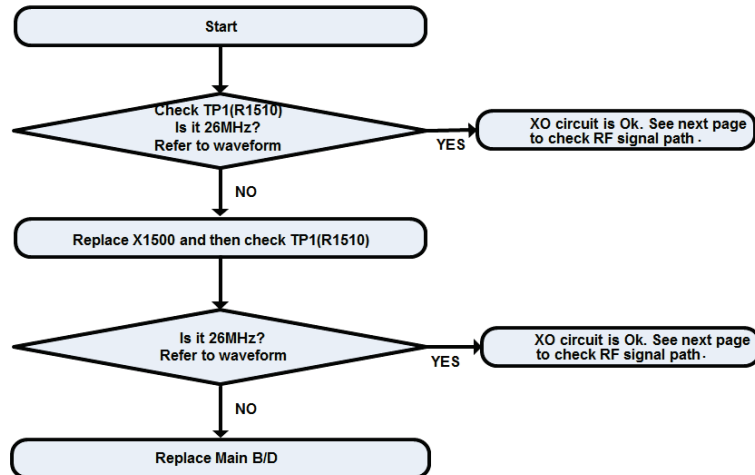
Circuit Diagram



3.1 WCDMA Rx Part

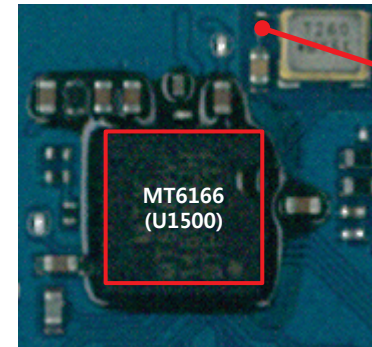
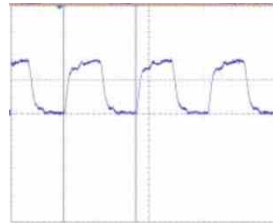
Checking MT6166 XO circuit

Checking Flow



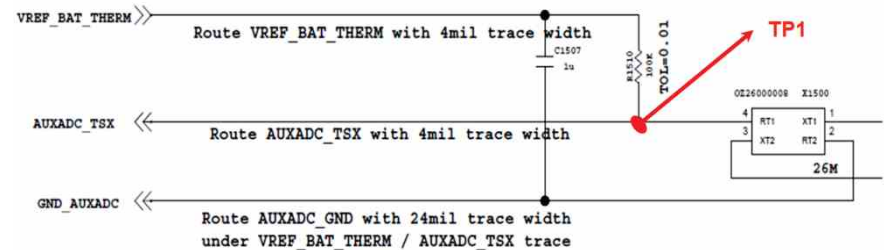
Image

Waveform



TP1: R1510
(MAIN_CLK_26MHz out)

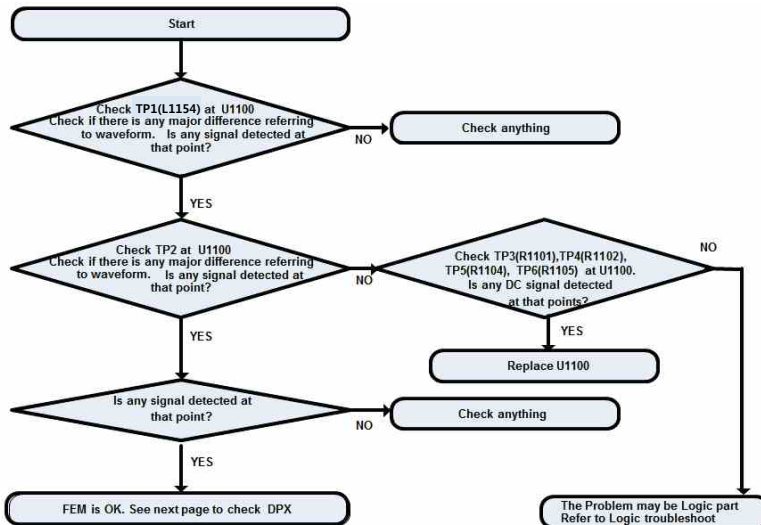
Circuit Diagram



3.1 WCDMA Rx Part

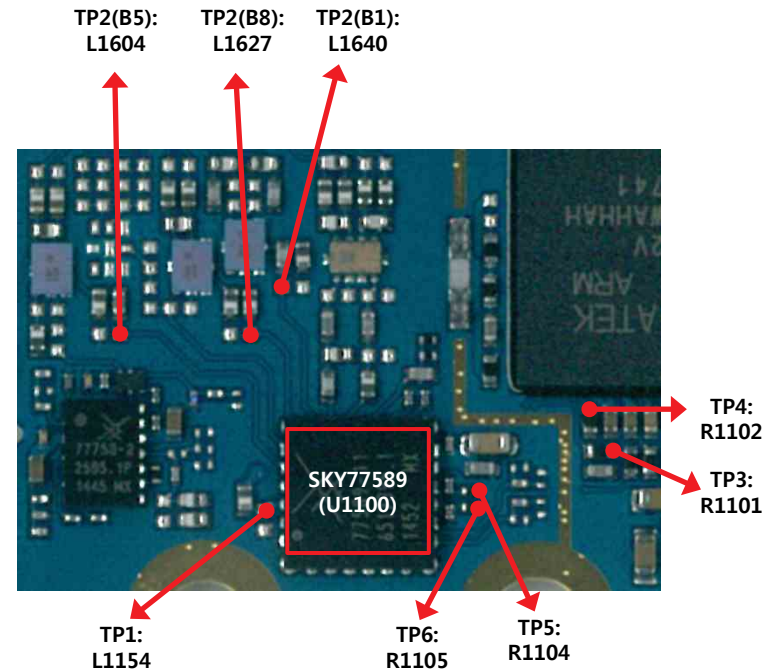
Checking RF signal path (FEM)

Checking Flow



Image

Main Top

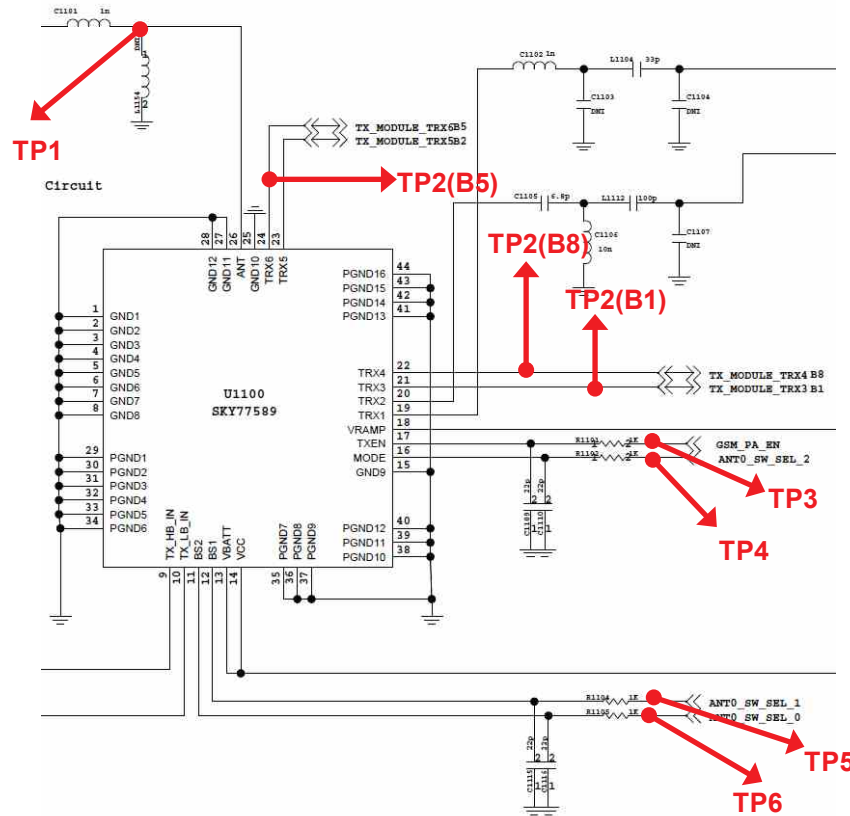


3. TROUBLE SHOOTING

3.1 WCDMA Rx Part

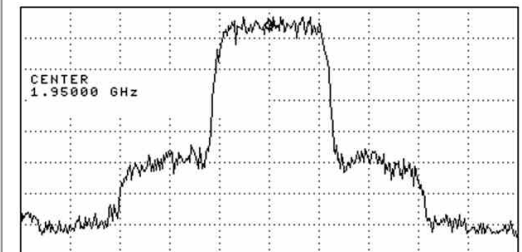
Checking RF signal path (FEM)

Circuit Diagram



Image

Waveform



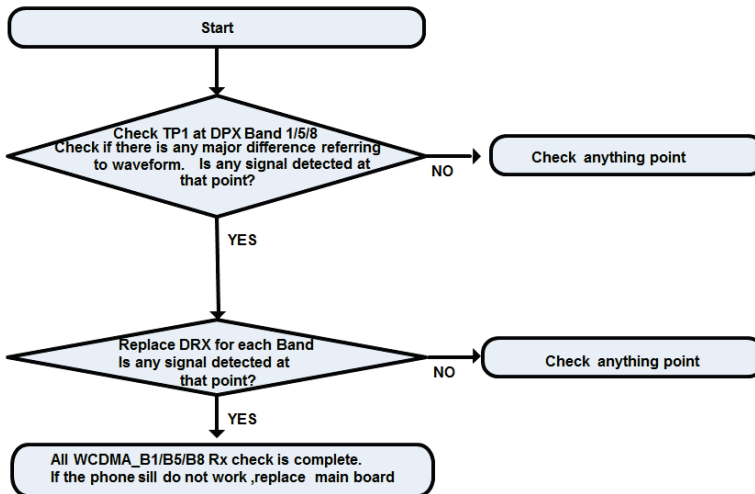
TP1, TP2 waveform frequency list

Band	TX Freq (MHz)	RX Freq (MHz)
B1	1920 ~ 1980	2110 ~ 2170
B5	824 ~ 849	869 ~ 894
B8	880 ~ 915	925 ~ 960

3.1 WCDMA Rx Part

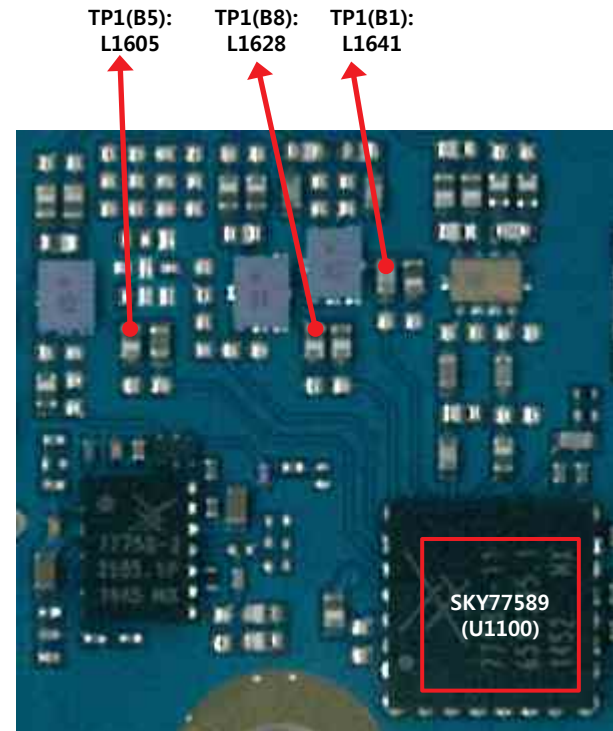
Checking RF signal path (DPX)

Checking Flow



Image

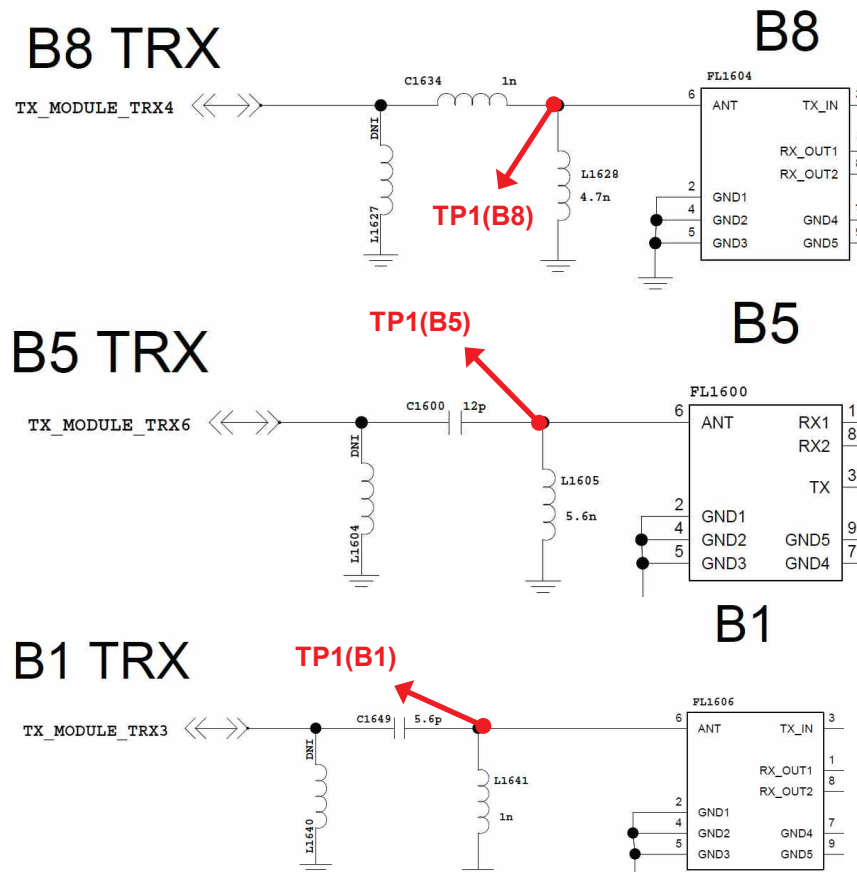
Main
Top



3.1 WCDMA Rx Part

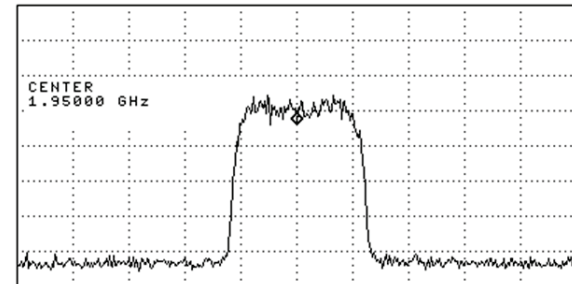
Checking RF signal path (DPX)

Circuit Diagram



Image

Waveform



TP1_RF IN

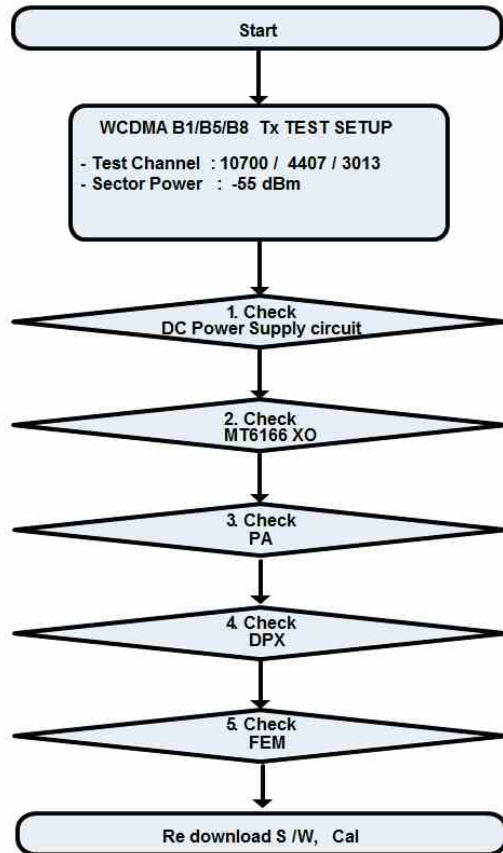
TP1 waveform frequency list

Band	RX Freq (MHz)
B1	2110 ~ 2170
B5	869 ~ 894
B8	925 ~ 960

3.2 WCDMA Tx Part

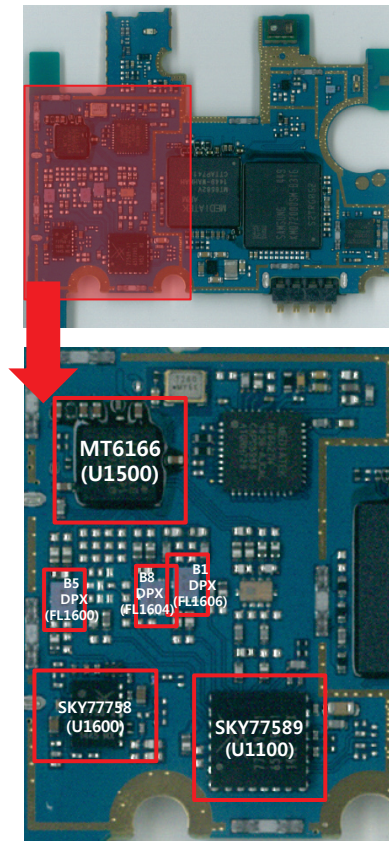
WCDMA B1/B5/B8 Tx

Checking Flow



Image

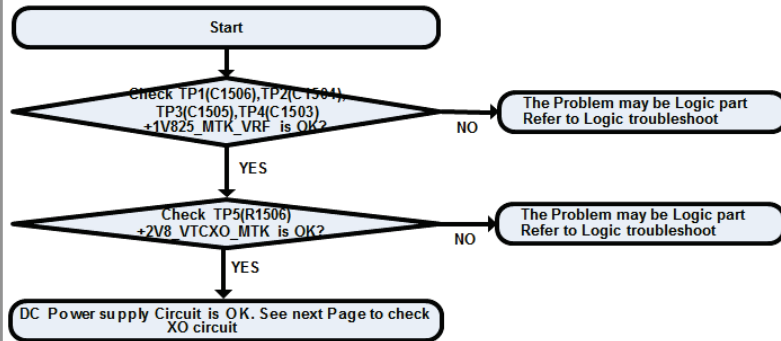
Main
Top



3.2 WCDMA Tx Part

Checking DC Power supply circuit

Checking Flow



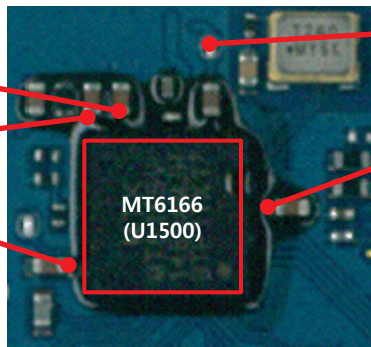
Image

Main Top

TP2: C1505
(+1V825MMTK_VRF)

TP4: C1503
(+1V825MMTK_VRF)

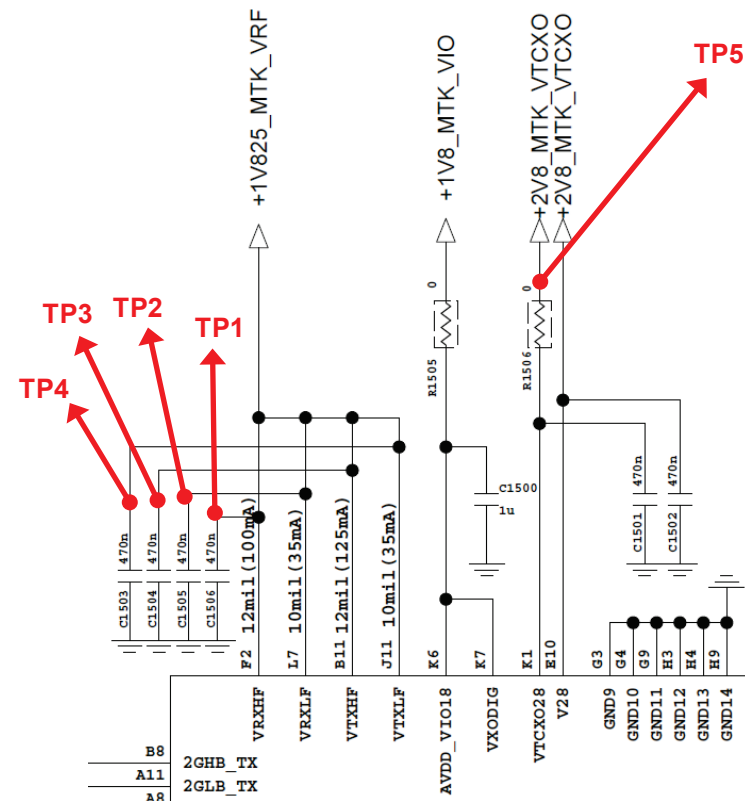
TP3: C1504
(+1V825MMTK_VRF)



TP5: R1506
(+2V8_MTK_VTCXO)

TP1: C1506
(+1V825MMTK_VRF)

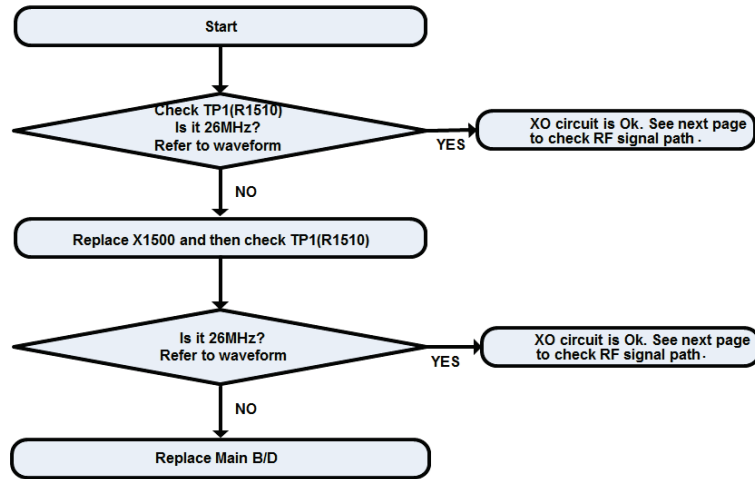
Circuit Diagram



3.2 WCDMA Tx Part

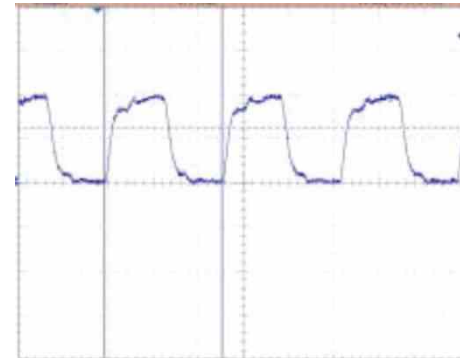
Checking MT6166 XO circuit

Checking Flow



Image

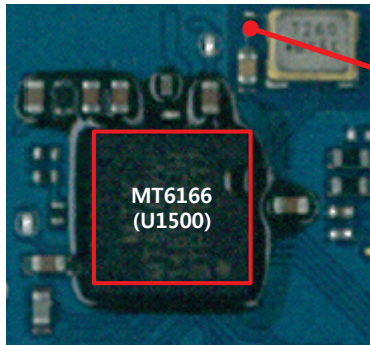
Waveform



3.2 WCDMA Tx Part

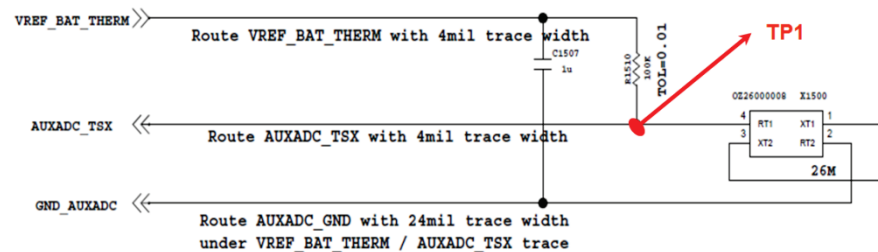
Checking MT6166 XO circuit

Image



TP1: R1510
(MAIN_CLK_26MHz out)

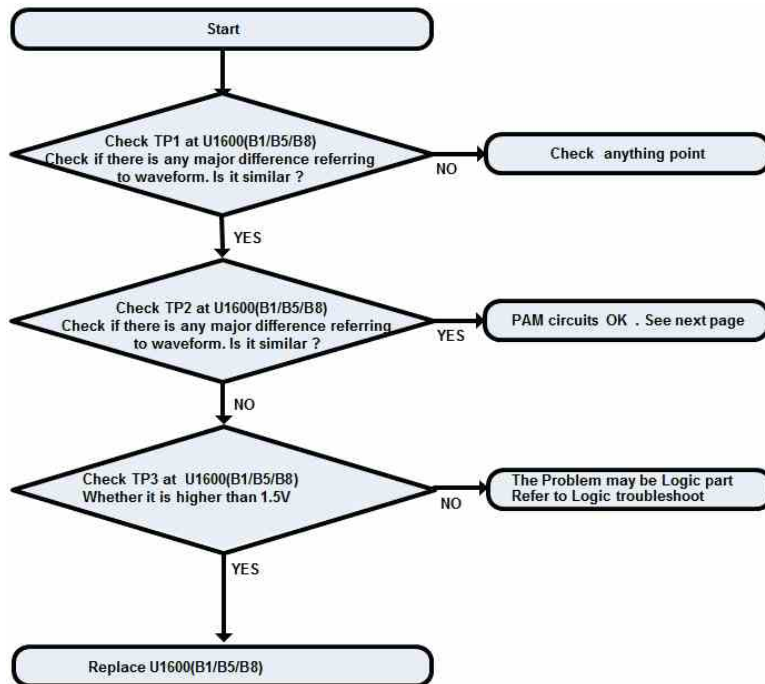
Circuit Diagram



3.2 WCDMA Tx Part

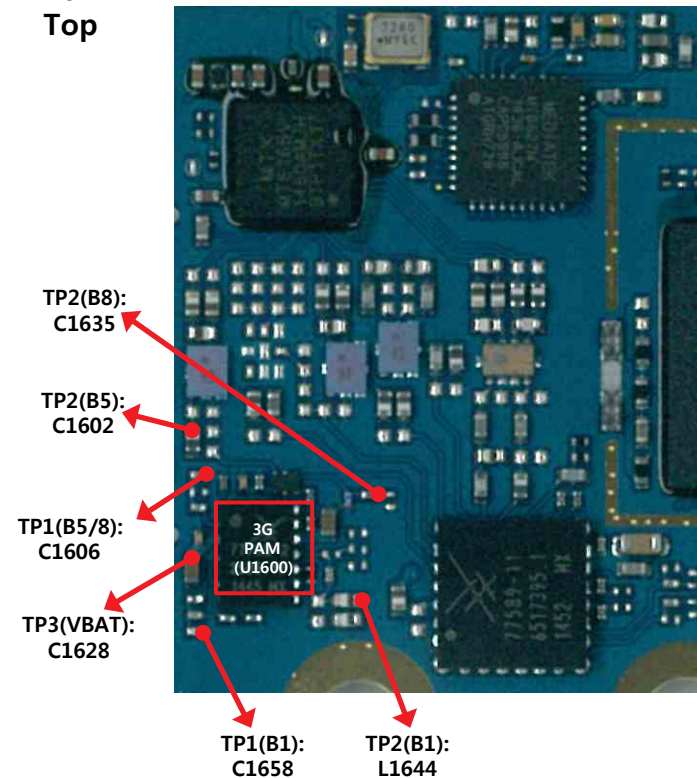
Checking B1/B5/B8 TX PA

Checking Flow



Image

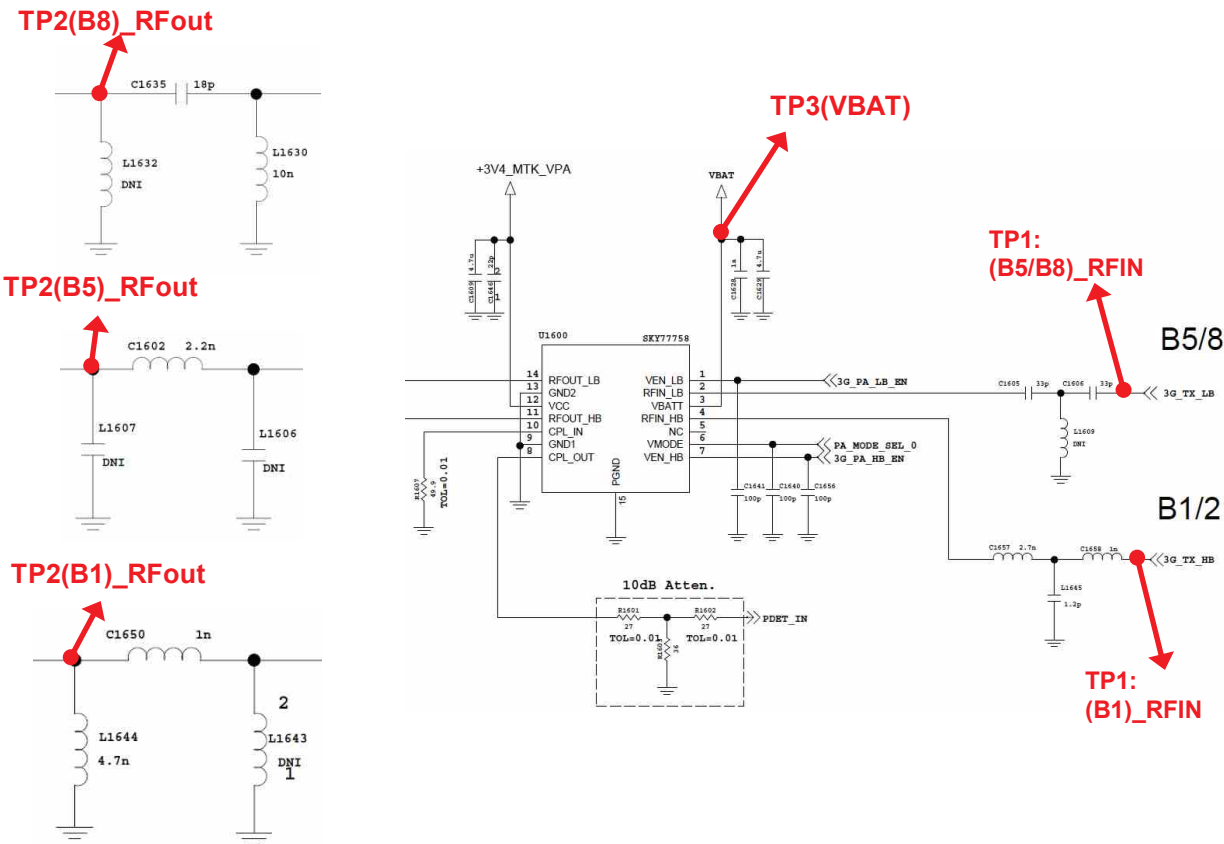
Main Top



3.2 WCDMA Tx Part

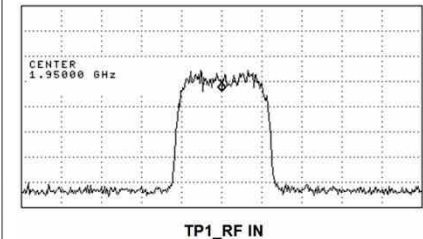
Checking B1/B5/B8 TX PA

Circuit Diagram



Image

Waveform



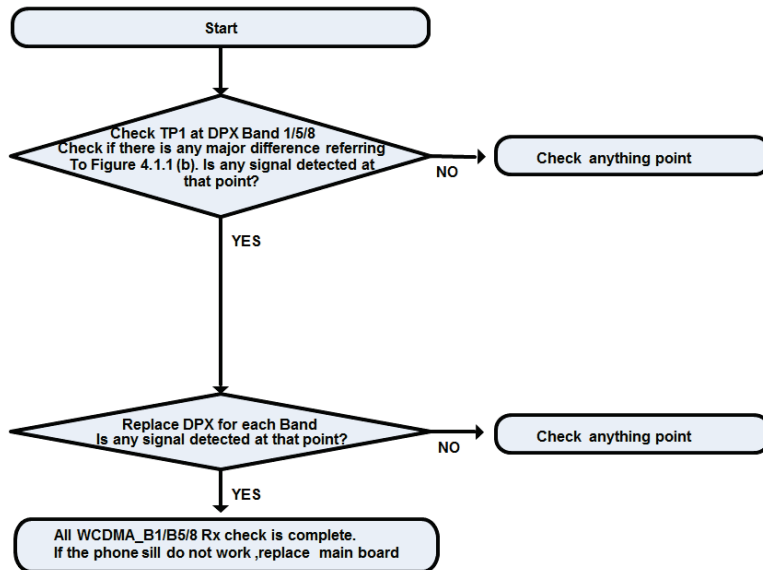
TP1 waveform frequency list

Band	TX Freq (MHz)
B1	1920 ~ 1980
B5	824 ~ 849
B8	880 ~ 915

3.2 WCDMA Tx Part

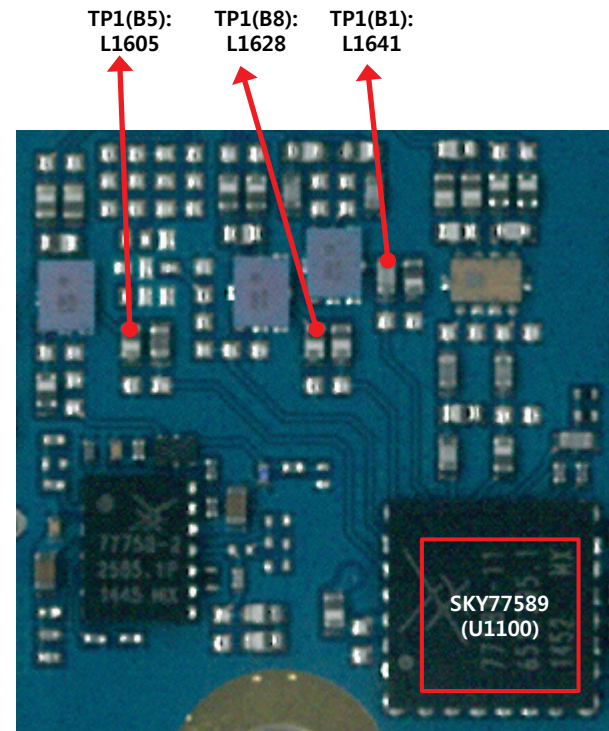
Checking RF signal path (DPX)

Checking Flow



Image

Main
Top



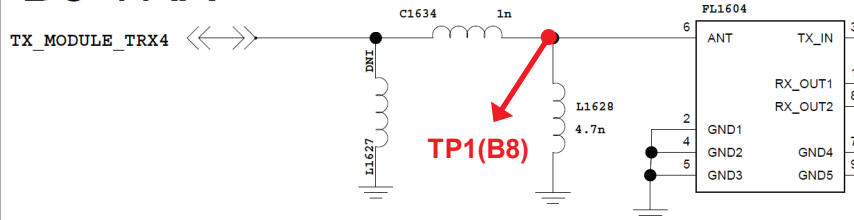
3.2 WCDMA Tx Part

Checking RF signal path (DPX)

Circuit Diagram

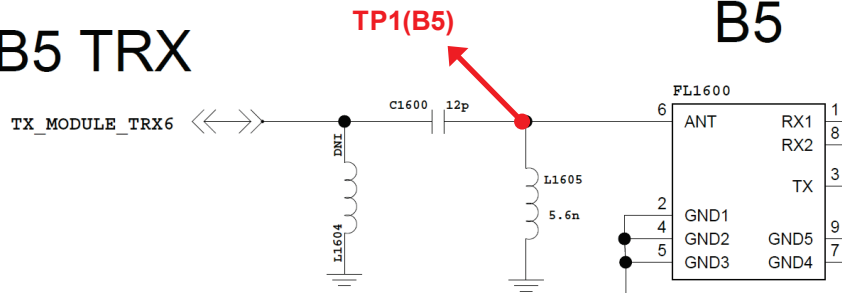
B8 TRX

B8



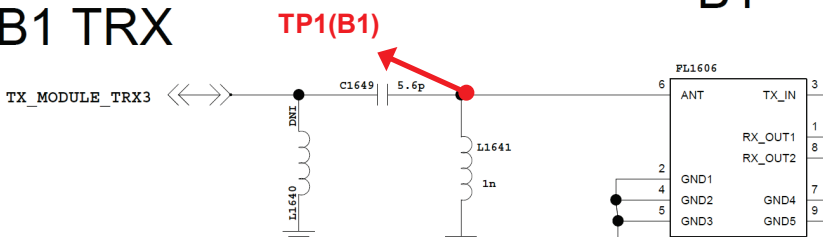
B5 TRX

B5



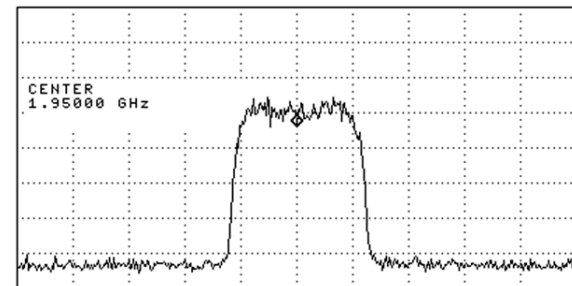
B1 TRX

B1



Image

Waveform

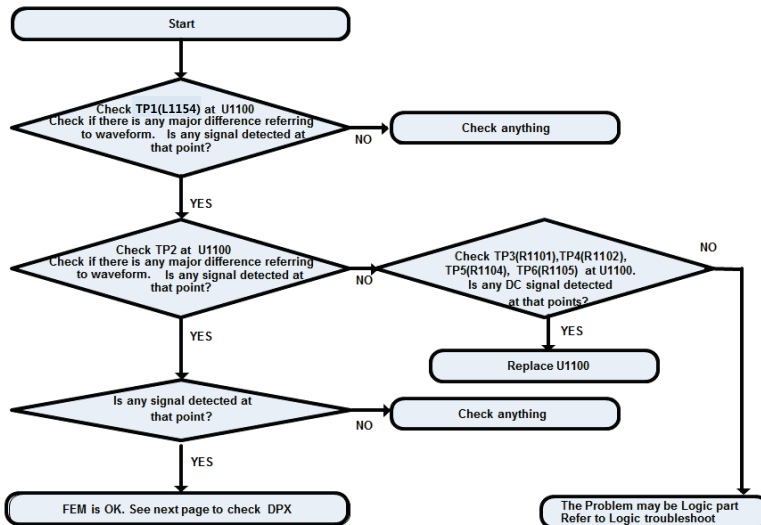


TP1_RF IN

3.2 WCDMA Tx Part

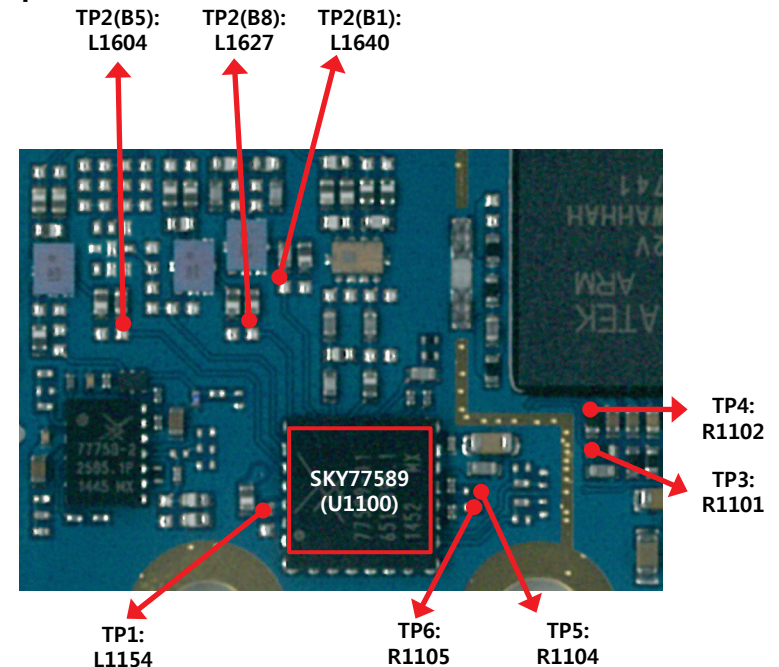
Checking RF signal path (FEM)

Checking Flow



Image

Main
Top

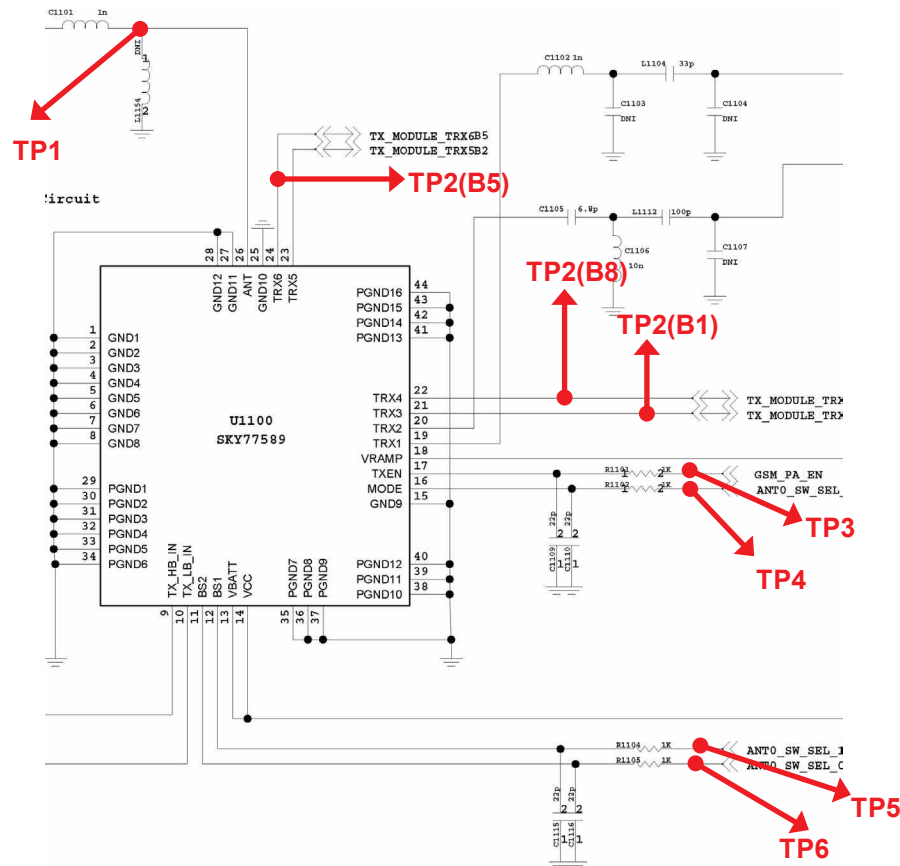


3. TROUBLE SHOOTING

3.2 WCDMA Tx Part

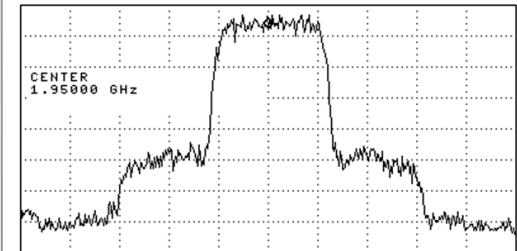
Checking RF signal path (FEM)

Circuit Diagram



Image

Waveform



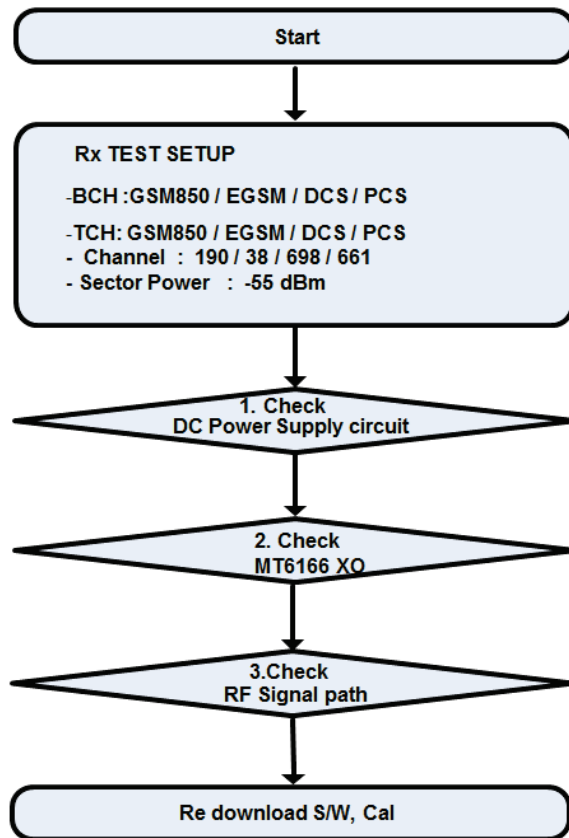
TP1 waveform frequency list

Band	TX Freq (MHz)
B1	1920 ~ 1980
B5	824 ~ 849
B8	880 ~ 915

3.3 GSM Rx Part

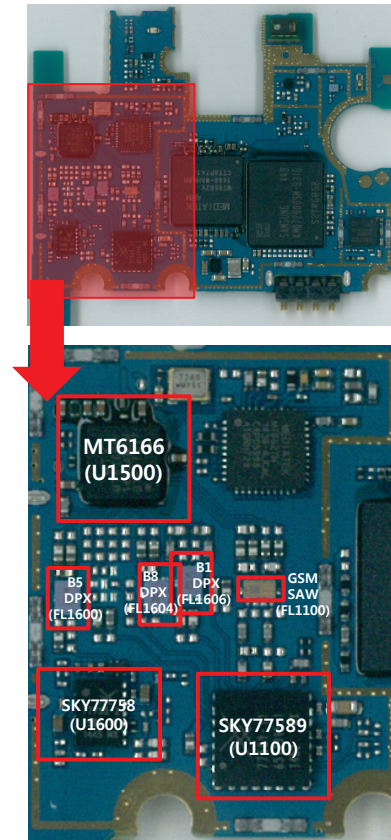
GSM850/900/DCS/PCS Rx

Checking Flow



Image

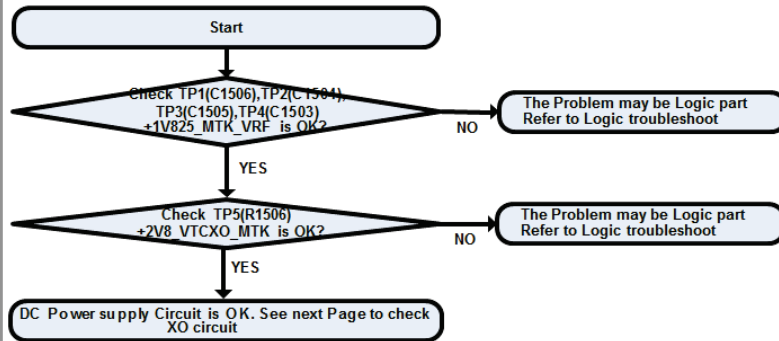
Main
Top



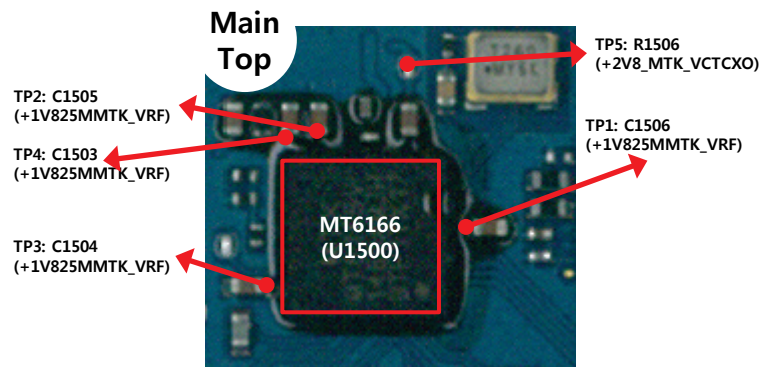
3.3 GSM Rx Part

Checking DC Power supply circuit

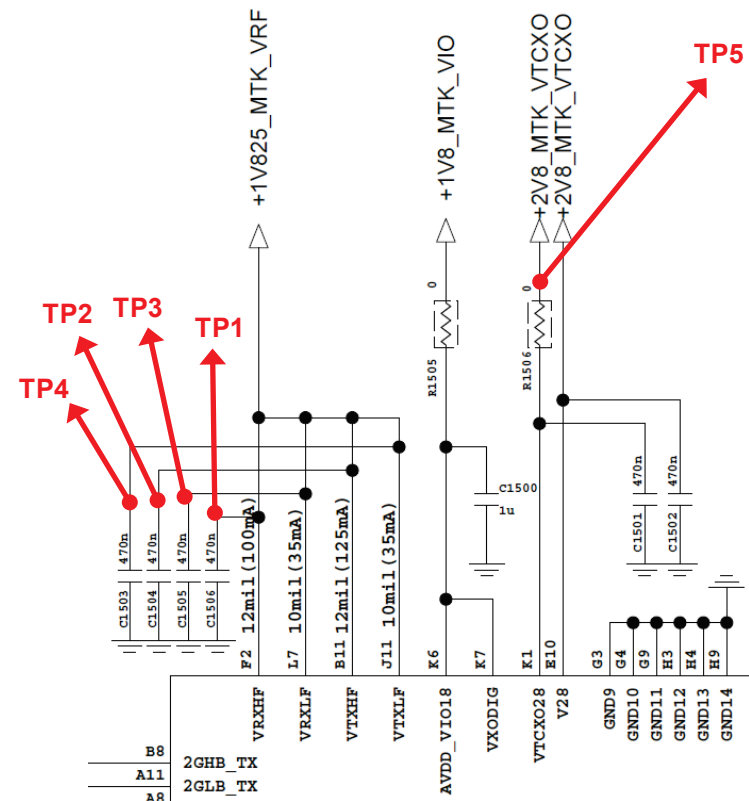
Checking Flow



Image



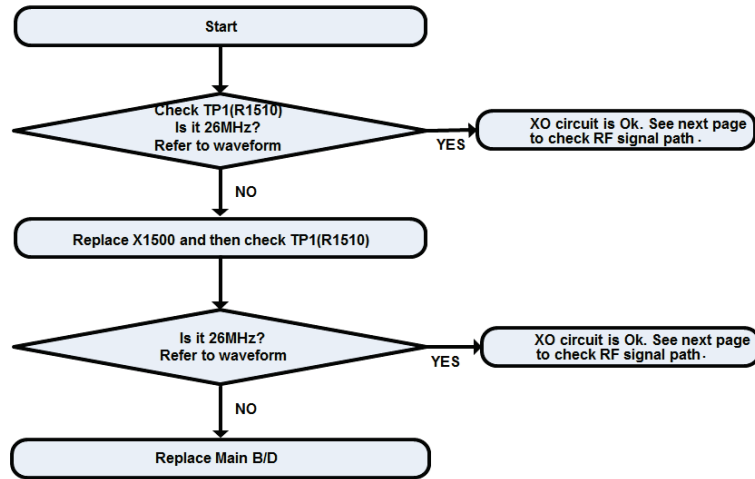
Circuit Diagram



3.3 GSM Rx Part

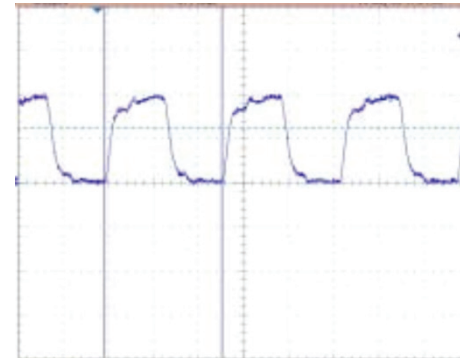
Checking MT6166 XO circuit

Checking Flow



Image

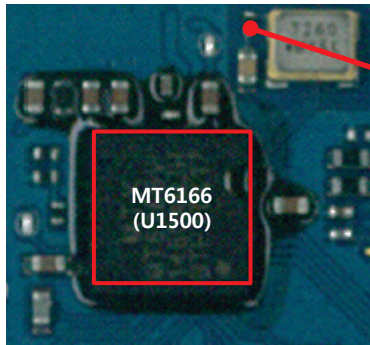
Waveform



3.3 GSM Rx Part

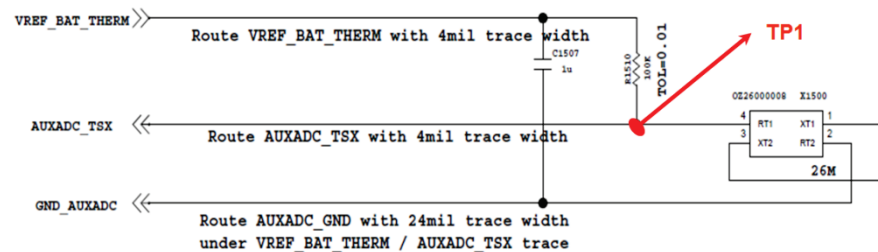
Checking MT6166 XO circuit

Image



TP1: R1510
(MAIN_CLK_26MHz out)

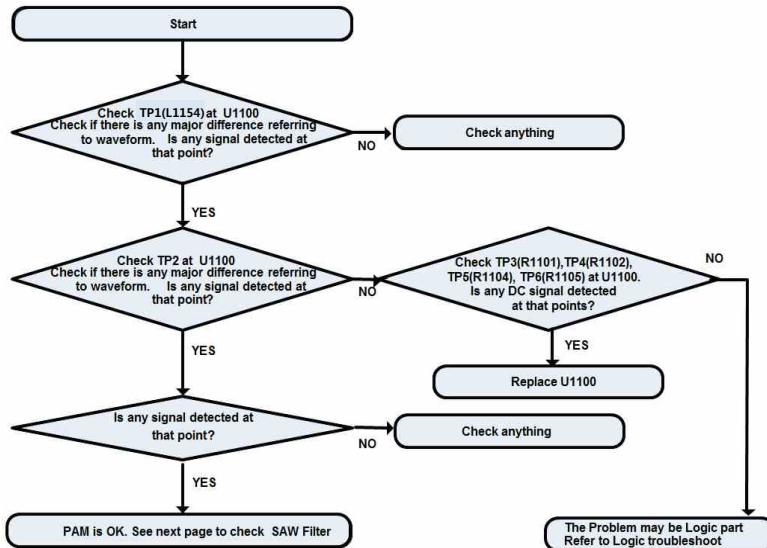
Circuit Diagram



3.3 GSM Rx Part

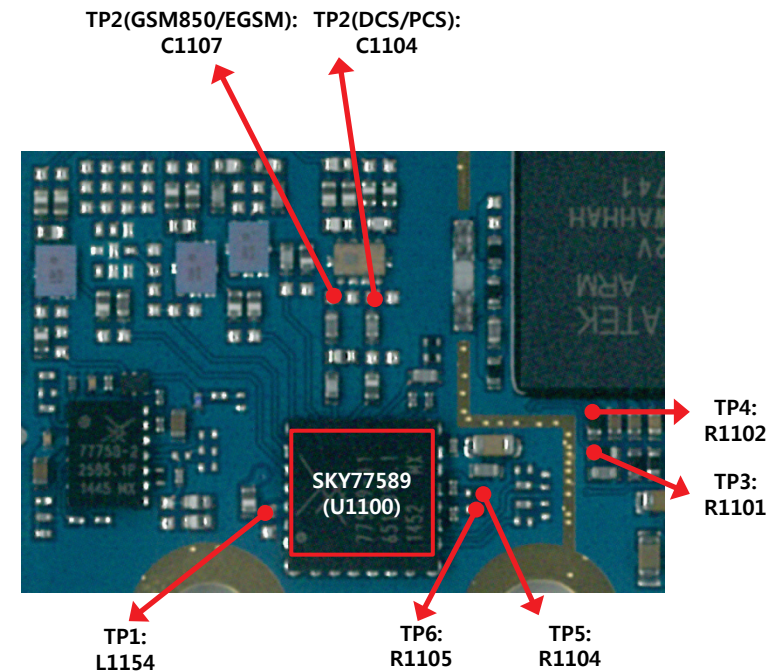
Checking RF signal path (FEM)

Checking Flow



Image

Main Top

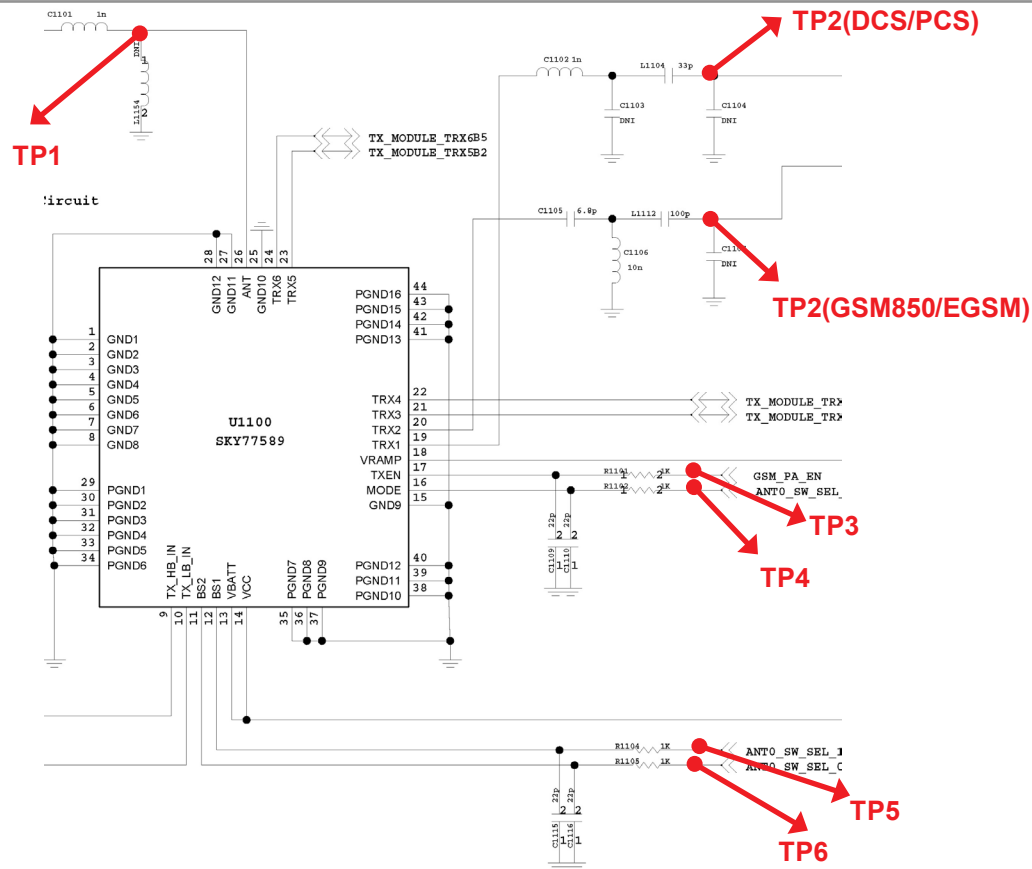


3. TROUBLE SHOOTING

3.3 GSM Rx Part

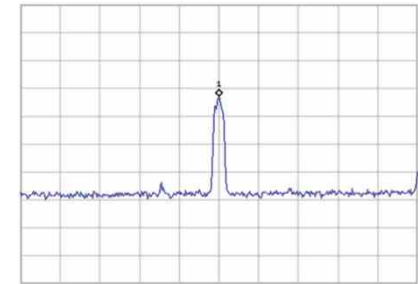
Checking RF signal path (FEM)

Circuit Diagram



Image

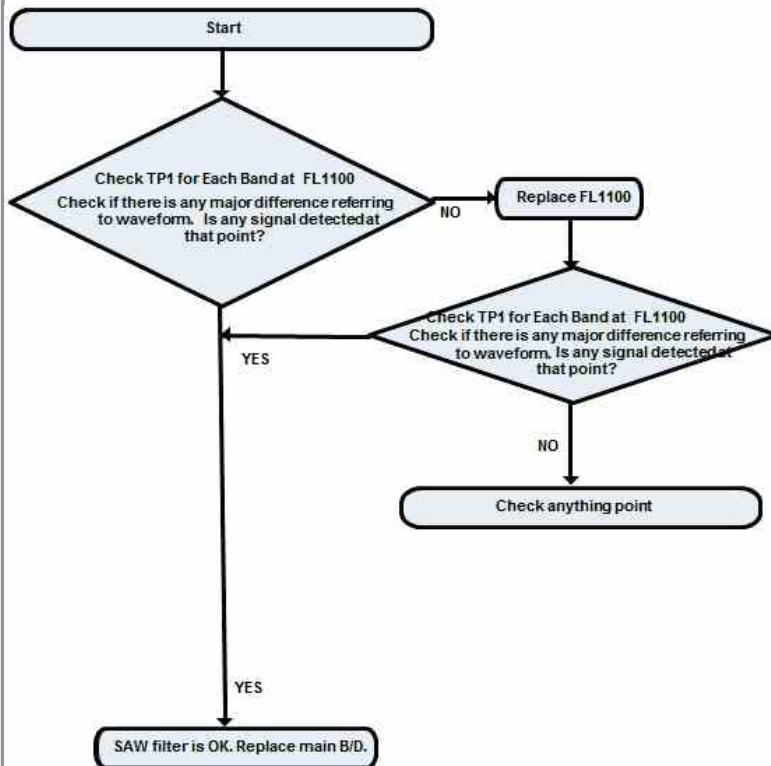
Waveform



3.3 GSM Rx Part

Checking RF Flow(SAW filter)

Checking Flow



Image

Main
Top

TP1(GSM850/EGSM):
L1111

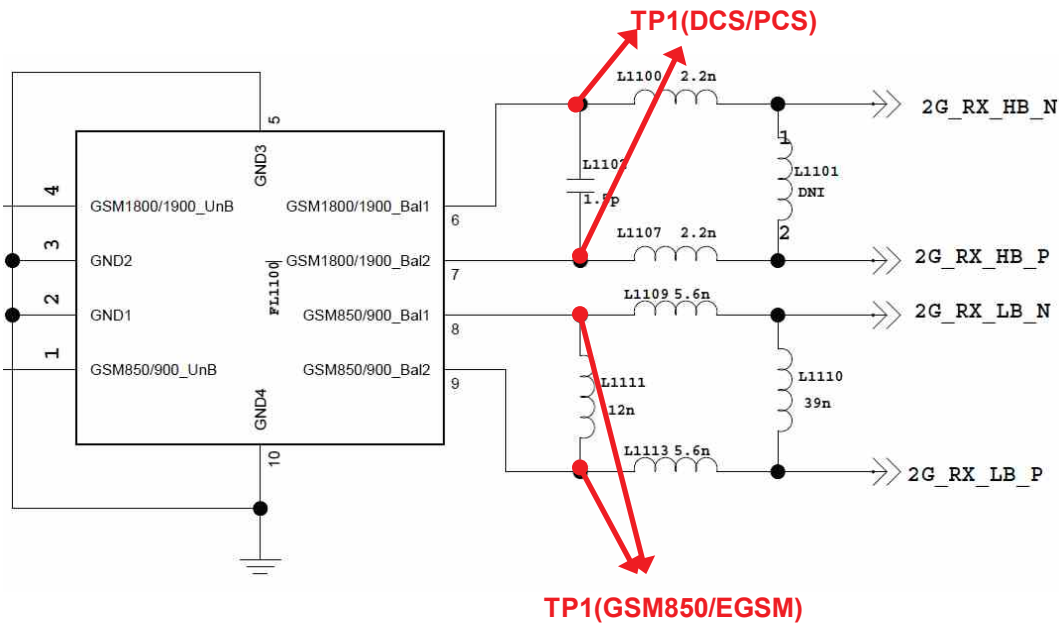
TP1(DCS/PCS):
L1102



3.3 GSM Rx Part

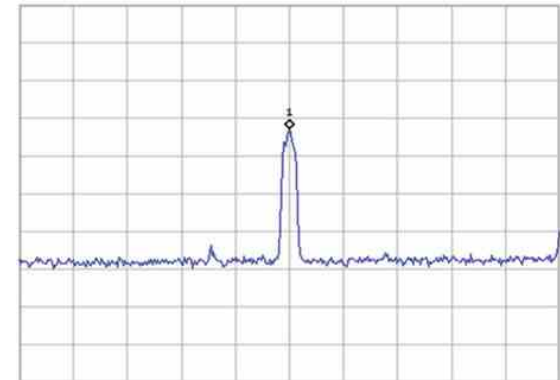
Checking RF signal path (FEM)

Circuit Diagram



Image

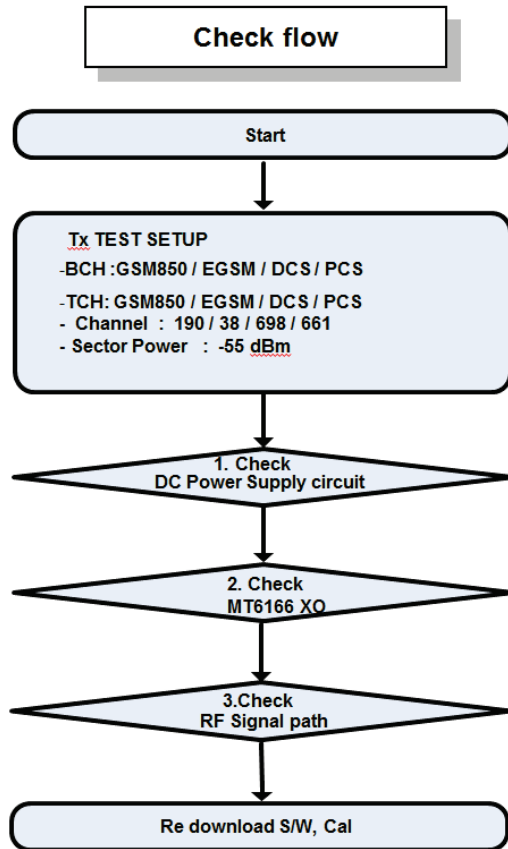
Waveform



3.4 GSM Tx Part

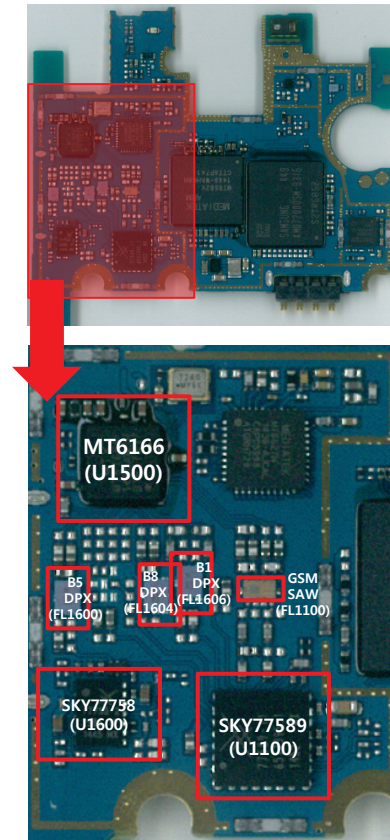
GSM850/900/DCS/PCS Tx

Checking Flow



Image

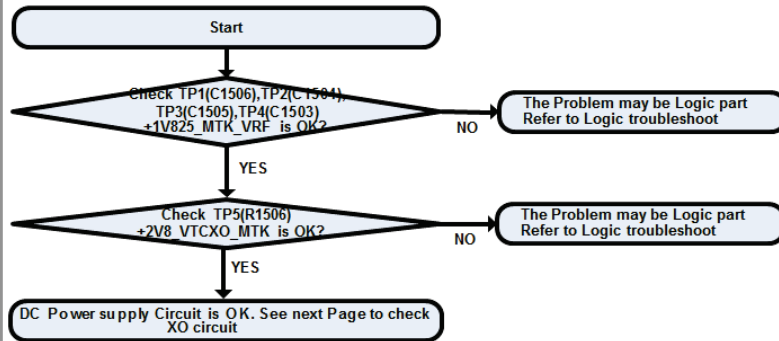
Main
Top



3.4 GSM Tx Part

Checking DC Power supply circuit

Checking Flow



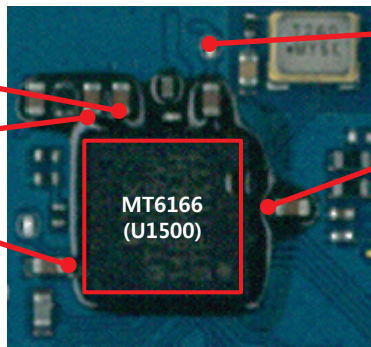
Image

Main Top

TP2: C1505
(+1V825MMTK_VRF)

TP4: C1503
(+1V825MMTK_VRF)

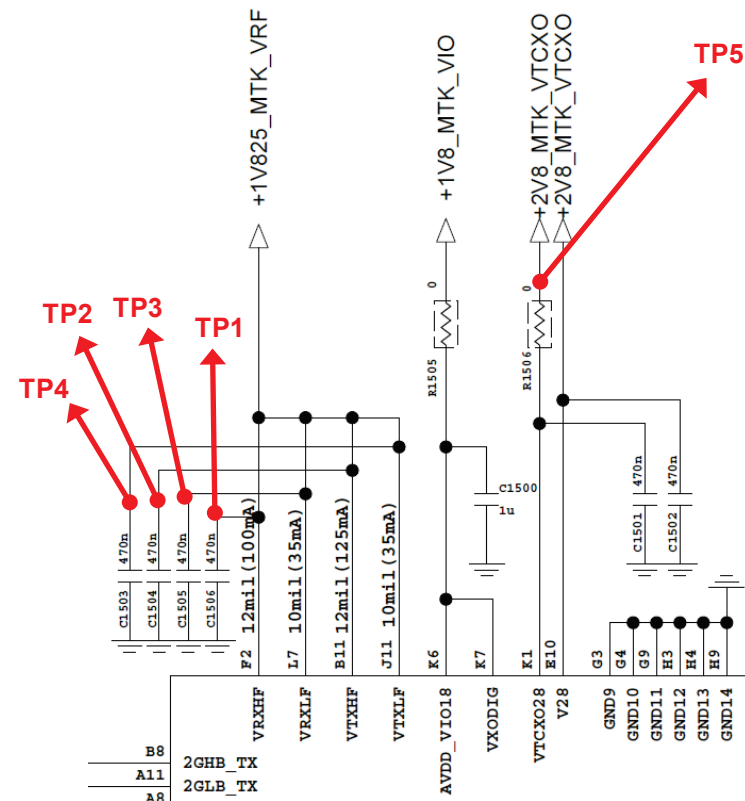
TP3: C1504
(+1V825MMTK_VRF)



TP5: R1506
(+2V8_MTK_VTCXO)

TP1: C1506
(+1V825MMTK_VRF)

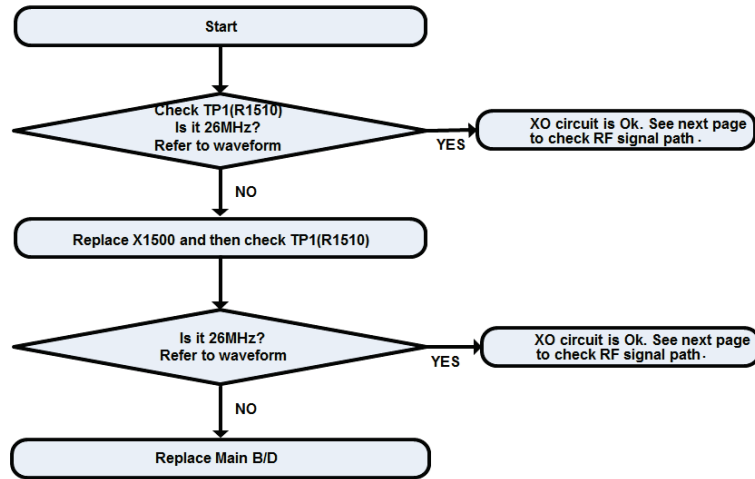
Circuit Diagram



3.4 GSM Tx Part

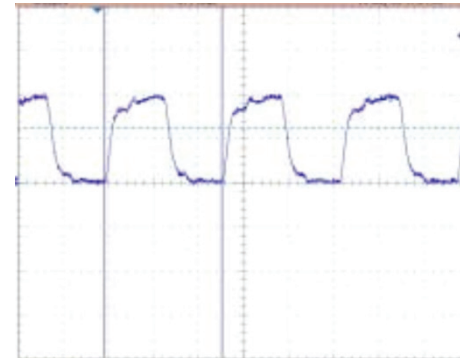
Checking MT6166 XO circuit

Checking Flow



Image

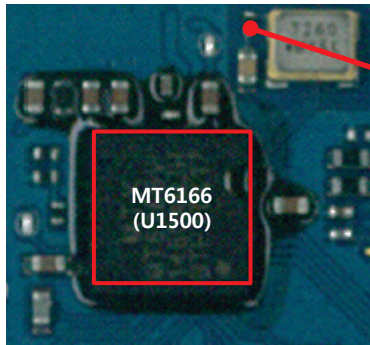
Waveform



3.4 GSM Tx Part

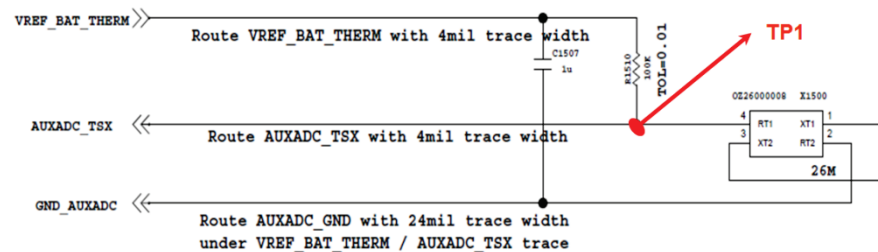
Checking MT6166 XO circuit

Image



TP1: R1510
(MAIN_CLK_26MHz out)

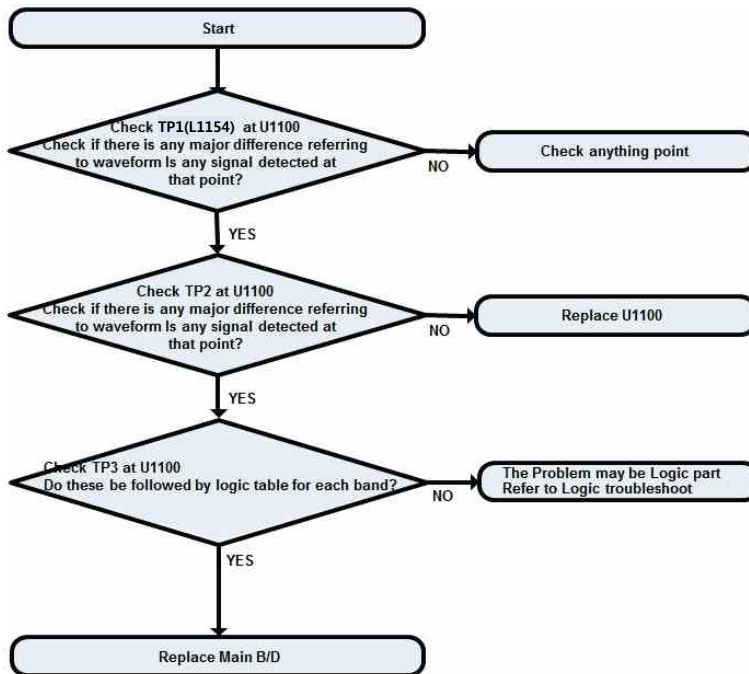
Circuit Diagram



3.4 GSM Tx Part

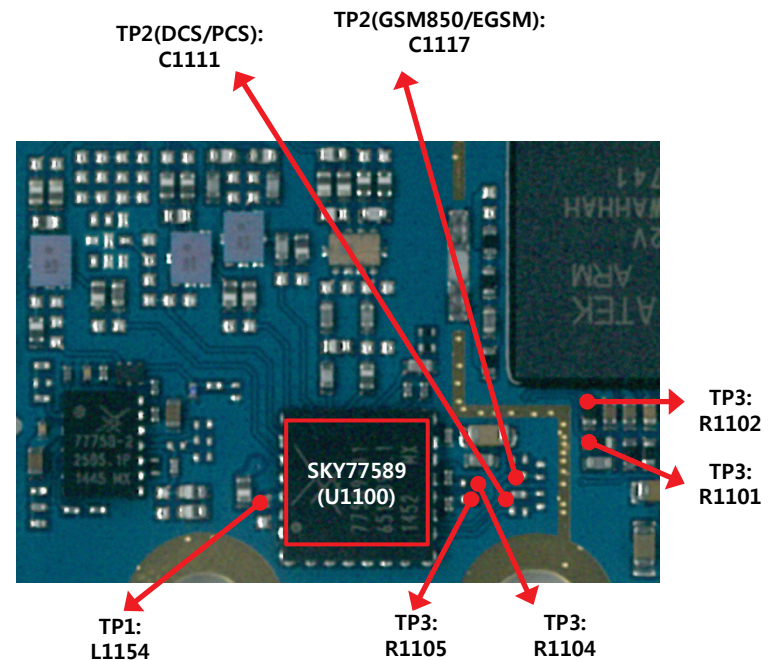
Checking RF signal path (FEM)

Checking Flow



Image

Main Top



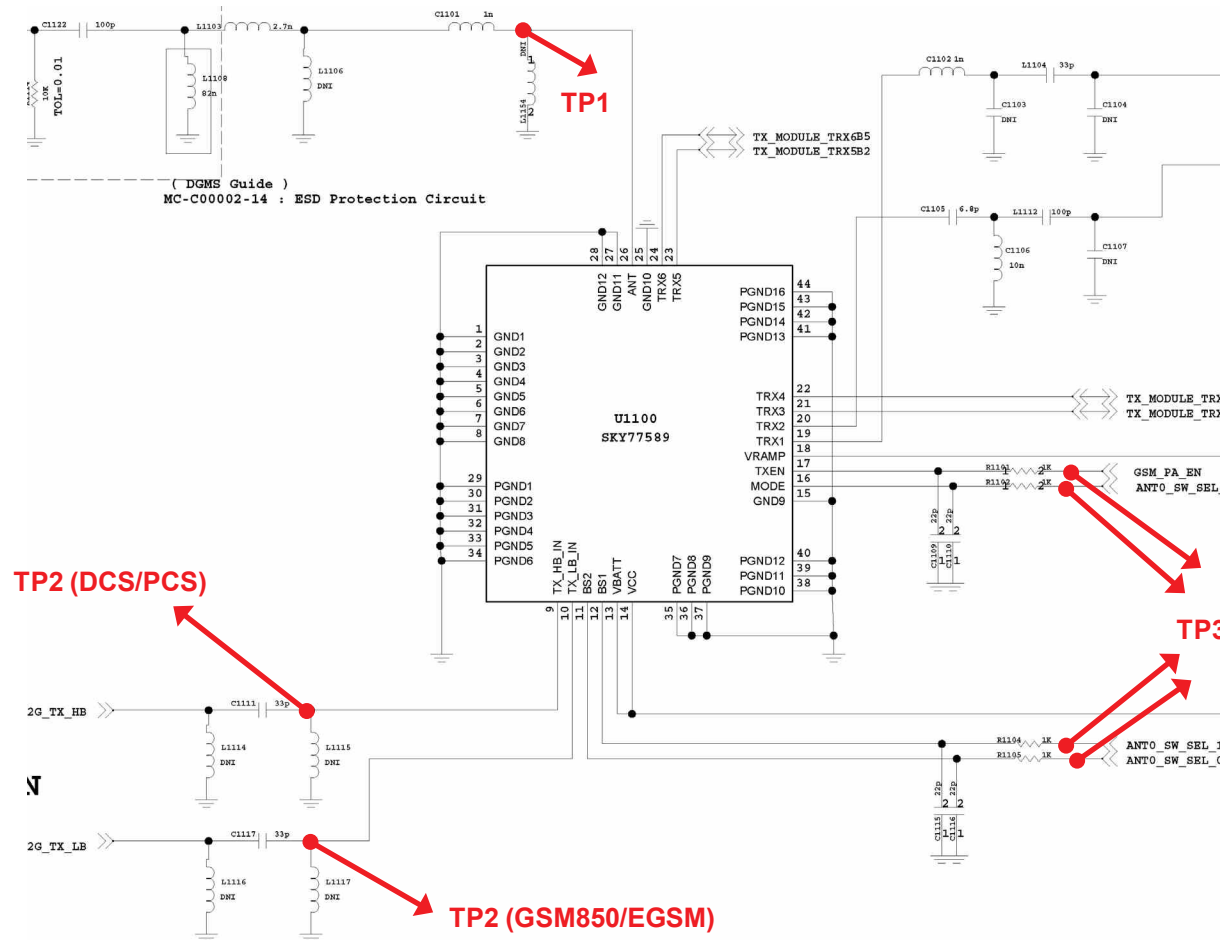
3. TROUBLE SHOOTING

3.4 GSM Tx Part

Checking RF signal path (FEM)

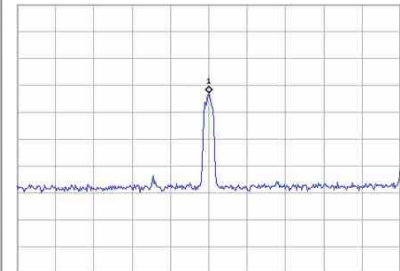
Circuit Diagram

Image



		TXEN	Mode	BS1	BS2
		GSM_PA_EN	ANT_SEL2	ANT_SEL1	ANT_SEL0
Stanby	Stanby	0	0	0	0
LB_TX	G_LB_TX	1	0	0	1
HB_TX	G_HB_TX	1	0	1	1
TRX1	B1	0	1	0	0
TRX2	B8	0	1	1	0
TRX3	B5	0	1	0	1
TRX4	B2	0	1	1	1
TRX5	G_LB_Rx	0	0	1	0
TRX6	G_HB_Rx	0	0	0	1

Waveform

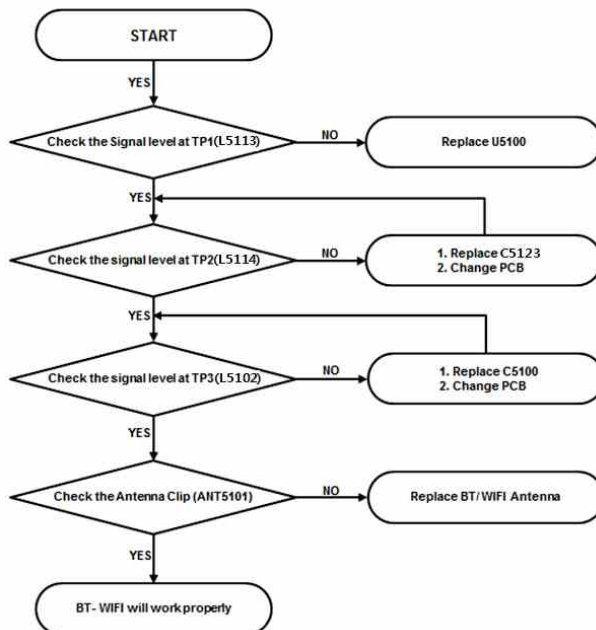


3. TROUBLE SHOOTING

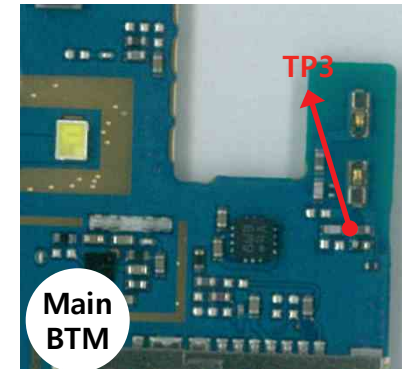
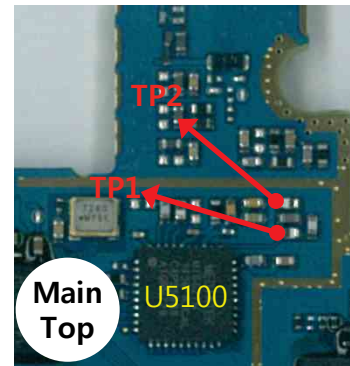
3.5 BT/WIFI Block

Check BT/WIFI path

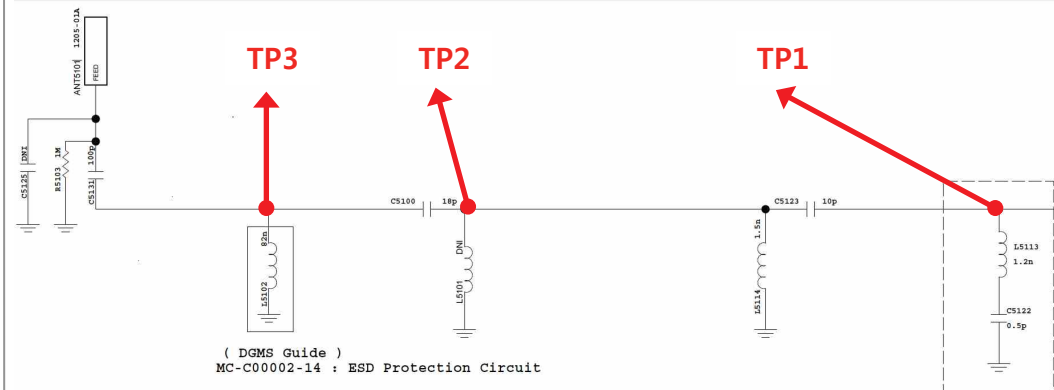
Checking Flow



Image



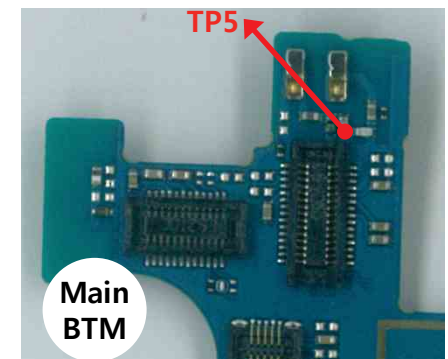
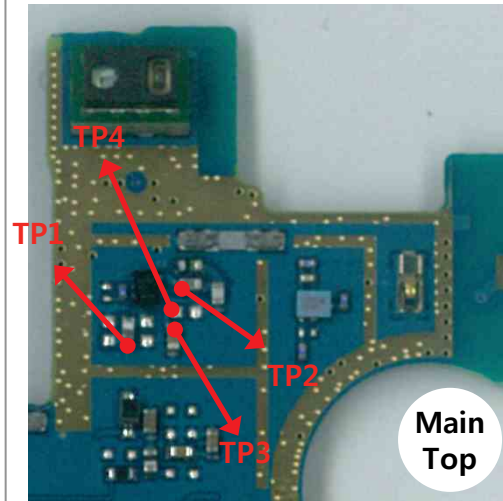
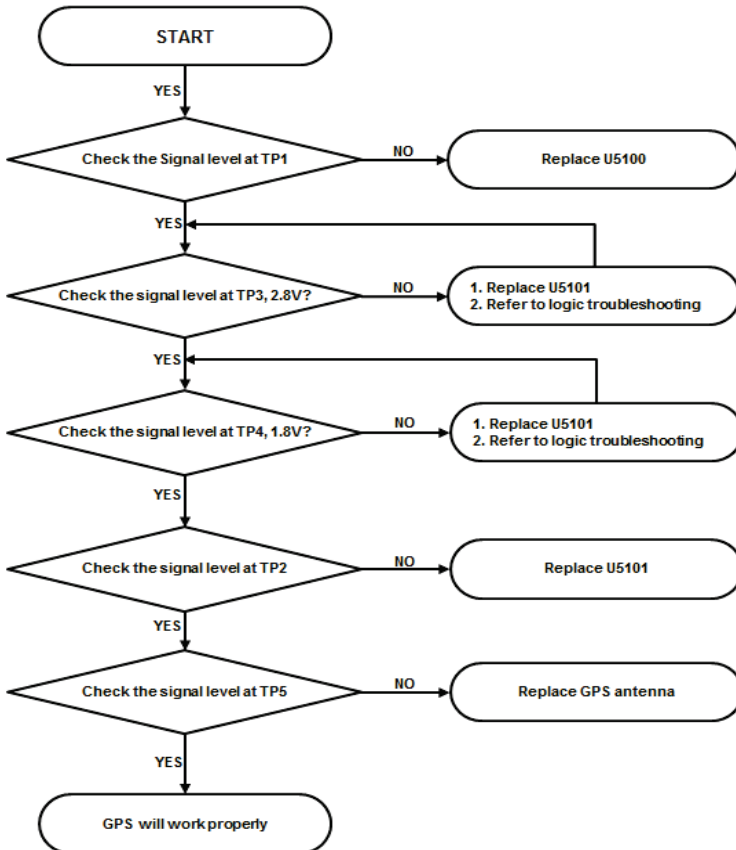
Circuit Diagram



3.6 GPS Block

Check GPS path

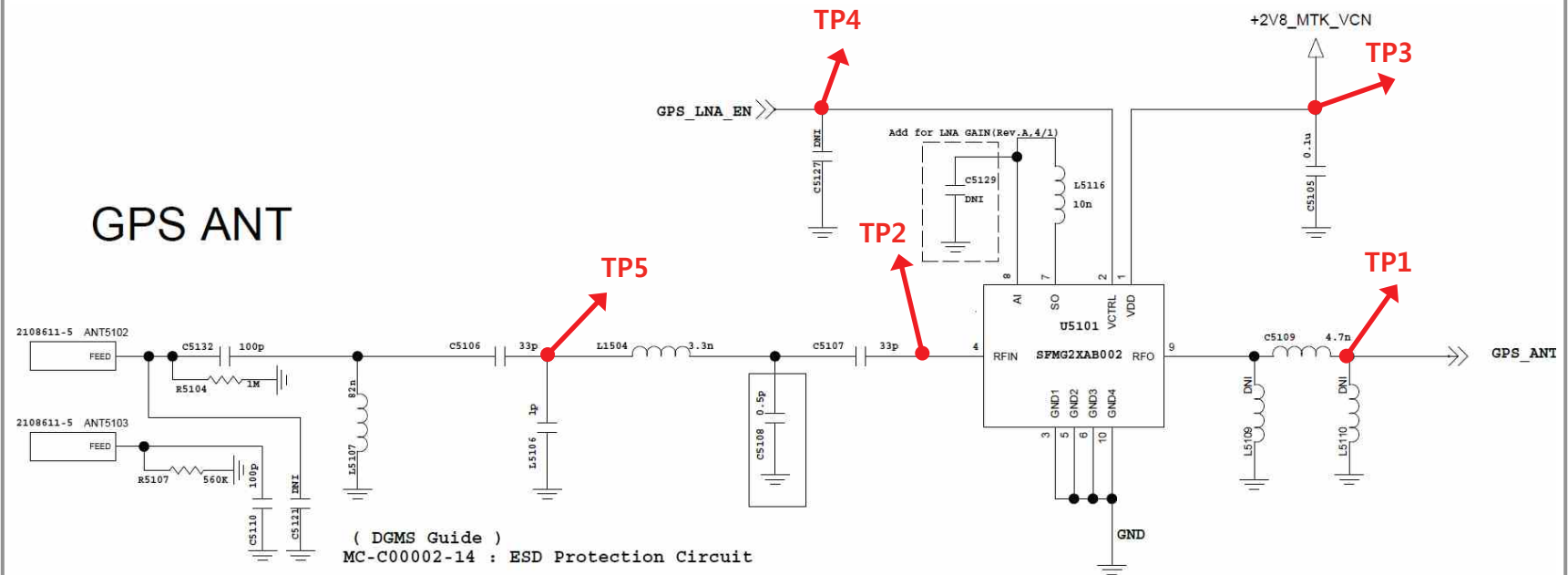
Checking Flow



3.6 GPS Block

Check GPS path

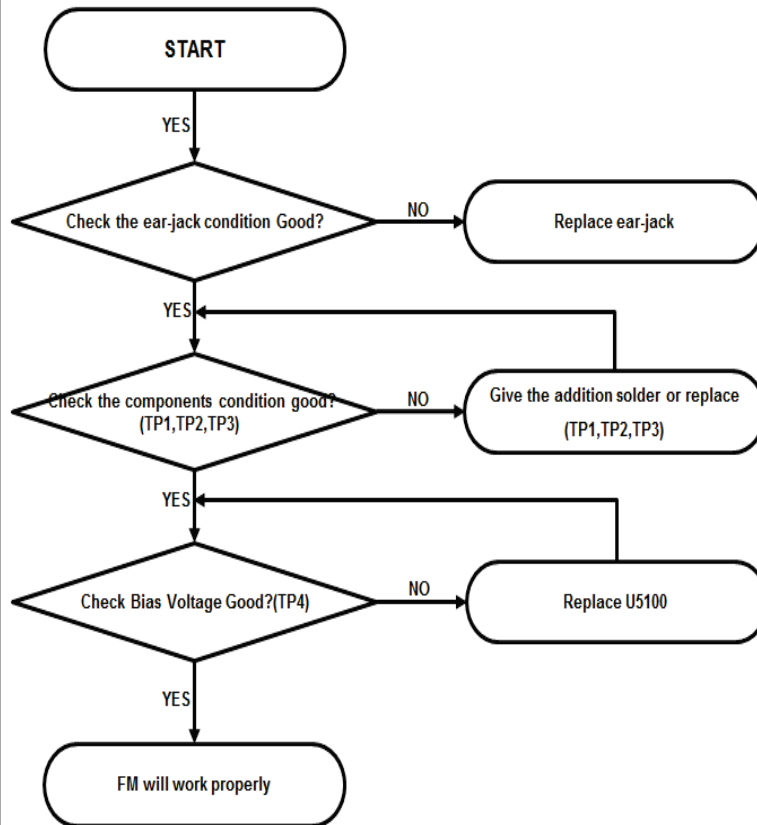
Circuit Diagram



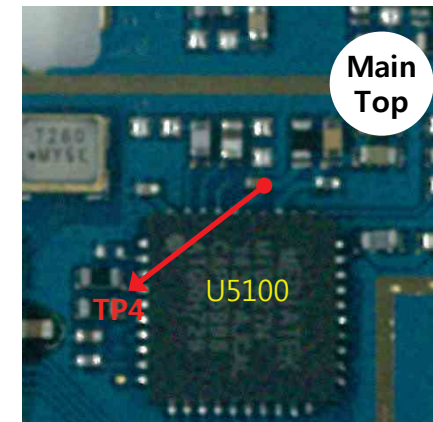
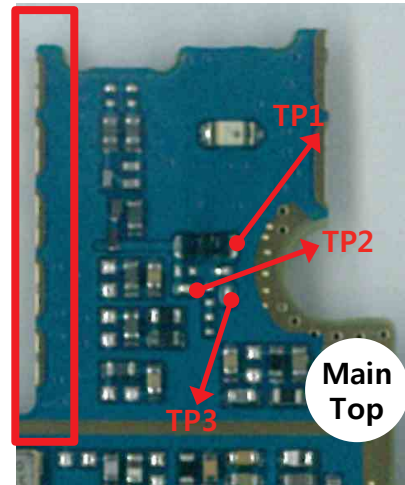
3.7 FM RADIO

Check FM radio path

Checking Flow



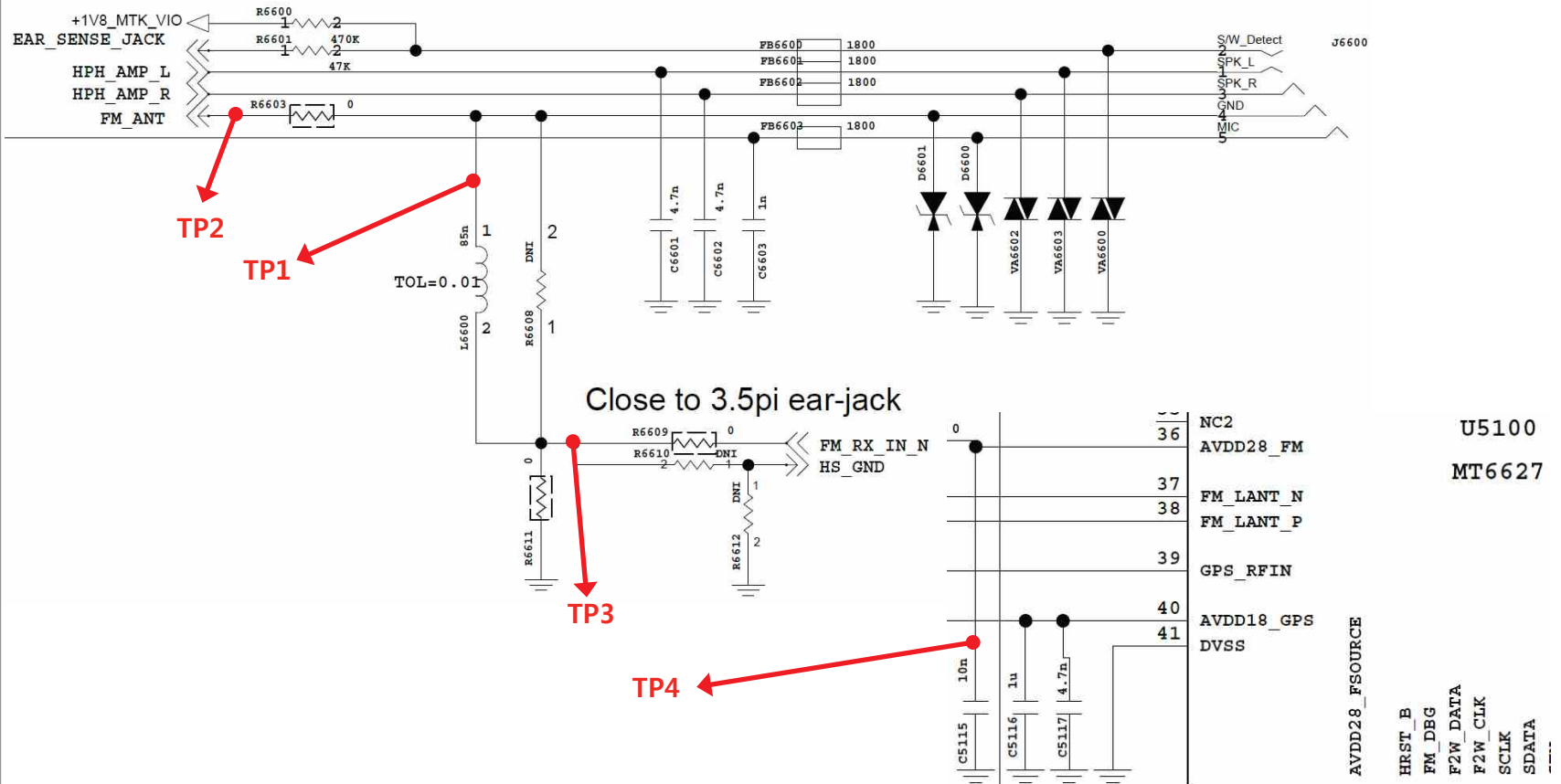
Image



3.7 FM RADIO

Check FM radio path

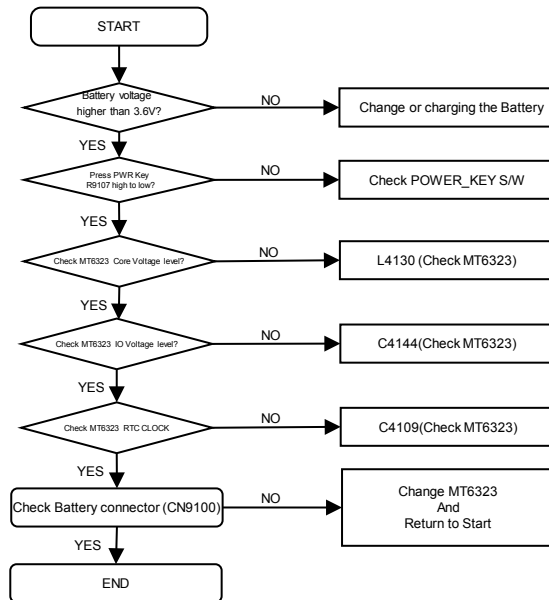
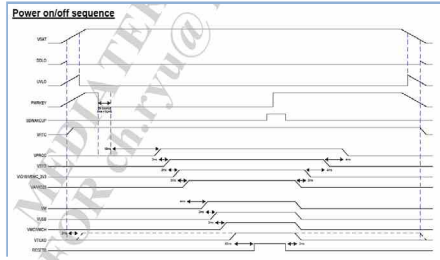
Circuit Diagram



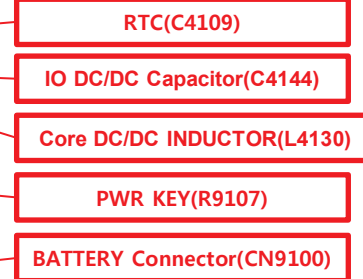
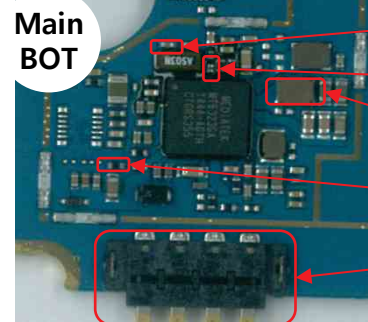
3.8 Power

The main power source of H402TV is provided by 1 chip which is MT6323

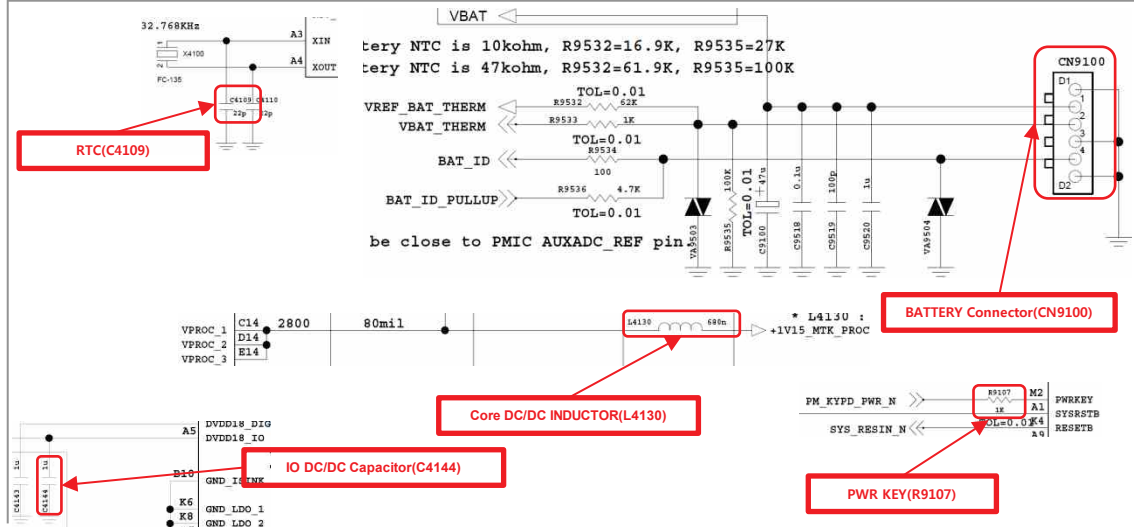
Checking Flow



Image



Circuit Diagram



3.9 Charging

H502TV uses Charger IC(linear Charger) for charging control. The Charger IC circuit includes the charging pass transistor detects the VDD and VBAT voltages, and its control driver. The Charger IC maximum charging current is set around 850mA.

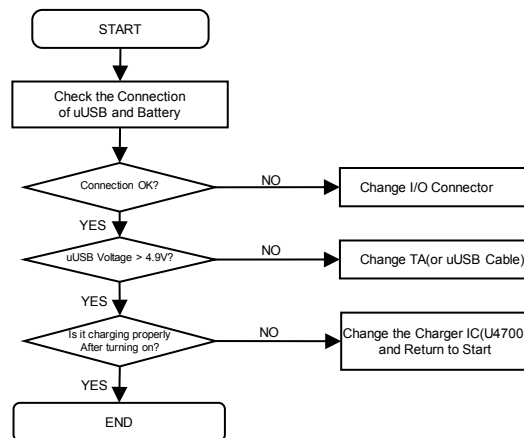
Checking Flow

Charging Procedure

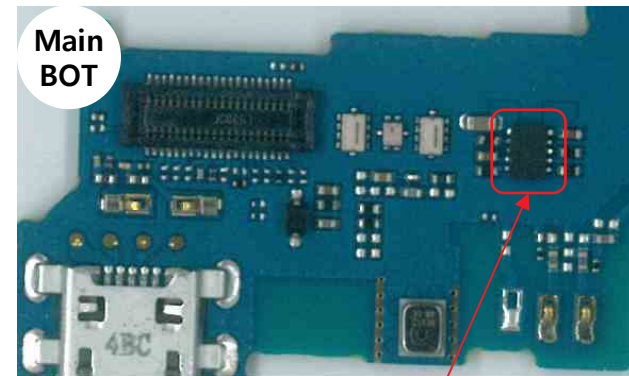
- Supply the VCHG use a TA or u-USB cable
- The Charger IC control the charging current
- Charging current flows into the main battery

Troubleshooting Check Point

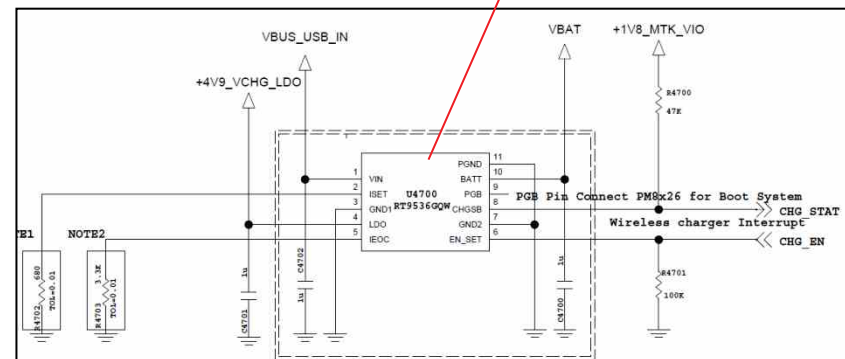
- Connection of the TA or u-USB cable
- Main battery



Image



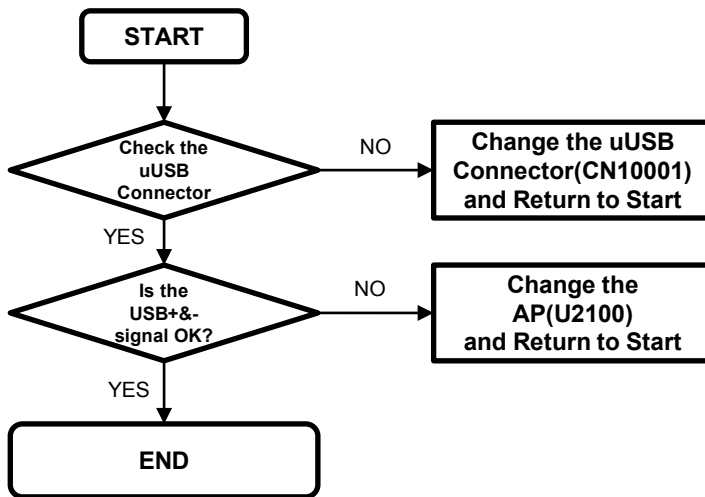
Circuit Diagram



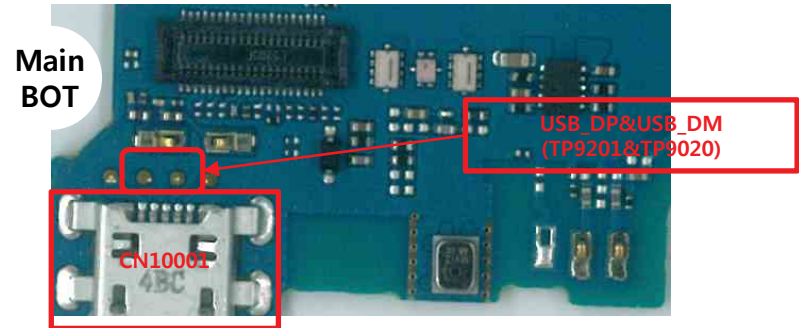
3.10 USB

USB

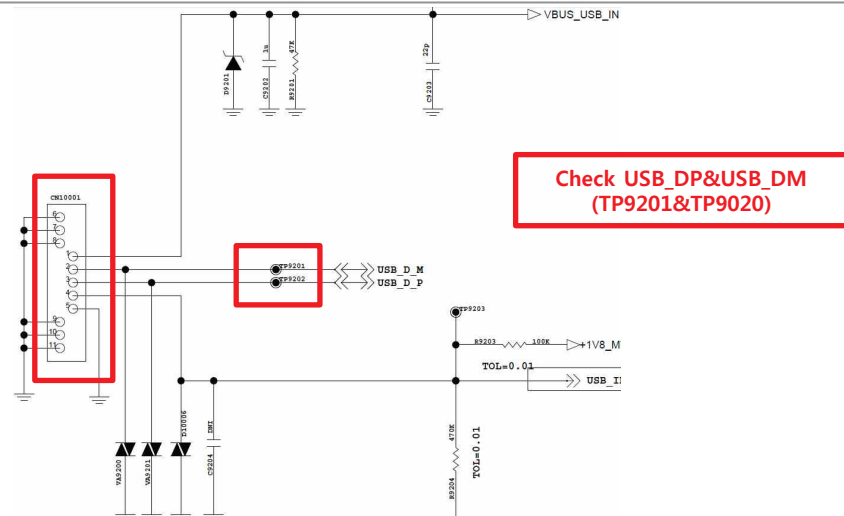
Checking Flow



Image



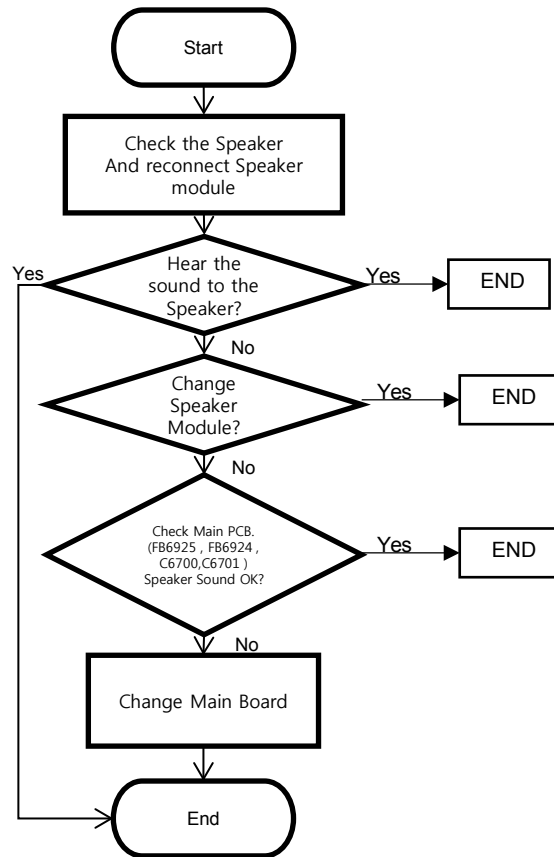
Circuit Diagram



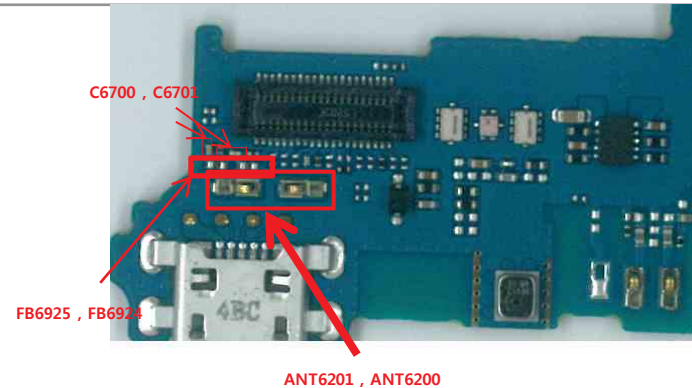
3.11.1 Speaker

It's trouble shooting guide for no sound case. Speaker control signals are generate by MT6323(U4100)
If low level volume or breaking the sound, change speaker module

Checking Flow

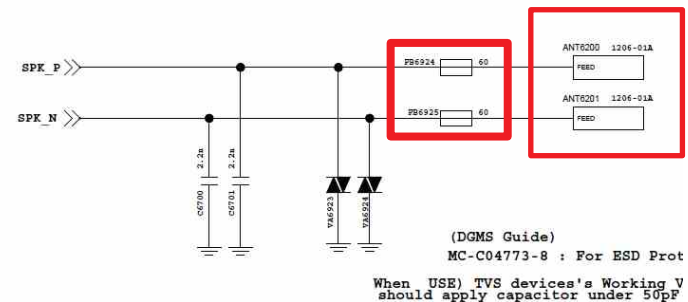


Image



Main
BOT

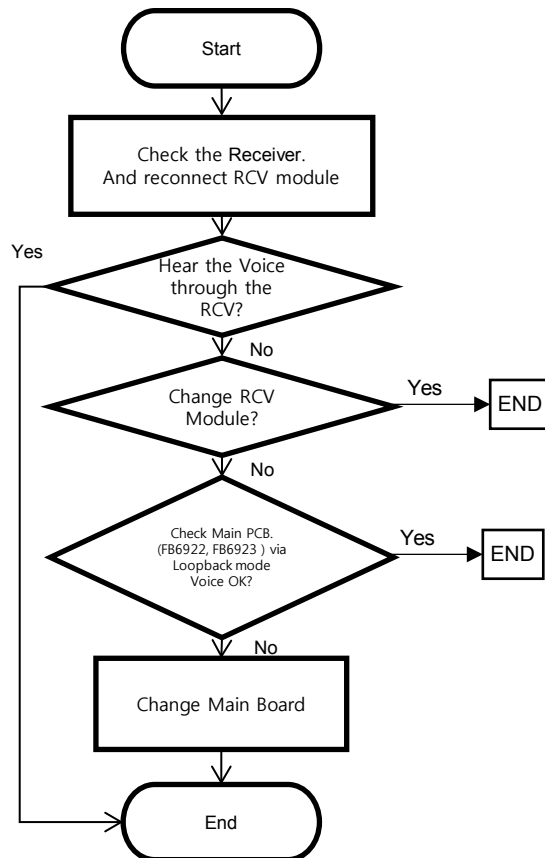
Circuit Diagram



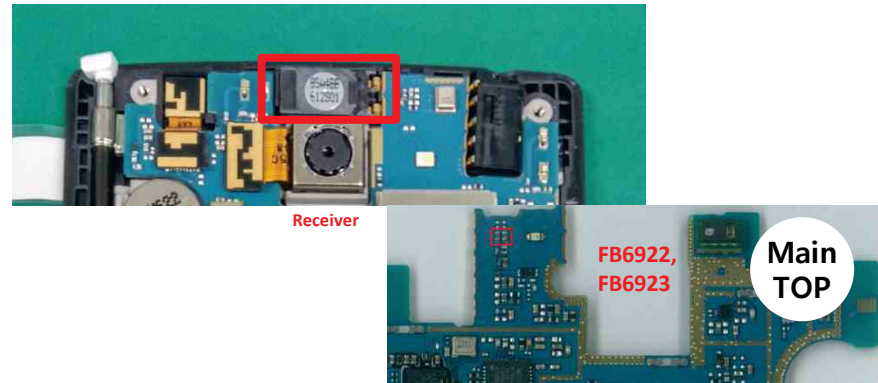
3.11.2 Receiver

It's trouble shooting guide for no sound case. Receiver control signals are generate by MT6323(U4100)
If low level volume or breaking the voice, change RCV module

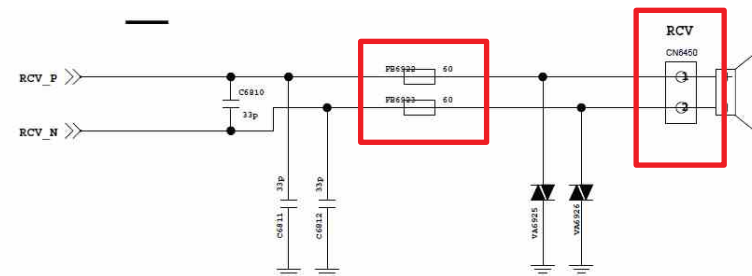
Checking Flow



Image



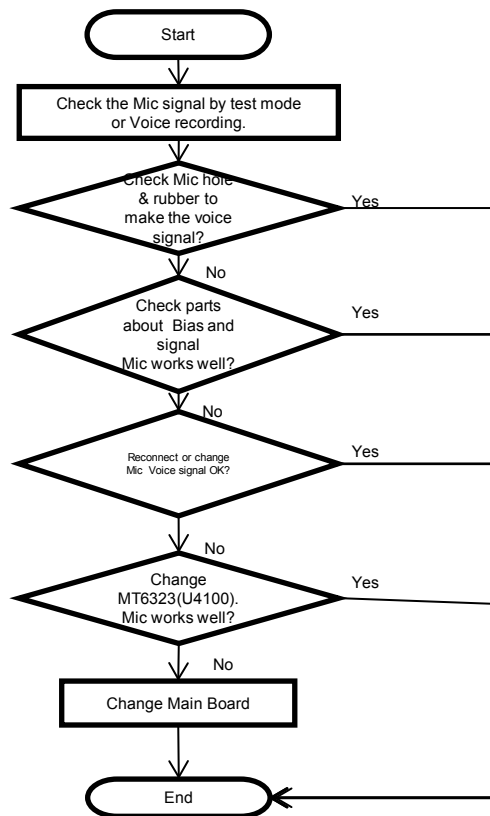
Circuit Diagram



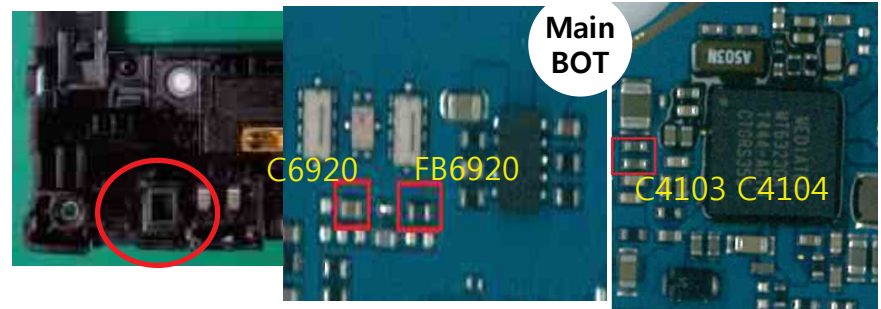
3.11.3 Main Mic

It's operating voice call(except speakerphone), voice recording, camcorder recording

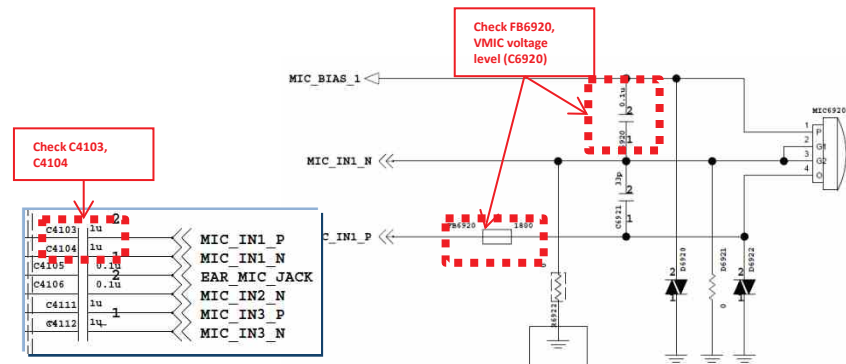
Checking Flow



Image



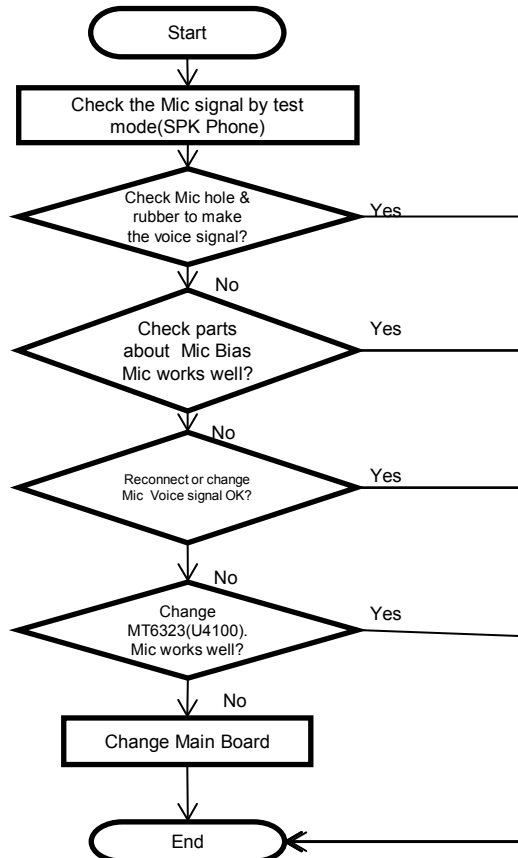
Circuit Diagram



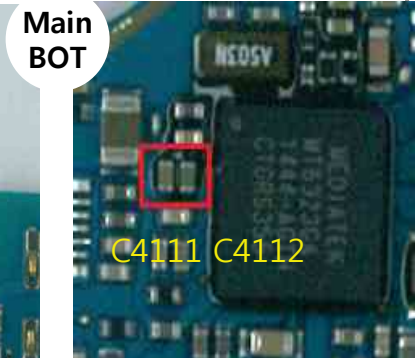
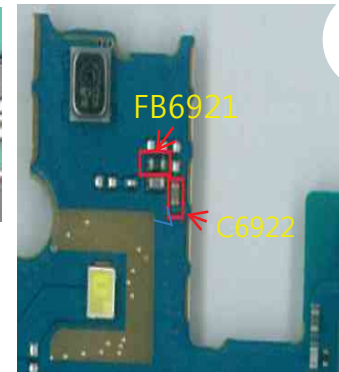
3.11.4 Sub Mic

It's operating Speakerphone call

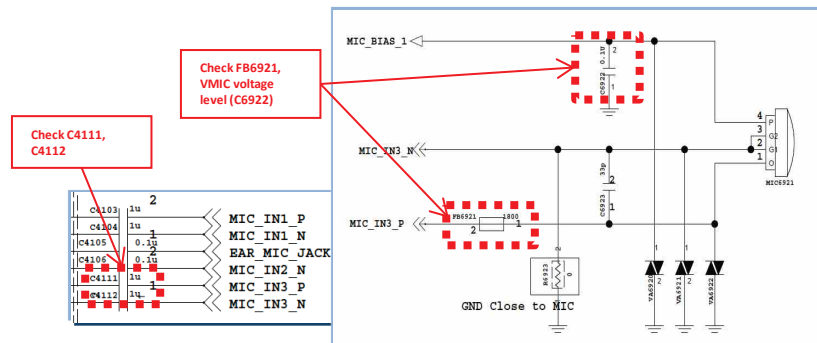
Checking Flow



Image



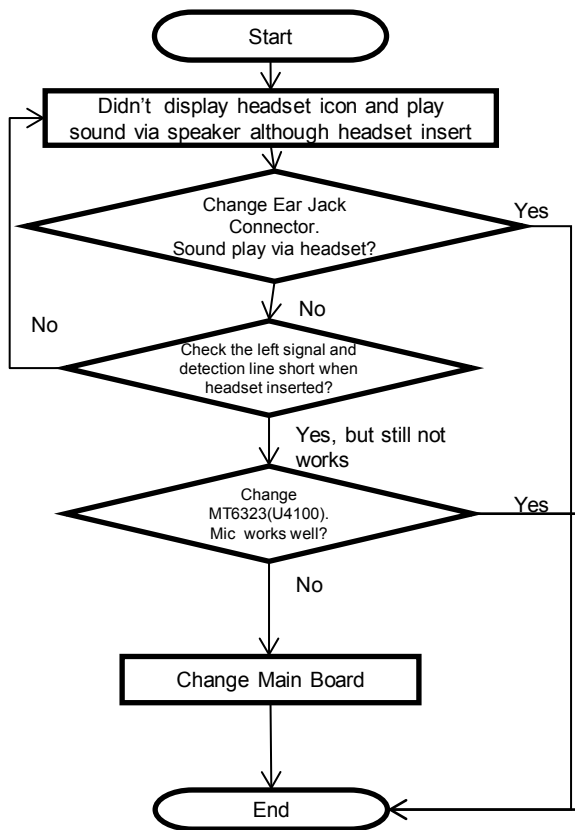
Circuit Diagram



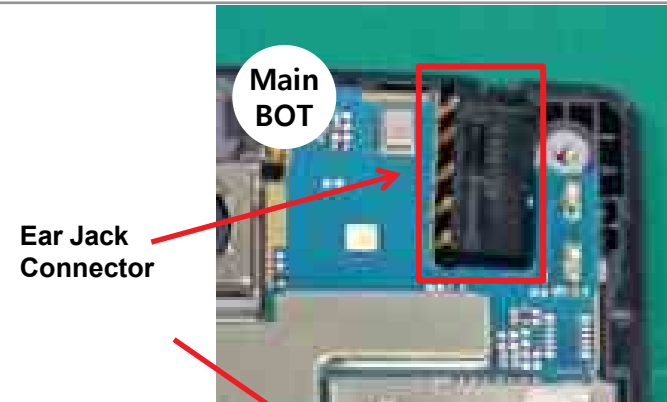
3.11.5 Ear-Mic -Case A

Ear MIC control signals are generate by MT6323(U4100)
Case A. Disable detected headset insert.

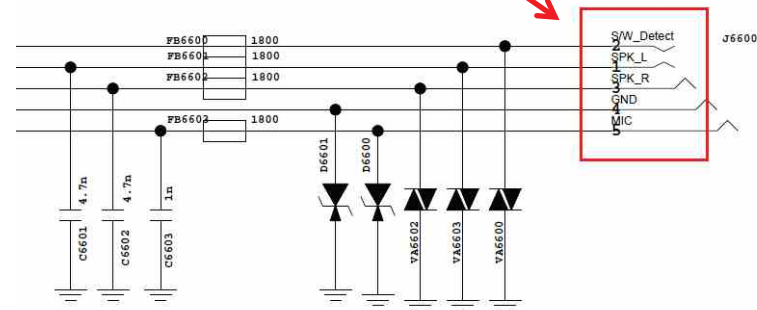
Checking Flow



Image



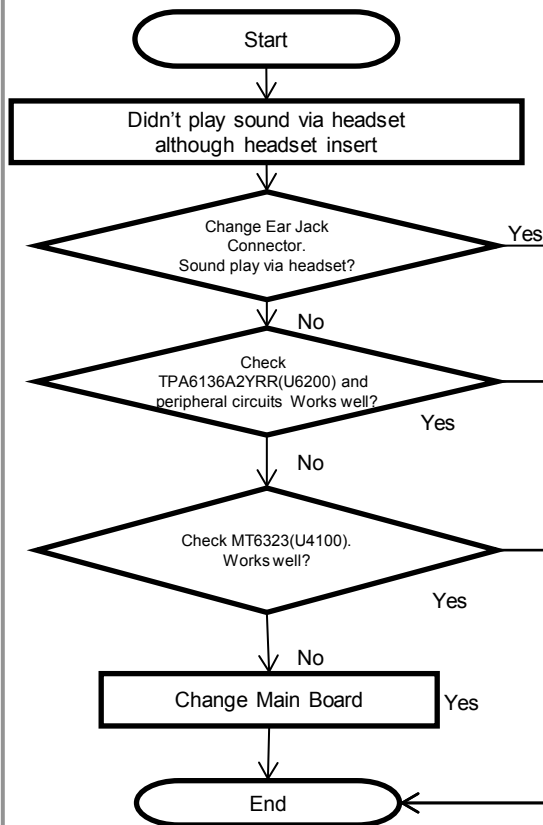
Circuit Diagram



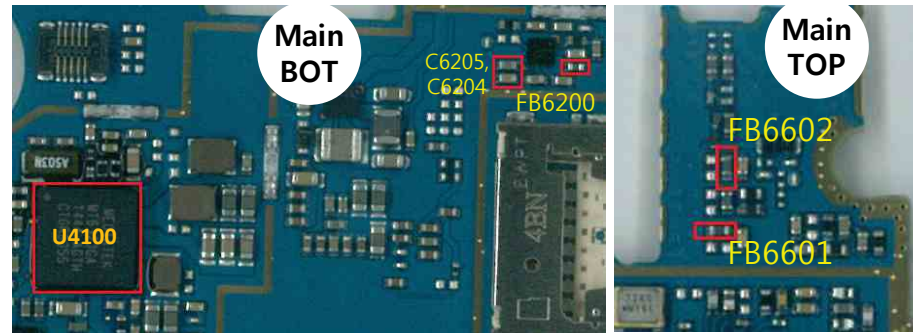
3.11.6 Ear-Mic- Case B

Case B. No Sound from Headset. Check the input power VBAT (FB6200)
In case of you can't heard anything, check C6204, C6205, FB6601, FB6602.

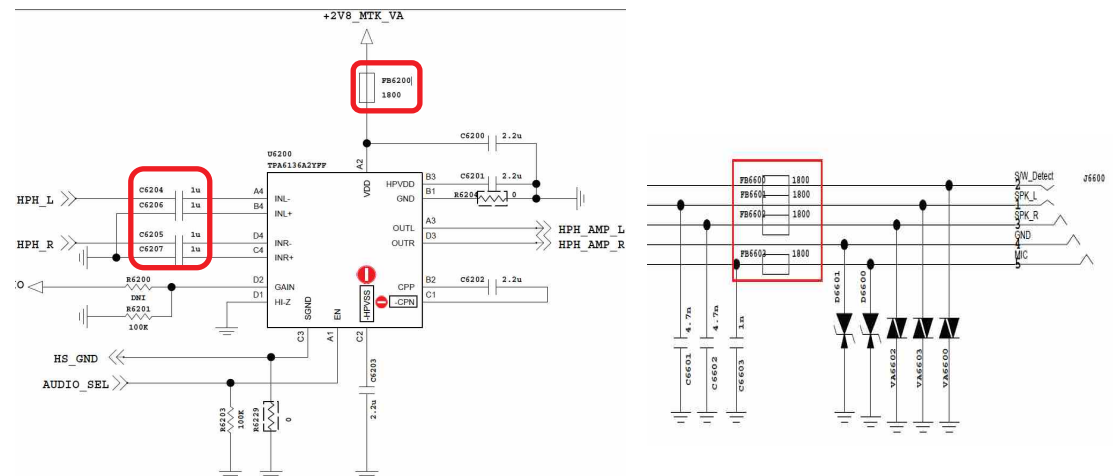
Checking Flow



Image

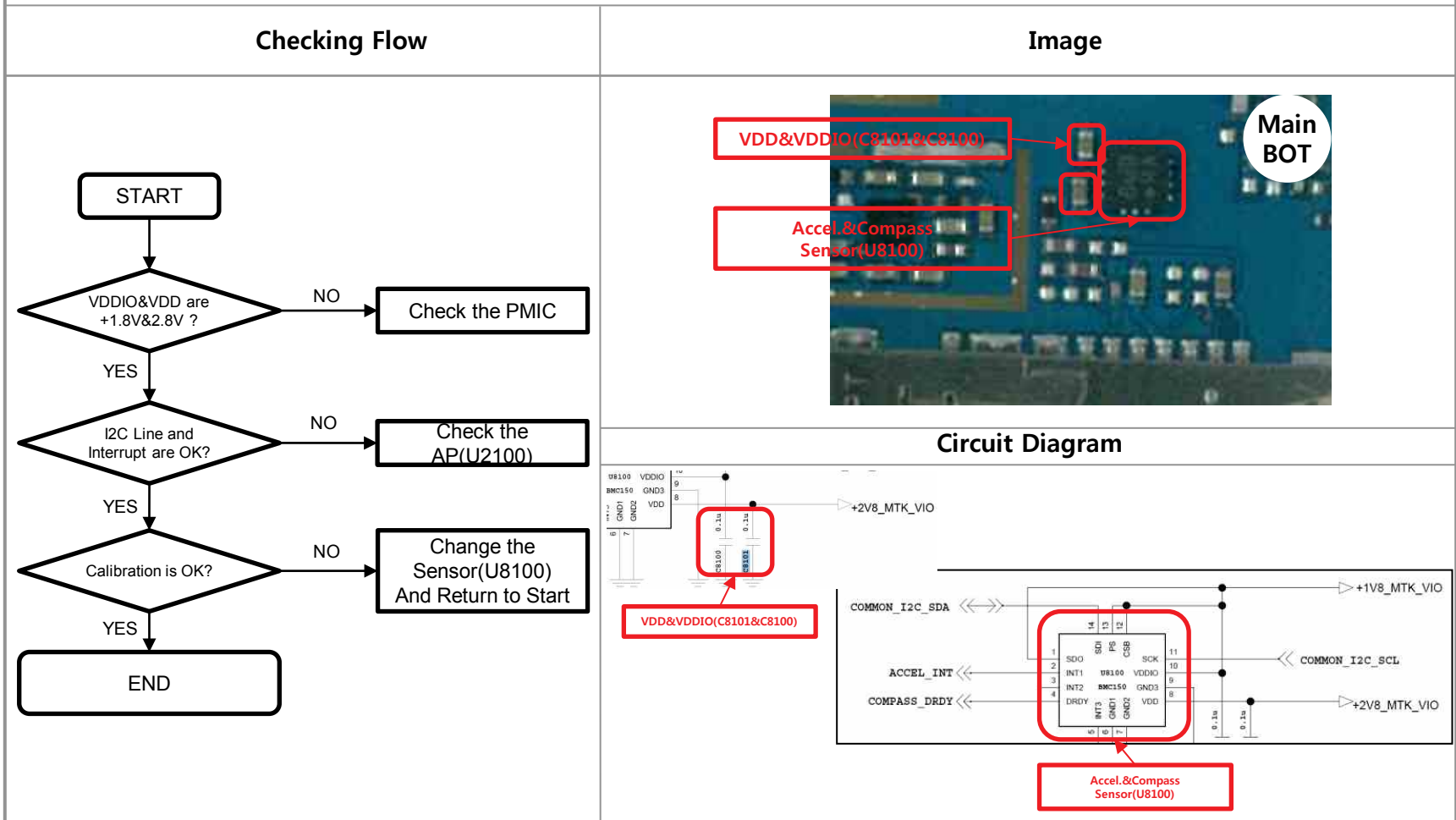


Circuit Diagram



3.12.1 Accelerometer & Compass Sensor on/off

The Accl. & compass sensors are calibrated by using SW algorithm.

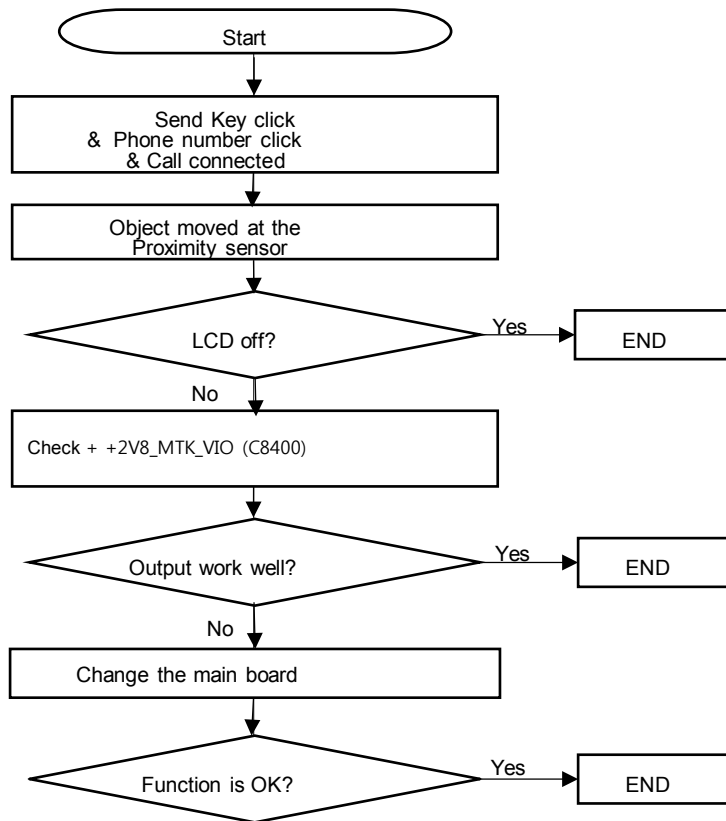


3.12.2 Proximity Sensor on/off

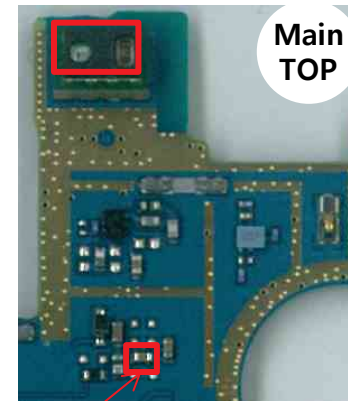
Proximity Sensor is worked as below

Control the screen's on/off operation automatically

Checking Flow

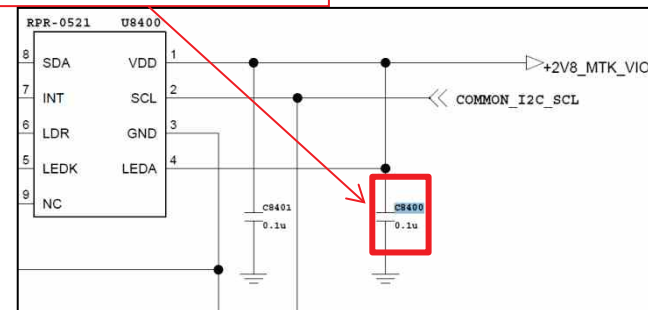


Image



Circuit Diagram

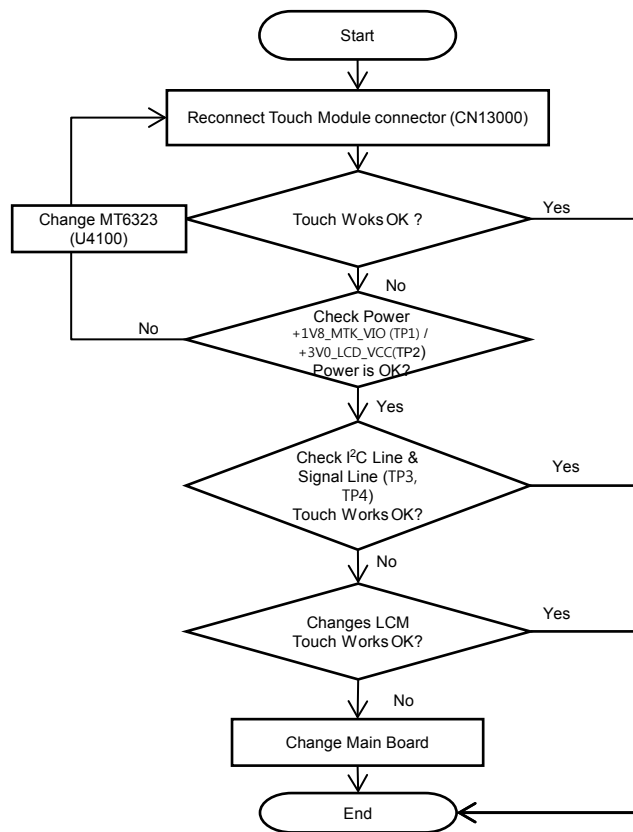
Sensor Power: C8400



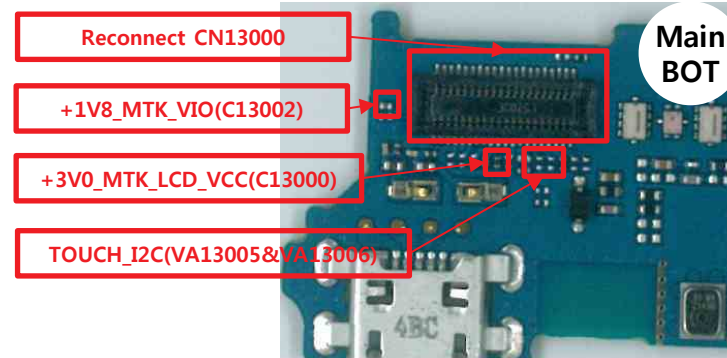
3.13 Touch

Touch Sensor

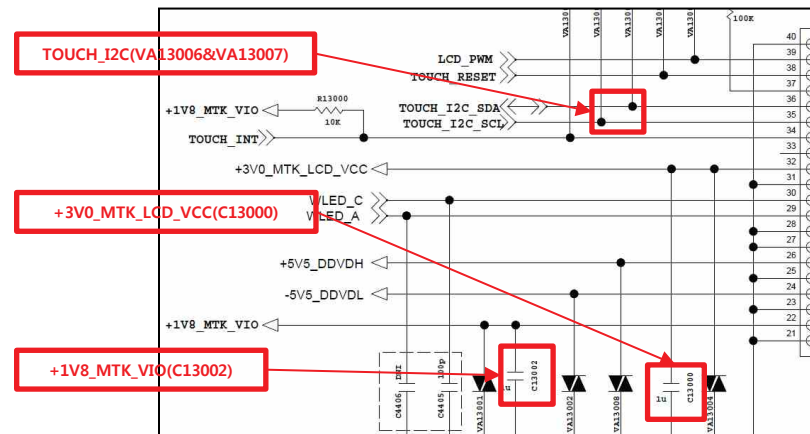
Checking Flow



Image



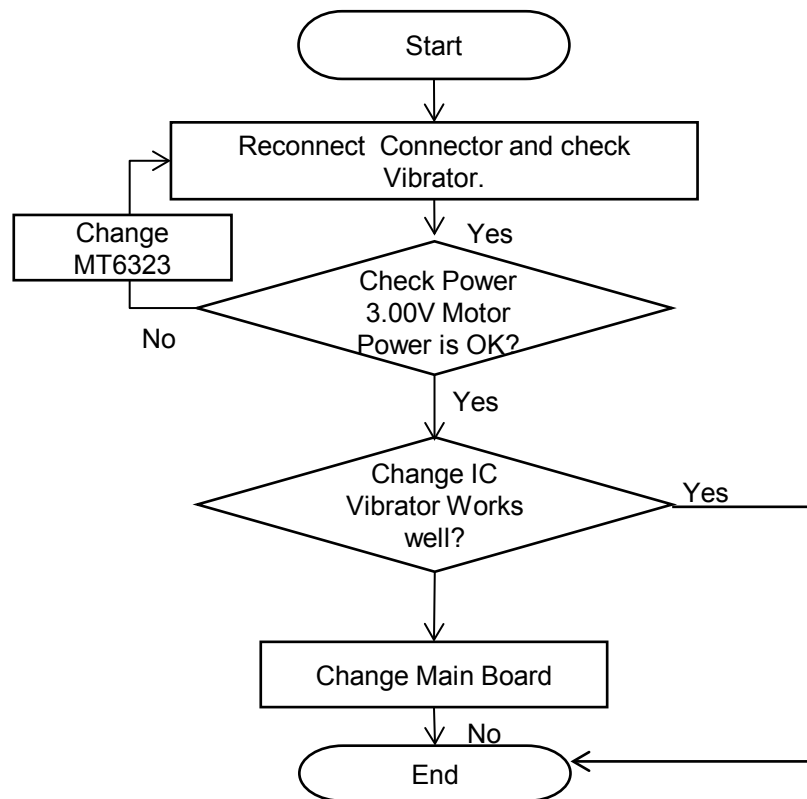
Circuit Diagram



3.14 Vibrator

Motor Type is small cylinder.

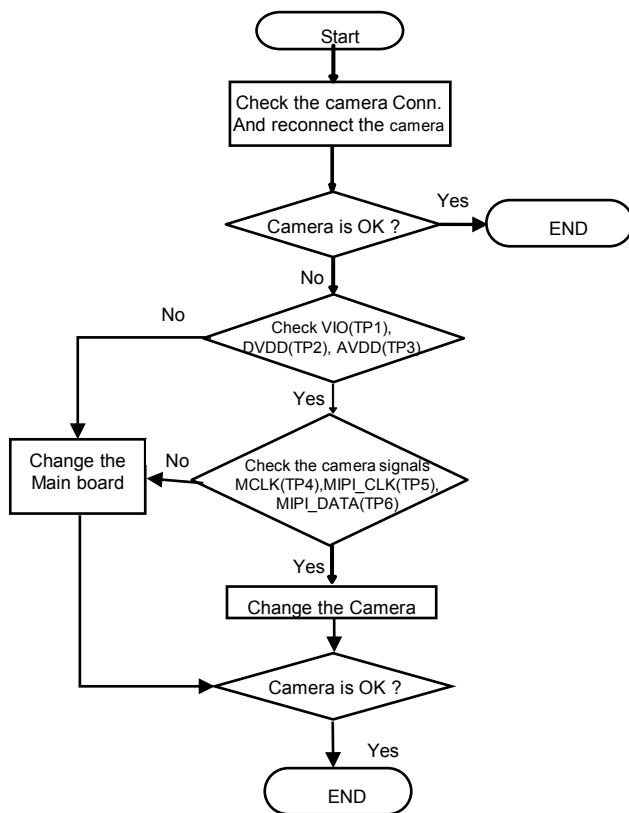
Checking Flow



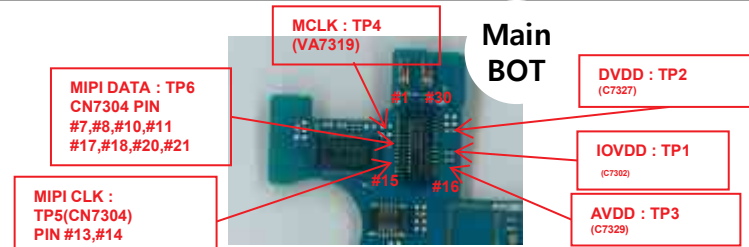
3.15.1 8M Main Camera

8M camera control signals are generated by MT6582 (U2100 : Main Chipset).
And powered by voltage regulator (U4100)

Checking Flow

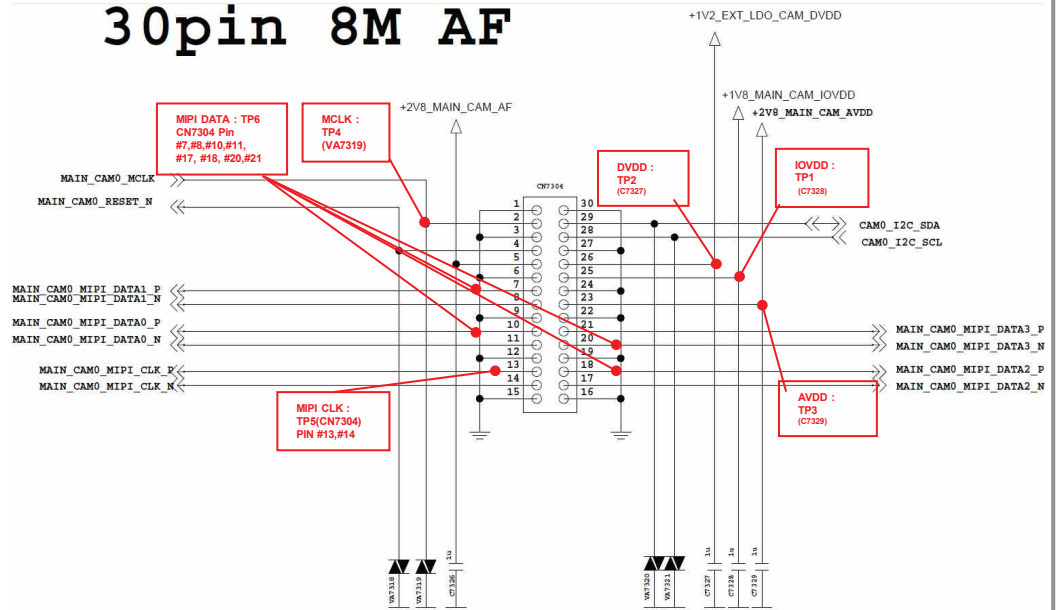


Image



Circuit Diagram

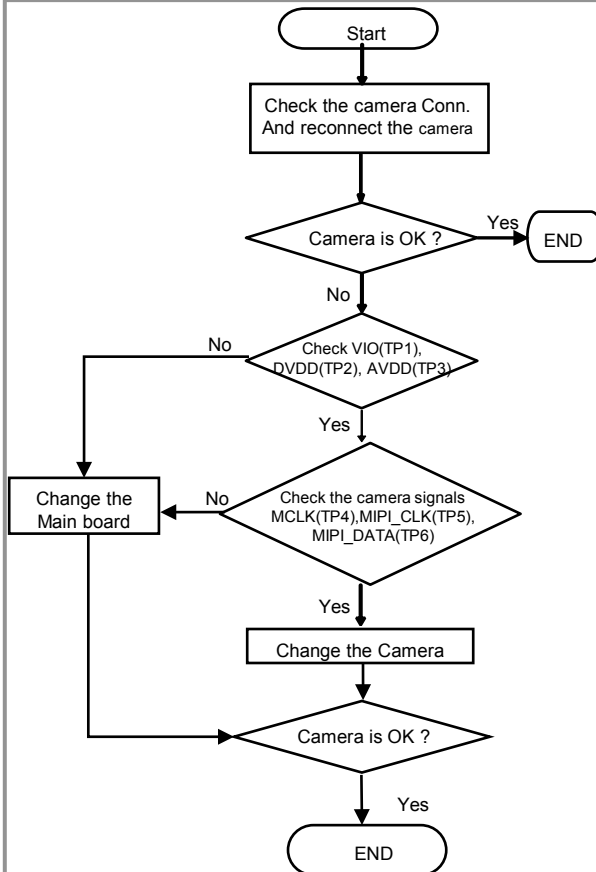
30pin 8M AF



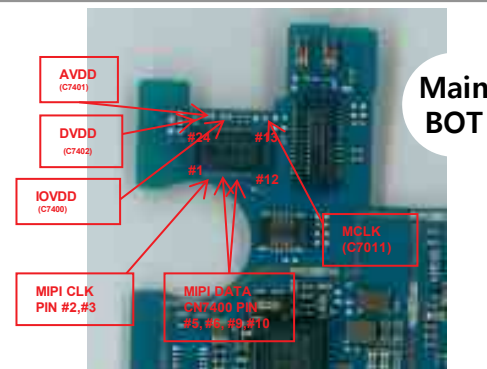
3.15.2 VT Camera

VGA camera control signals are generated by MT6582 (U2100 : Main Chipset).
And powered by voltage regulator (U4100)

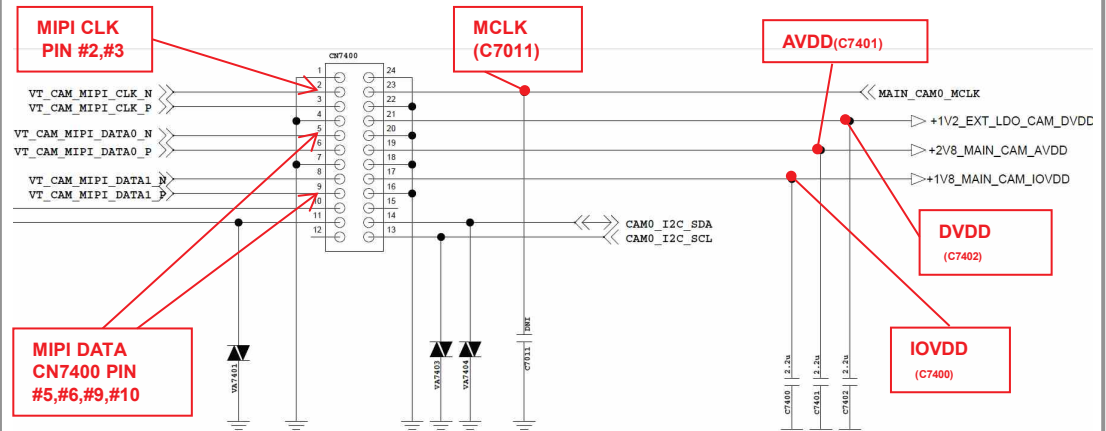
Checking Flow



Image



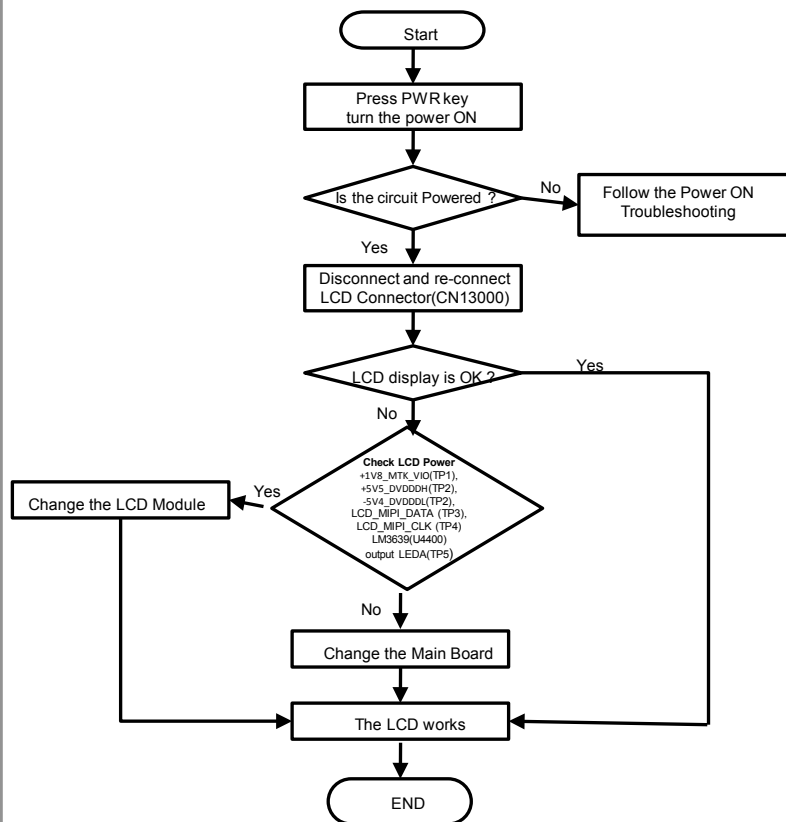
Circuit Diagram



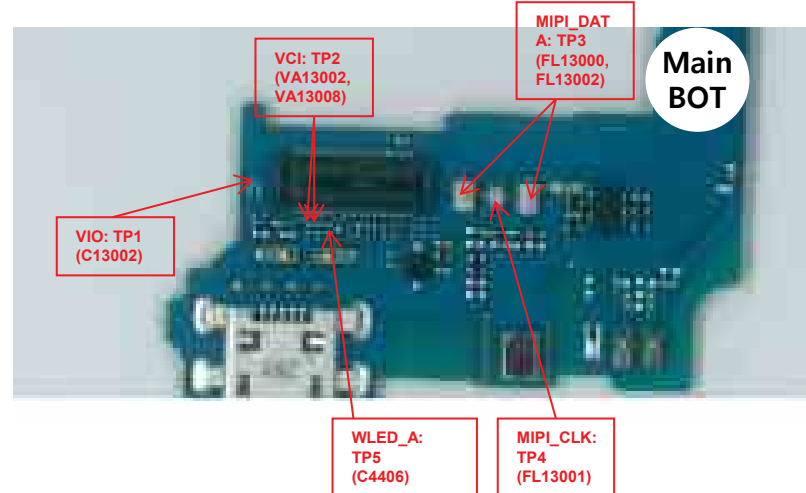
3.16 Main LCD

The LCD control signals are generated by MT6582. It needs three external powers like VIO, VCI. And its interface is MIPI having three data lanes and one clock lane.

Checking Flow



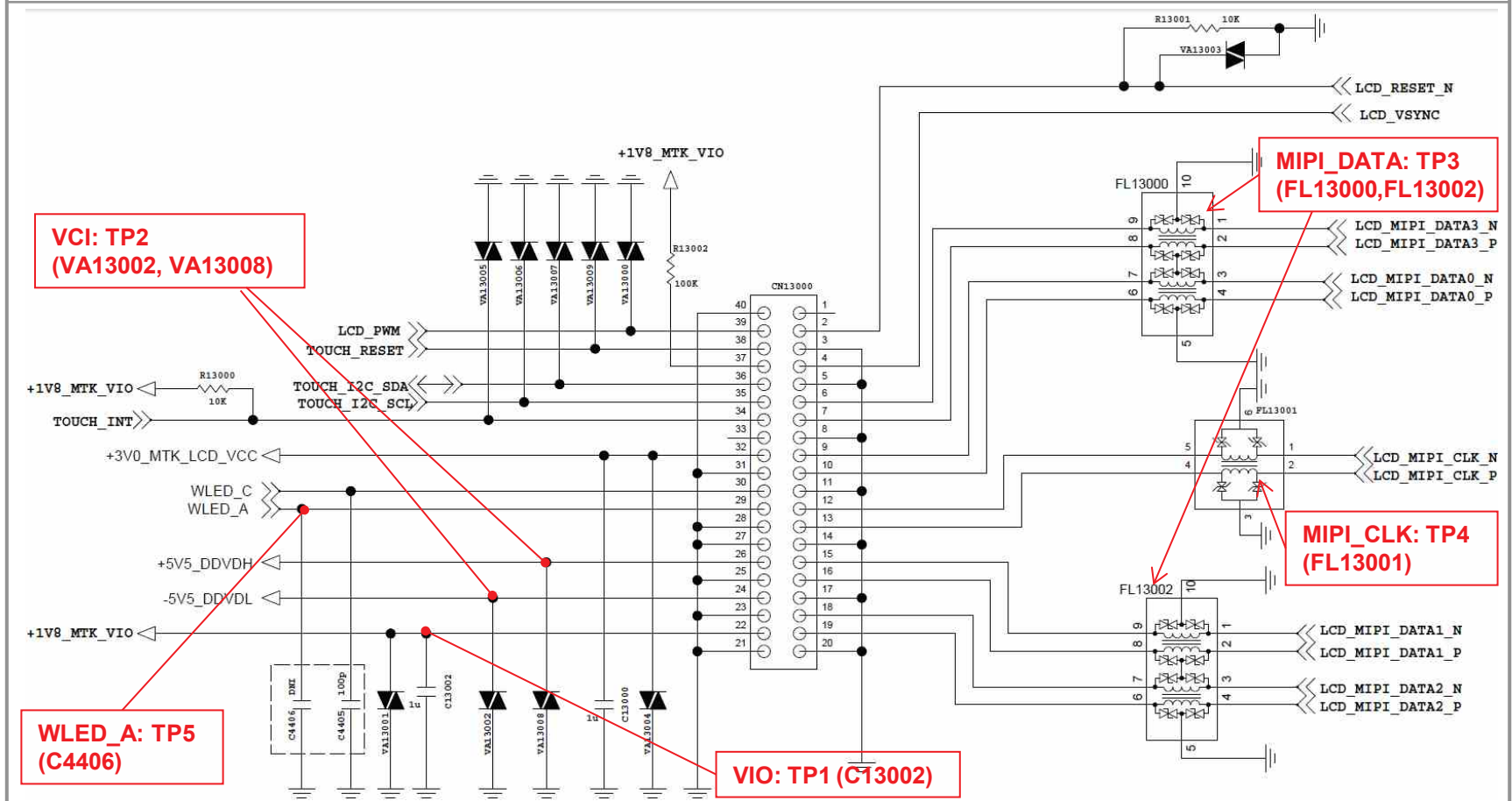
Image



3.16 Main LCD

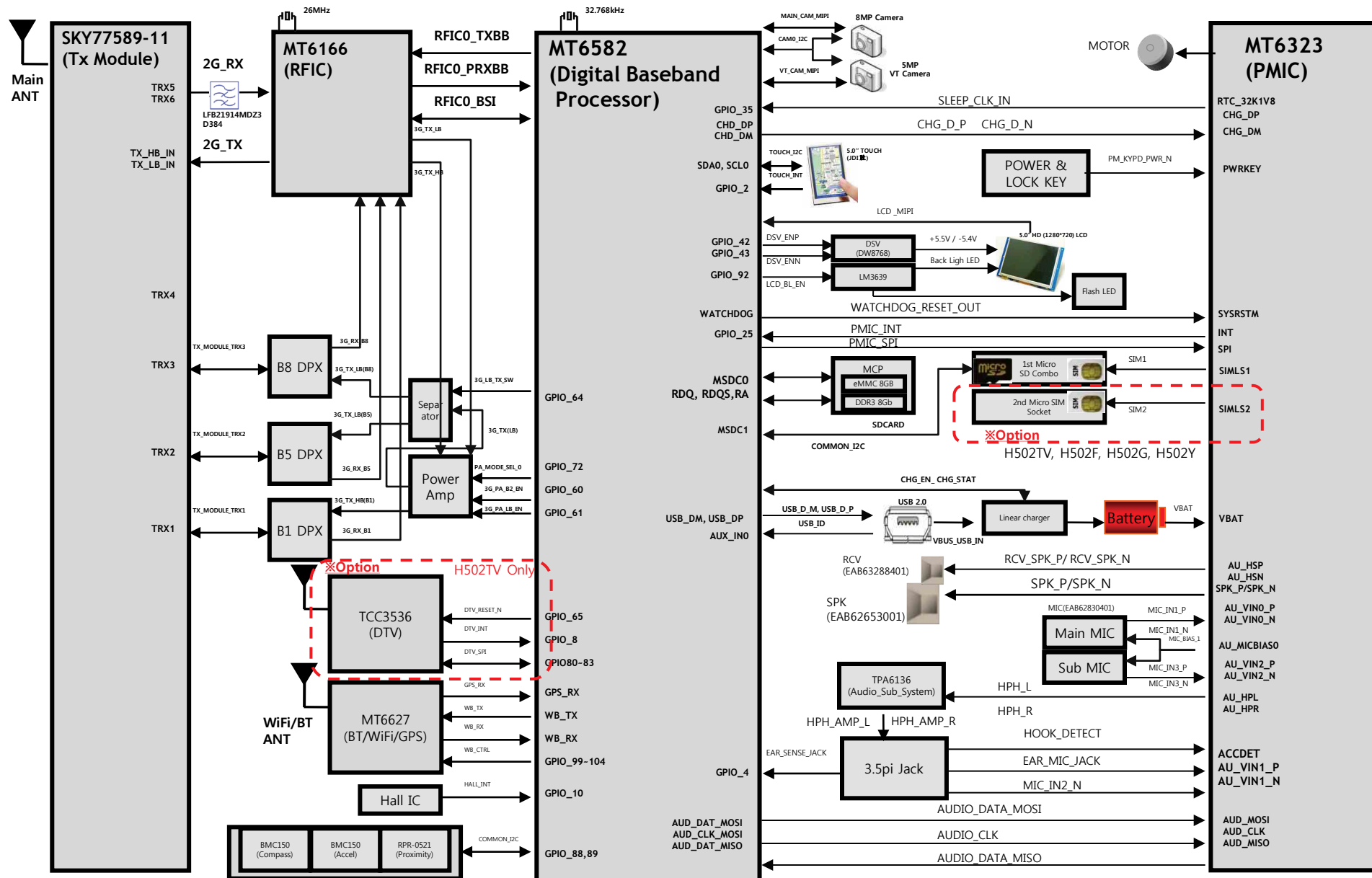
The LCD control signals are generated by MT6582. It needs three external powers like VIO, VCI. And its interface is MIPI having three data lanes and one clock lane.

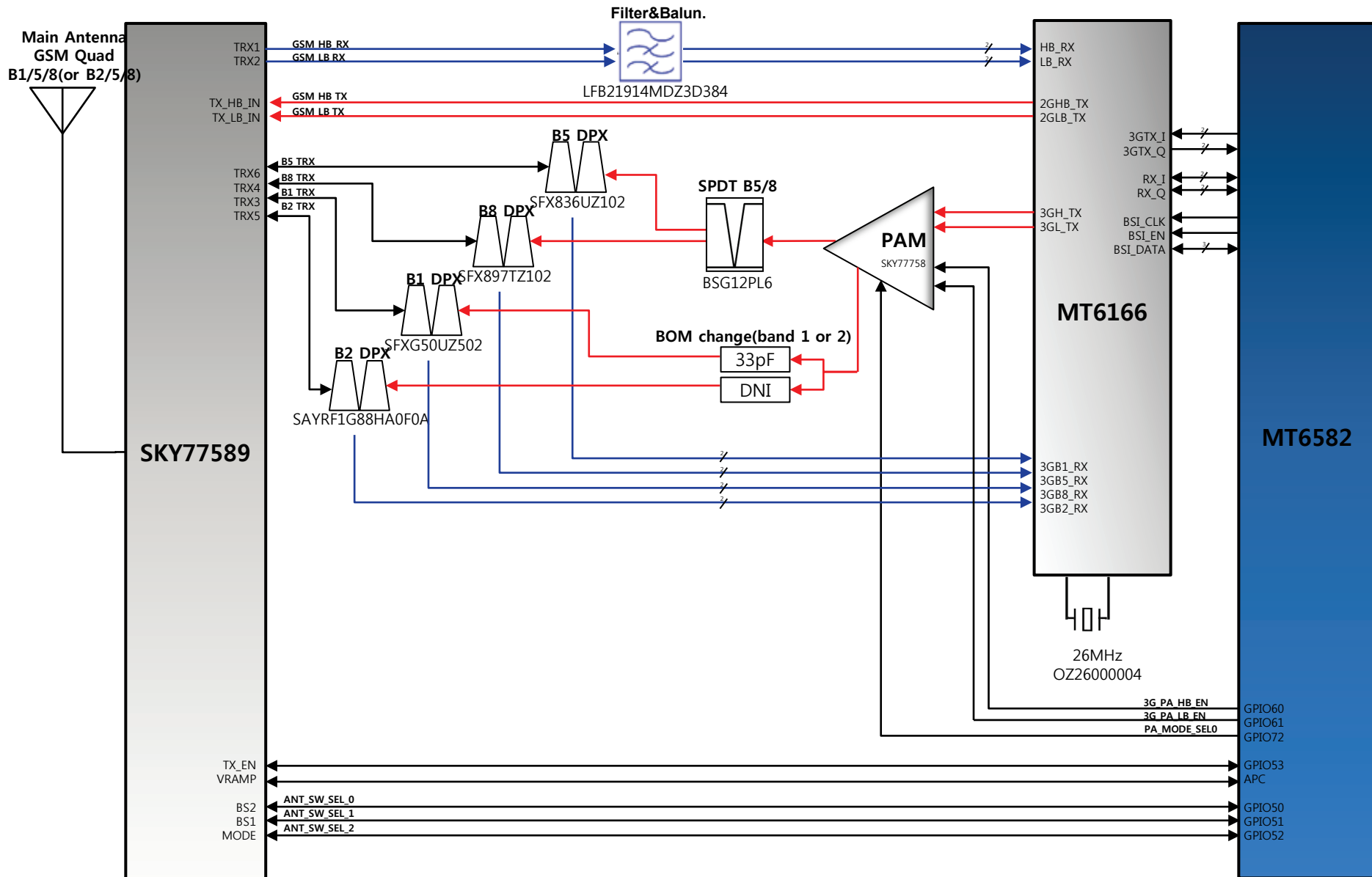
Circuit Diagram

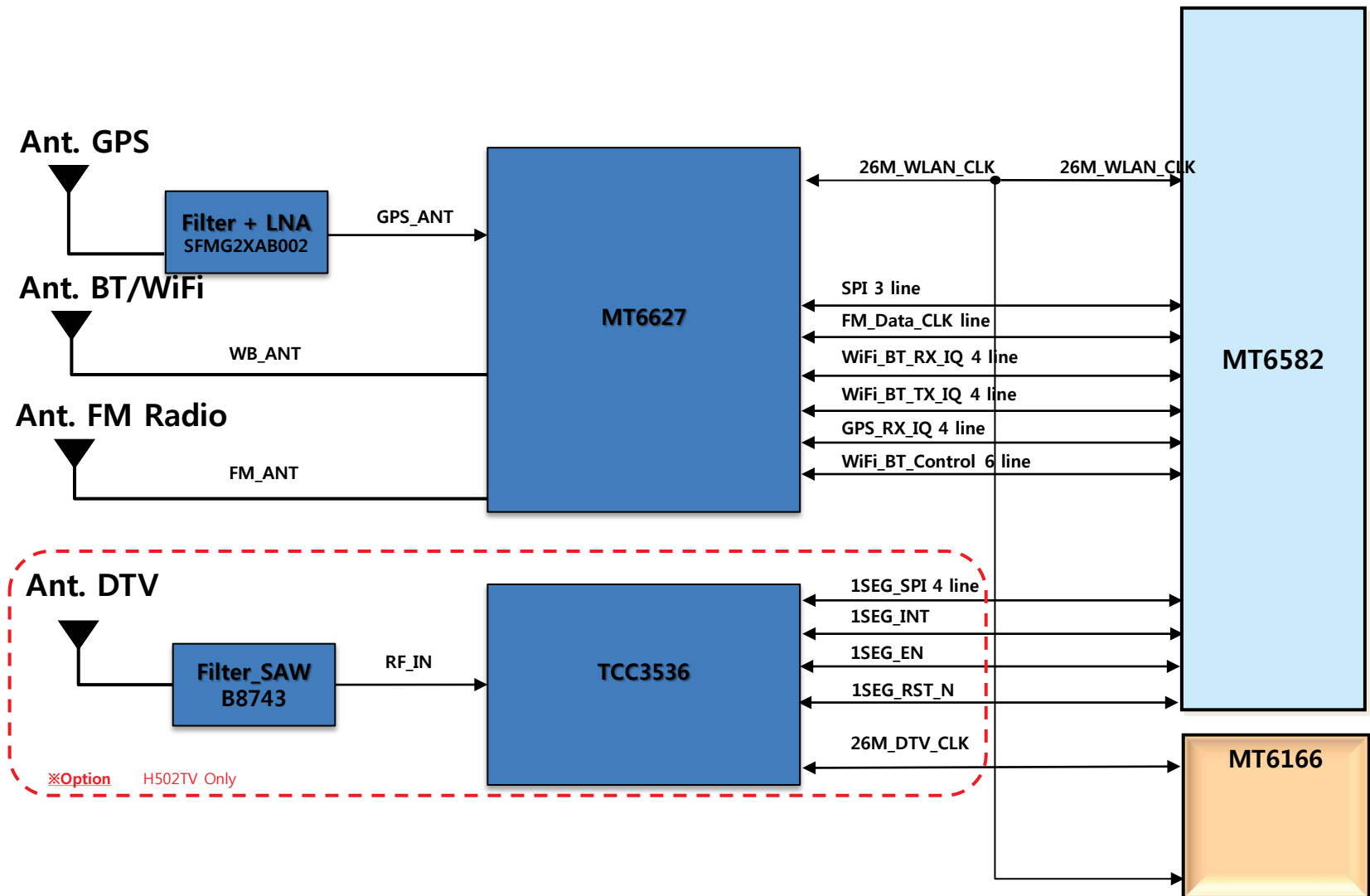


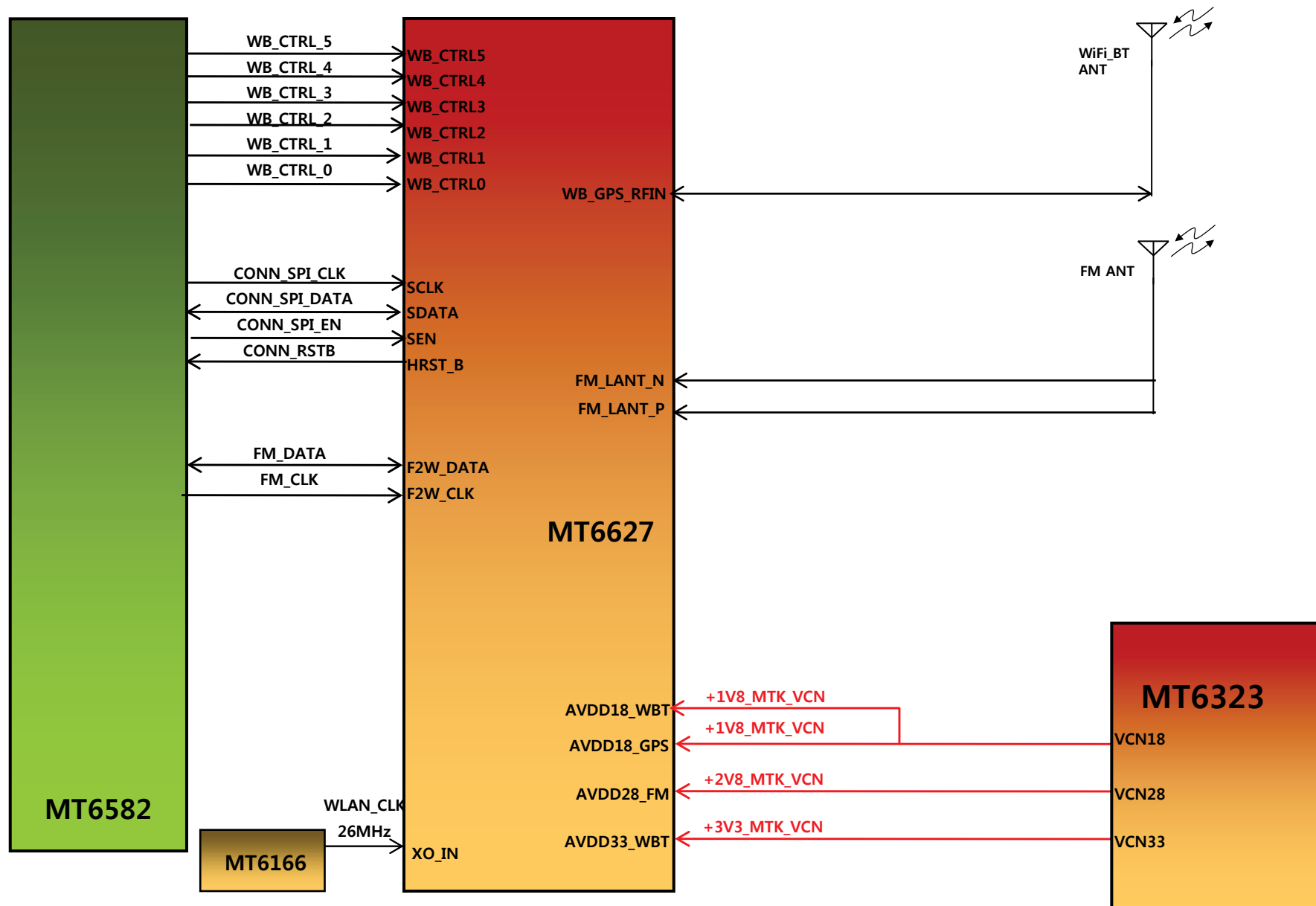
1. Total block diagram

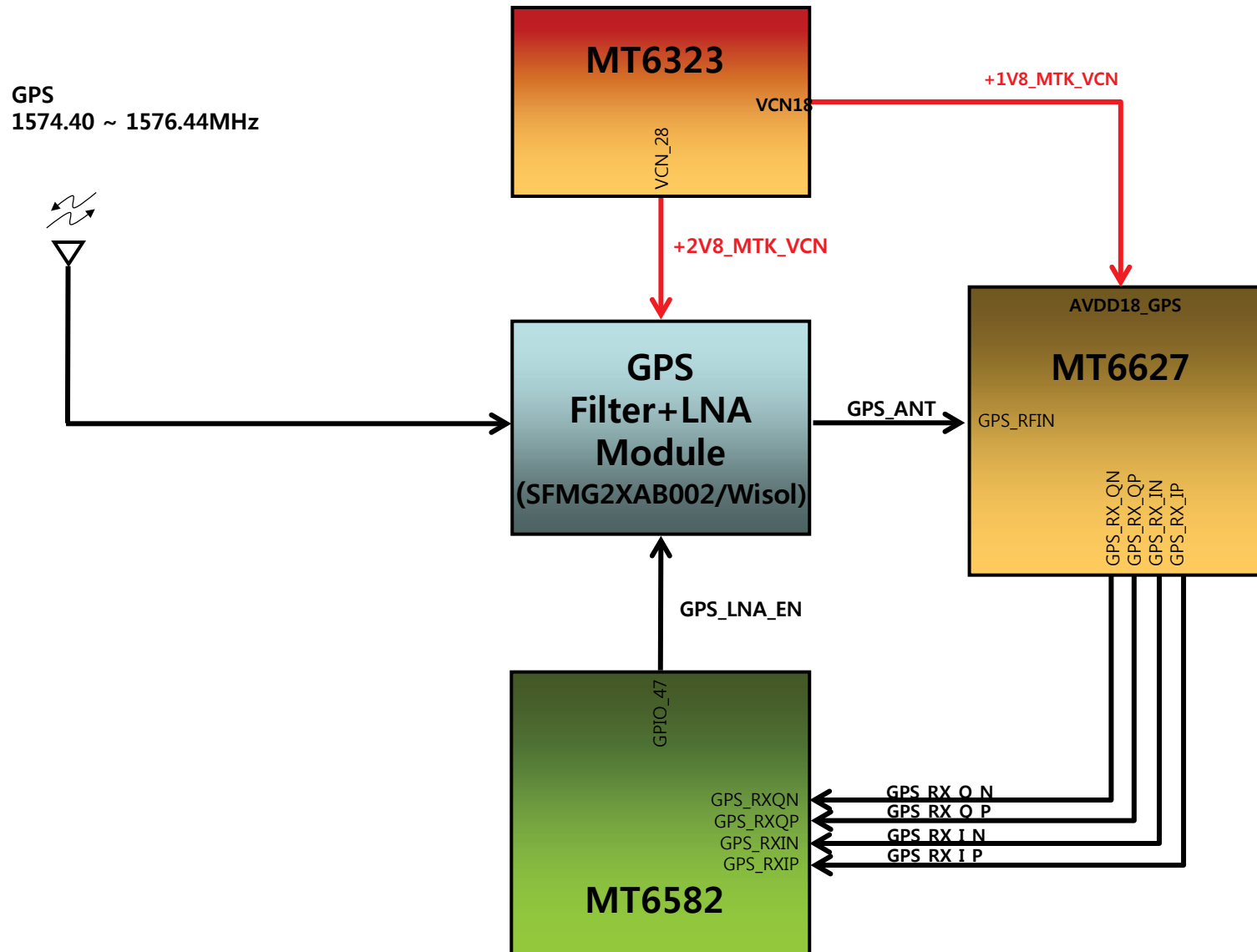
4. BLOCK DIAGRAM

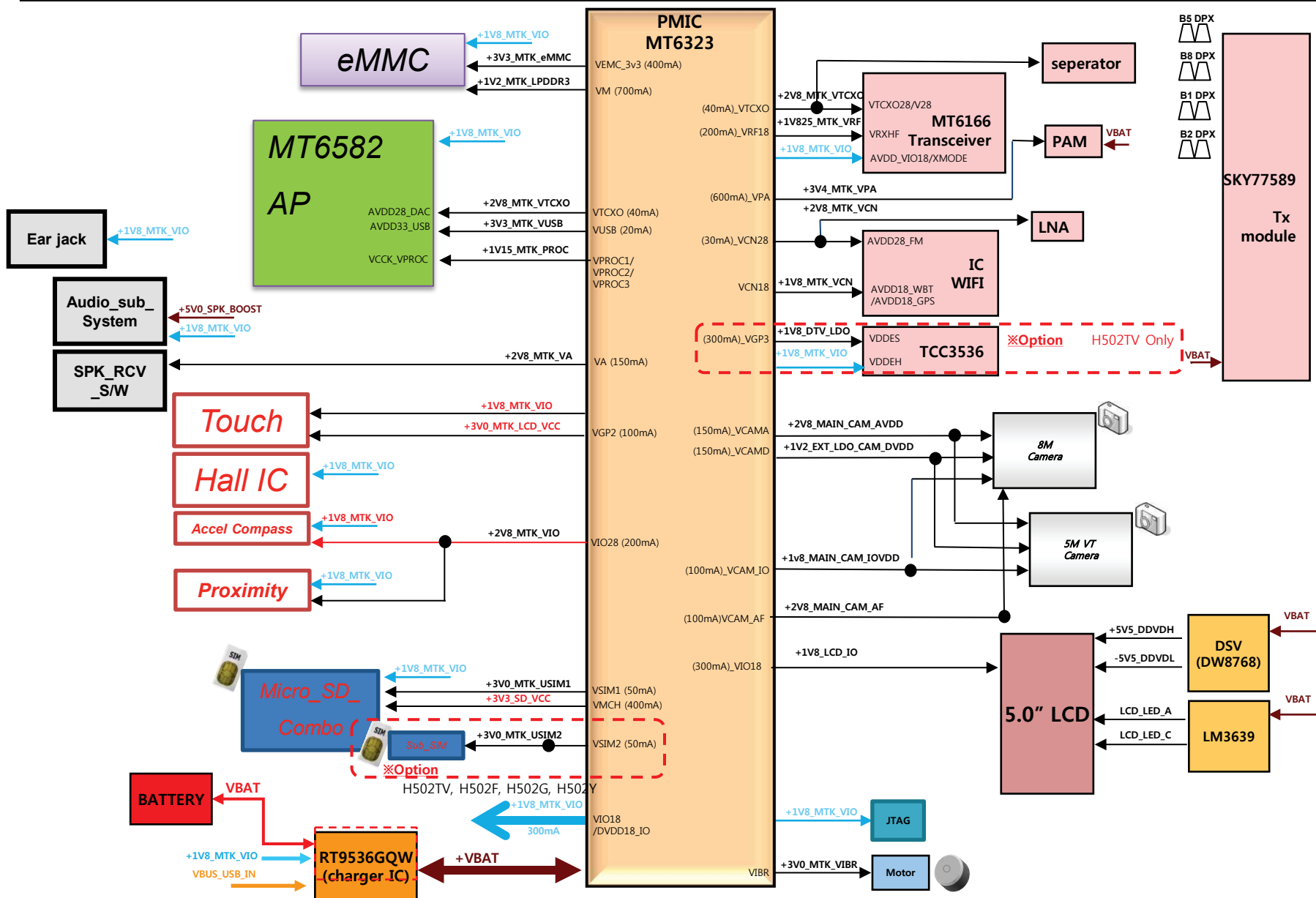


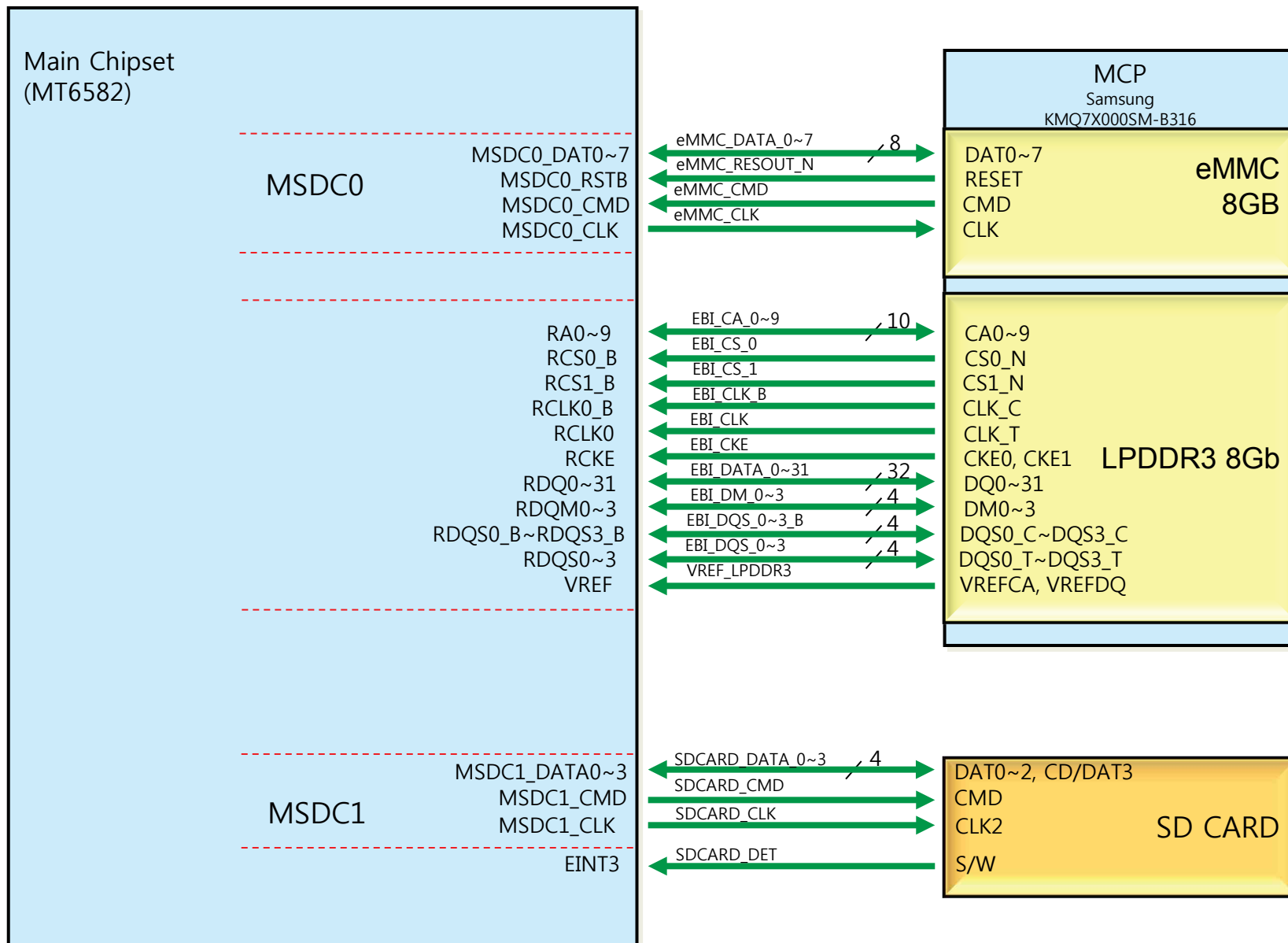


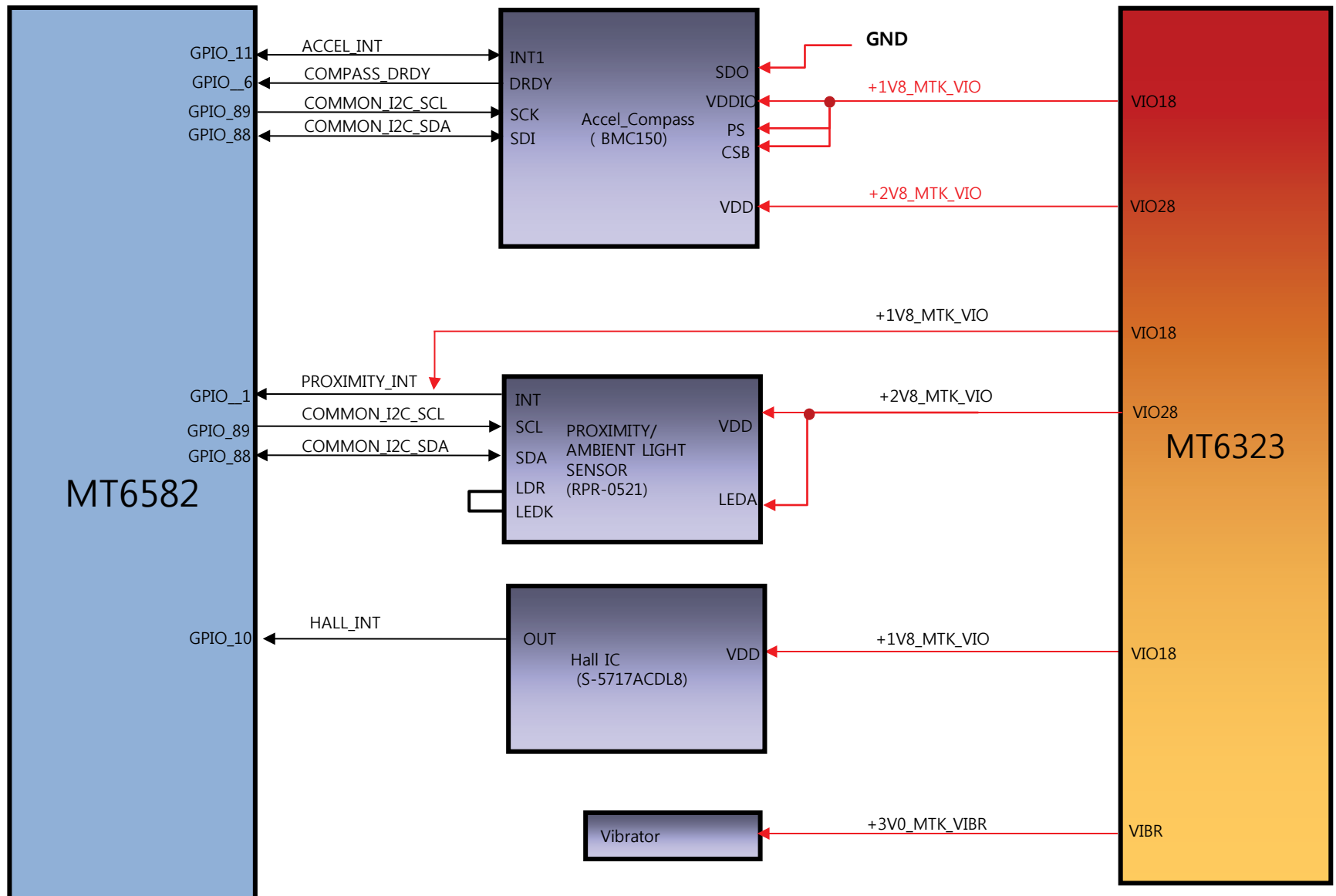


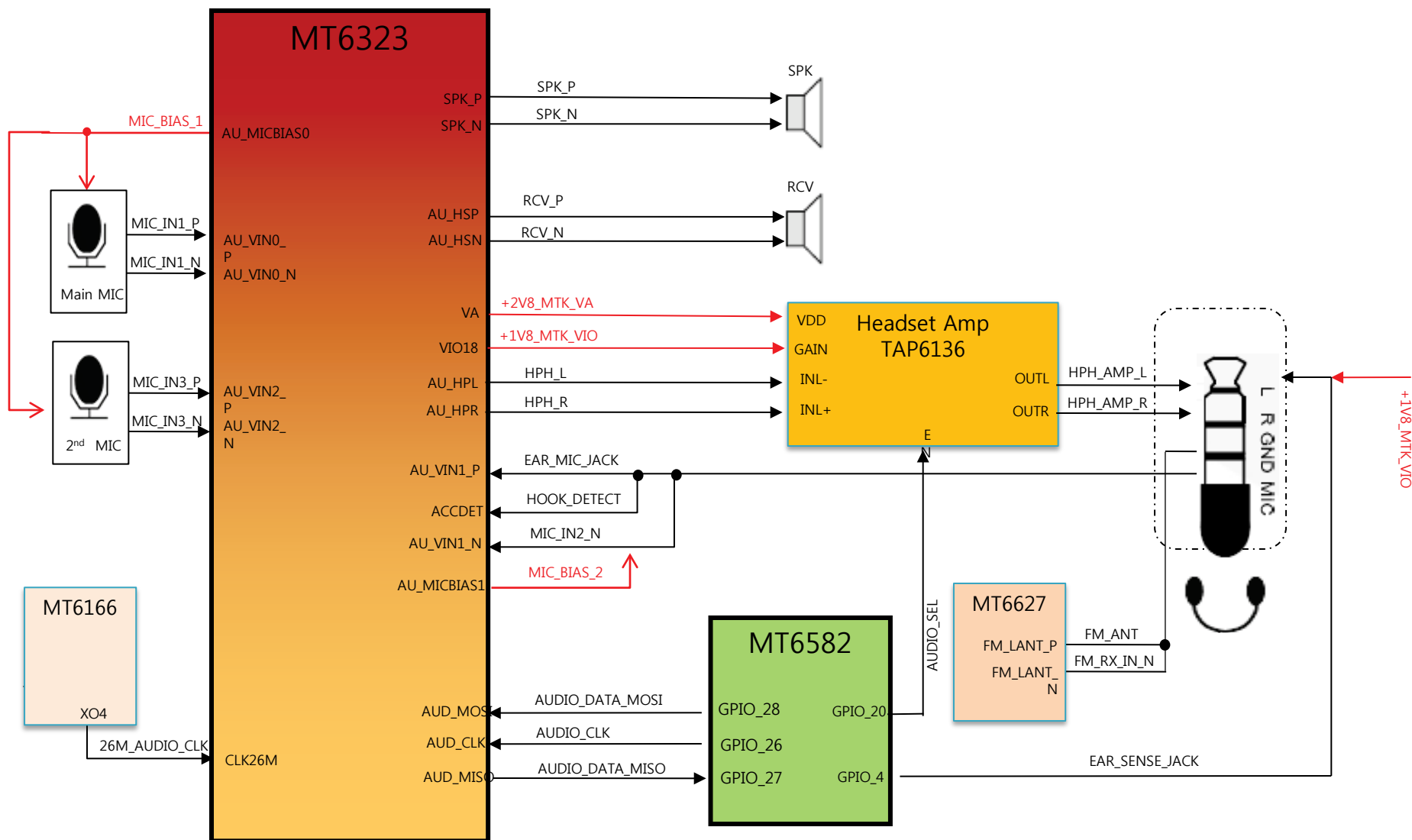


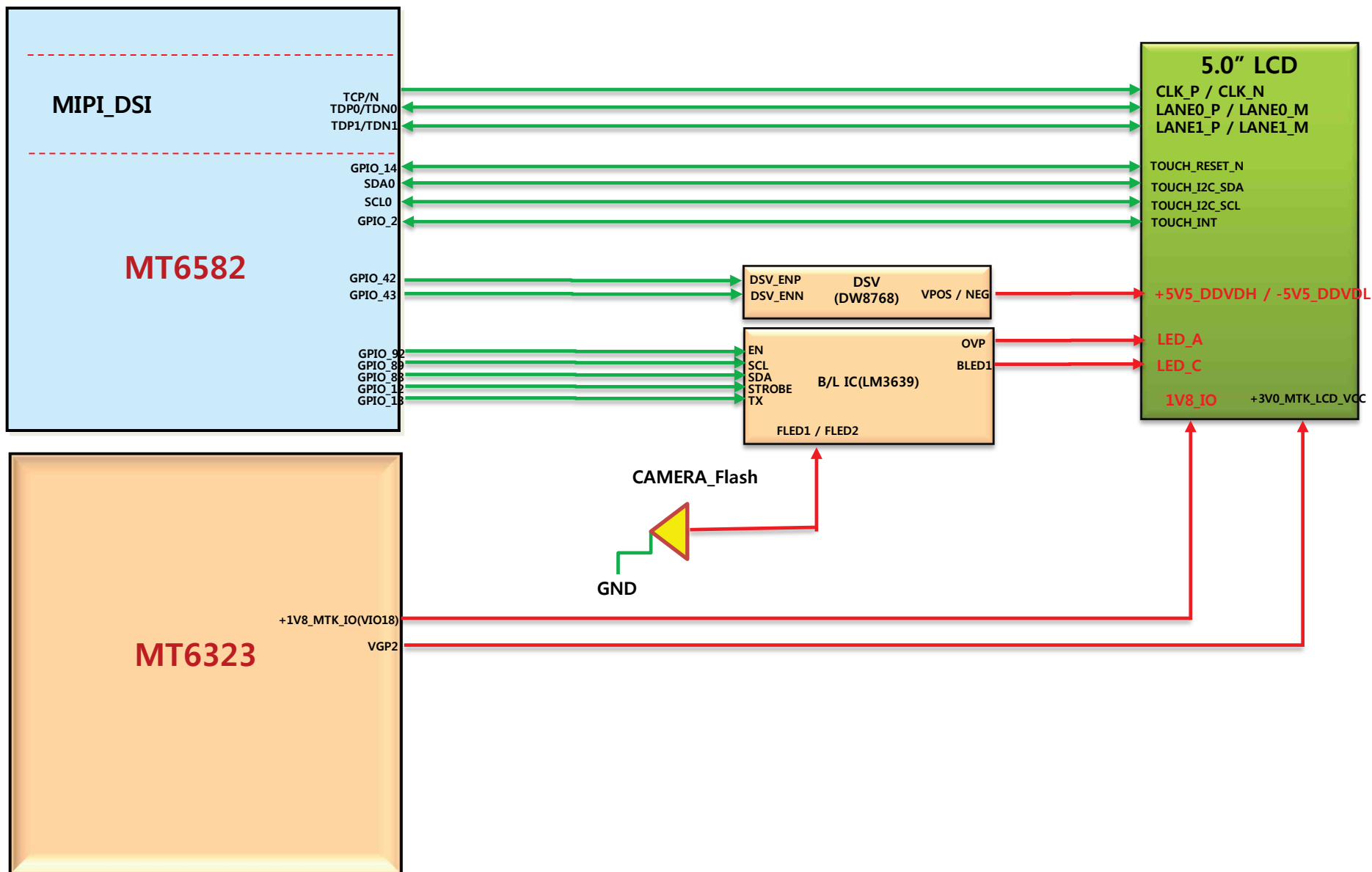


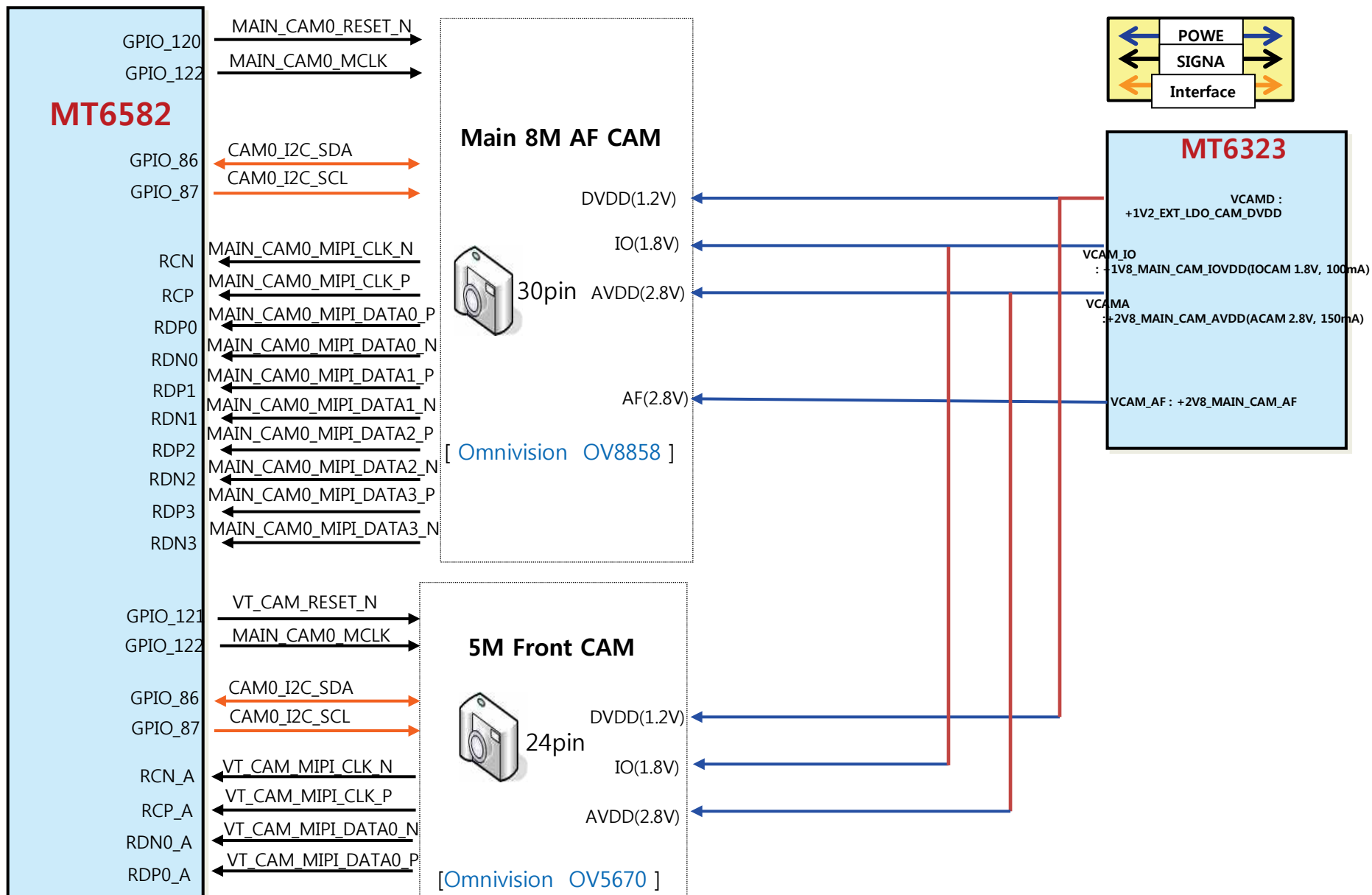








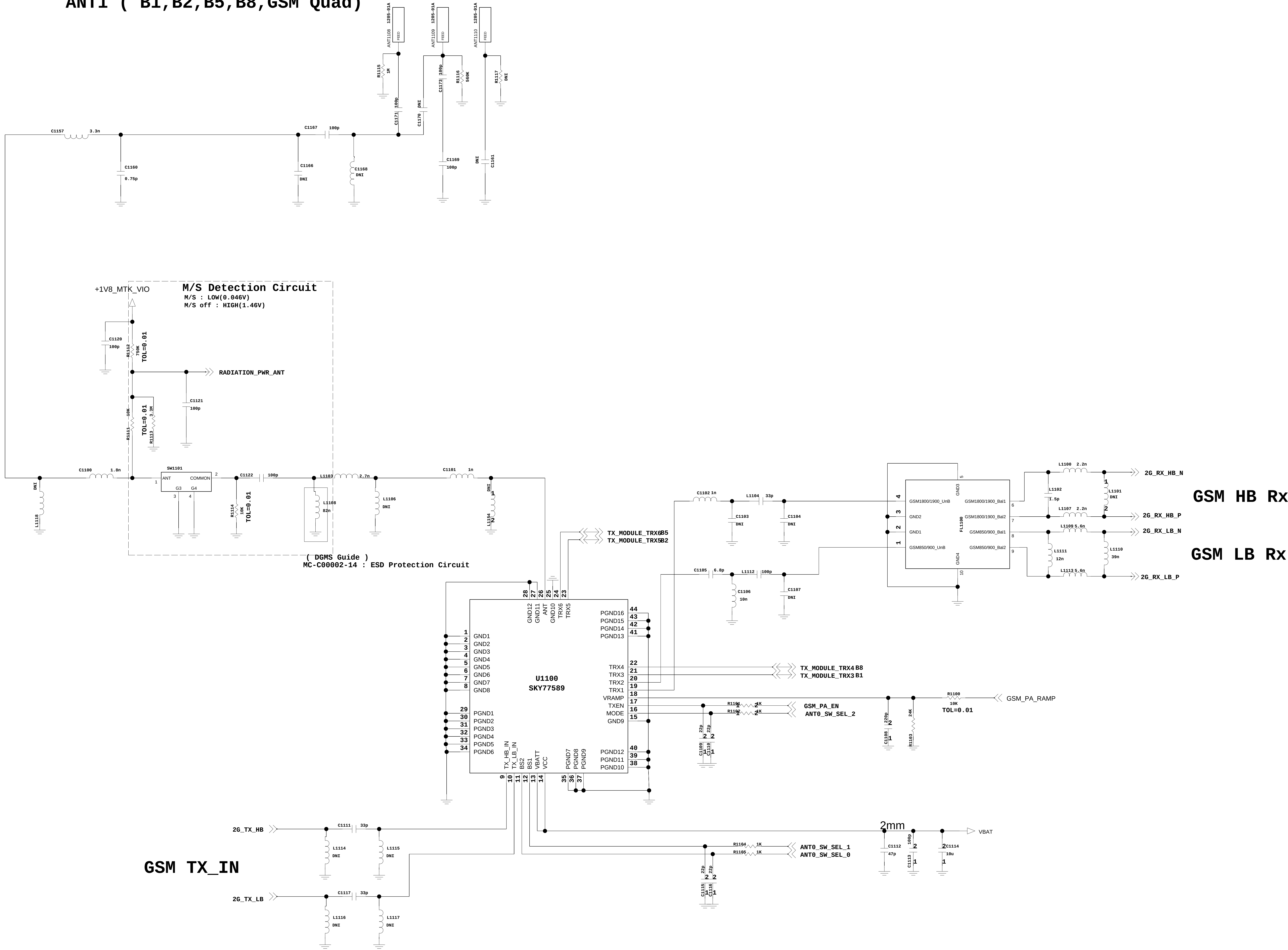




CIRCUIT DIAGRAM

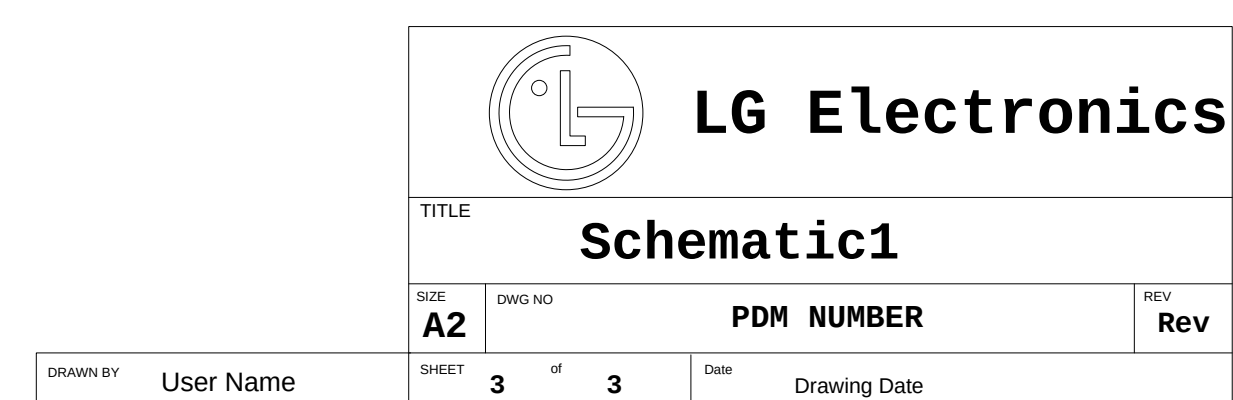
ANT1 (B1, B2, B5, B8, GSM Quad)

Ref. #	ANT1108	ANT1109	ANT1110
Default	Signal	GND	GND
Option (C1171=NC, C1170=100pF)	NC	Signal	GND

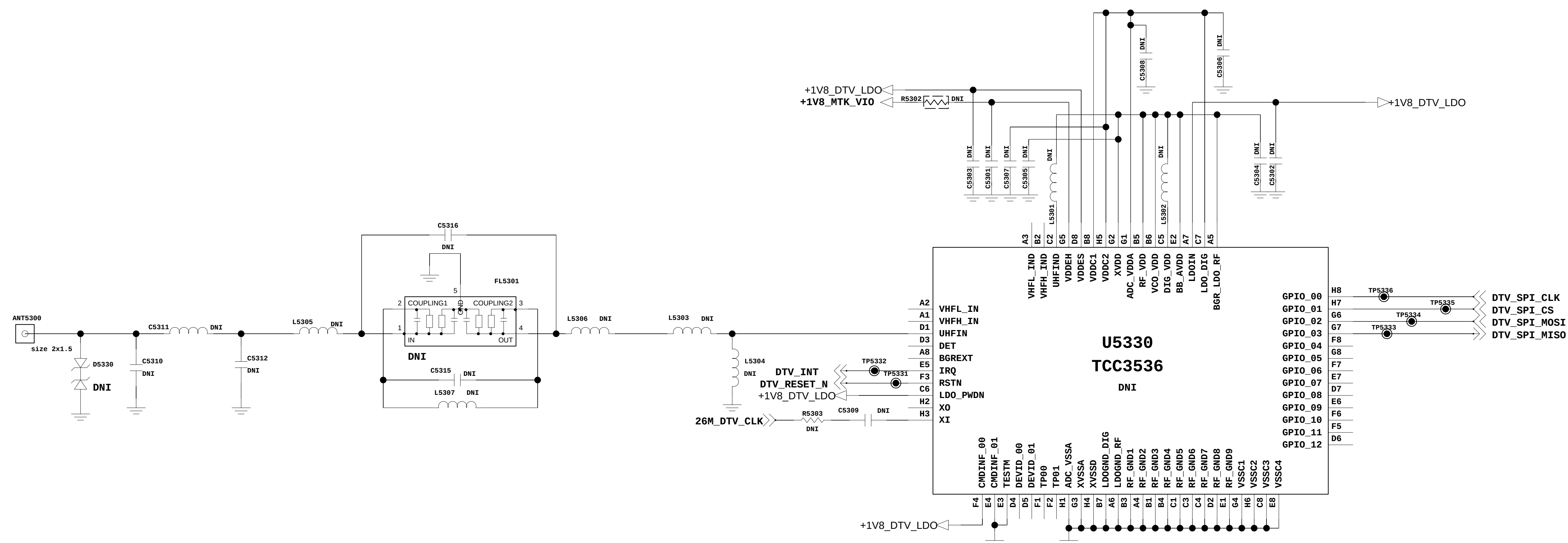


		TXEN	Mode	BS1	BS2
		GSM_PA	ENANT_SEL2	ANT_SEL1	ANT_SEL0
Stanby	Stanby	0	0	0	0
LB_TX	G_LB_TX	1	0	0	1
HB_TX	G_HB_TX	1	0	1	1
TRX1	G_HB_RX	0	1	0	0
TRX2	G_LB_RX	0	1	1	0
TRX3	B1	0	1	0	1
TRX4	B8	0	1	1	1
TRX5	B2	0	0	1	0
TRX6	B5	0	0	0	1

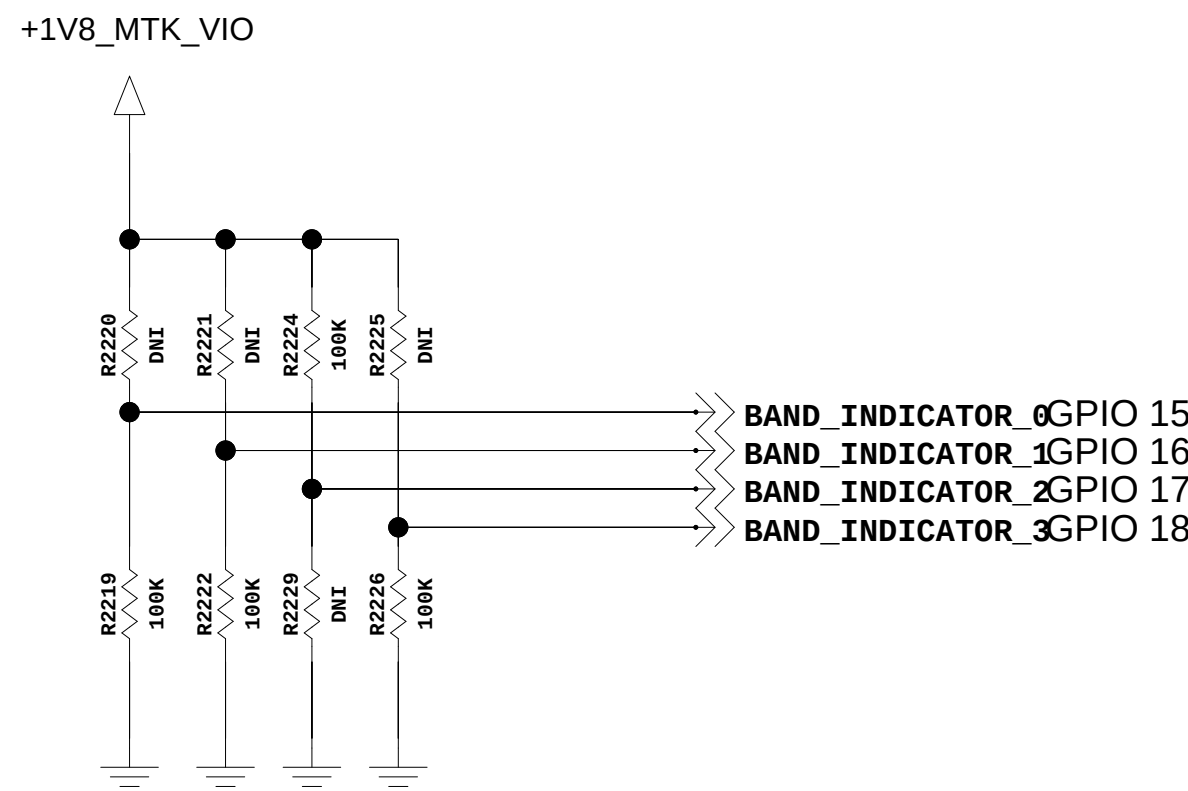
Rev_0.3



<5-3-1-10_Full-SEG_TCC3536> Rev_1.0

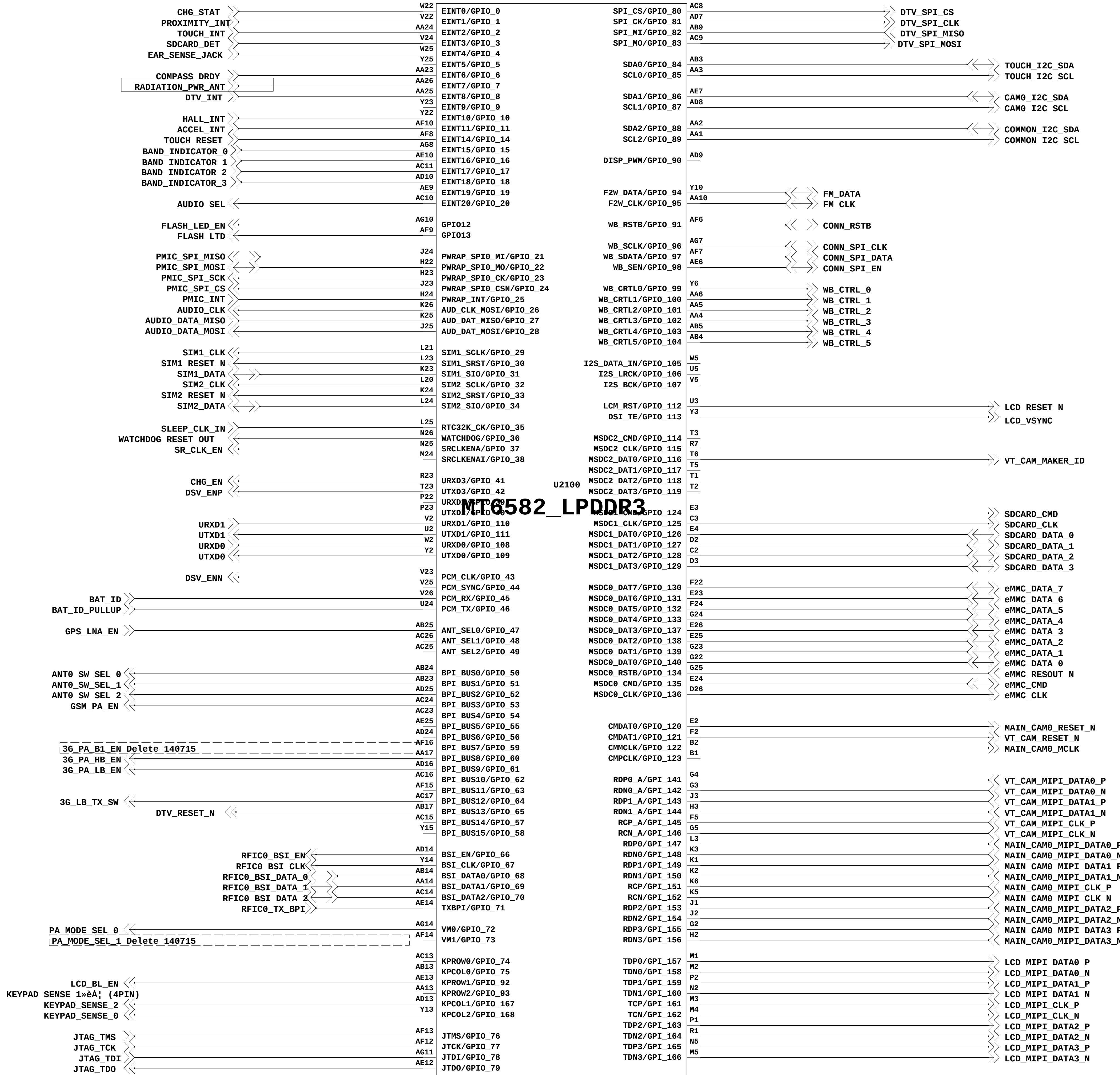


Model_Indicator

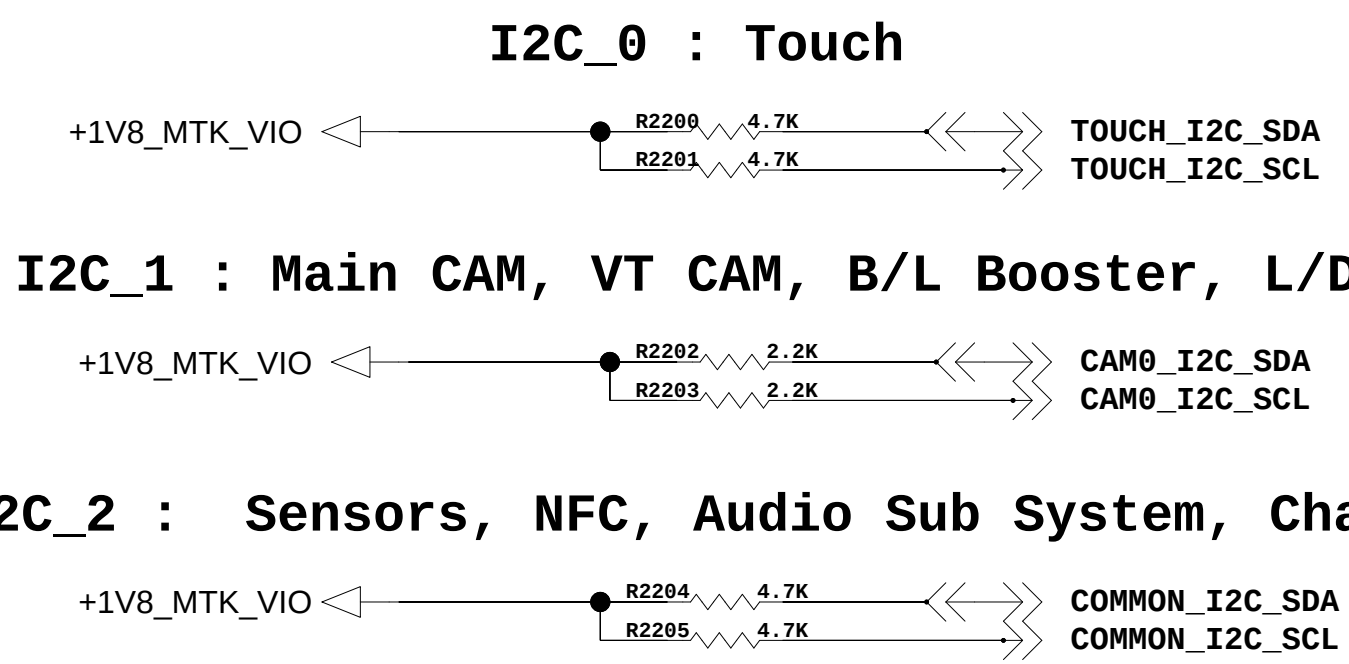


			GPIO 15	GPIO 16	GPIO 17	GPIO 18	SUM
H502TV	DUAL, 158	0	0	0	0	0	0
H502f	DUAL, 158	0	0	0	1	0	4
H502g	DUAL, 258	0	1	0	0	0	2
H500f	SINGLE, 158	1	0	0	0	0	1
H500TR	SINGLE, 158	1	0	1	0	0	5
H500g	SINGLE, 258	1	1	0	0	0	3

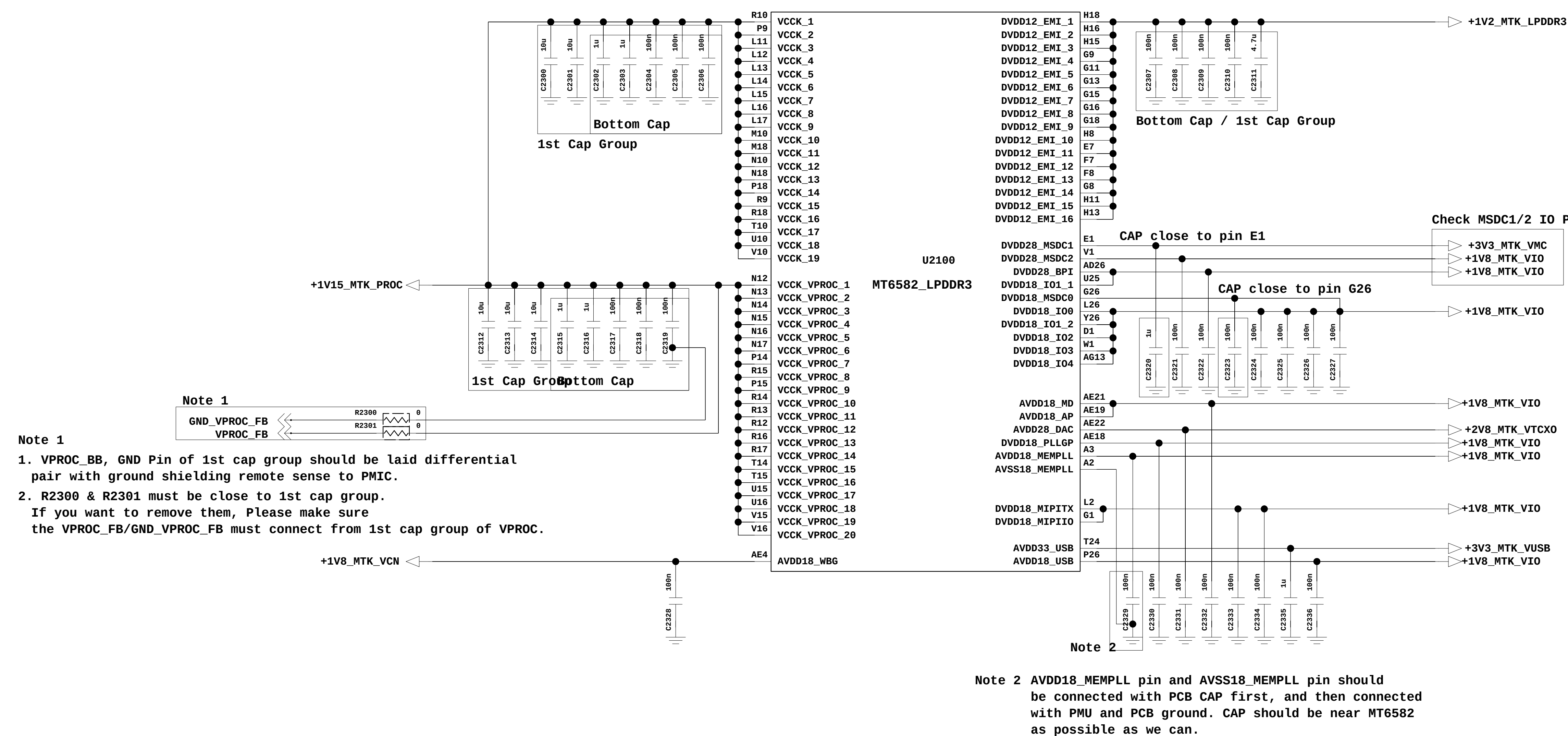
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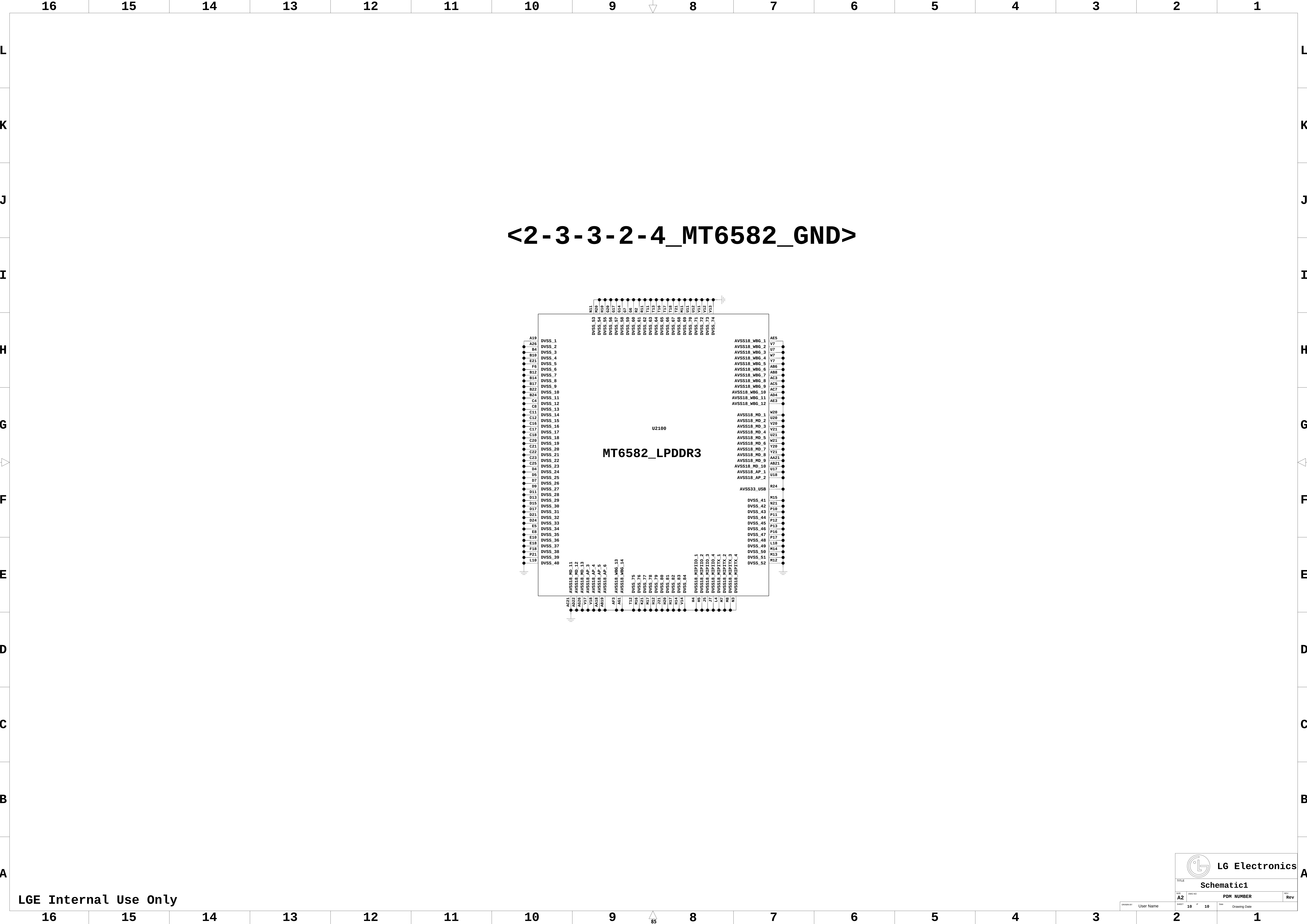


I2C Pull-Up



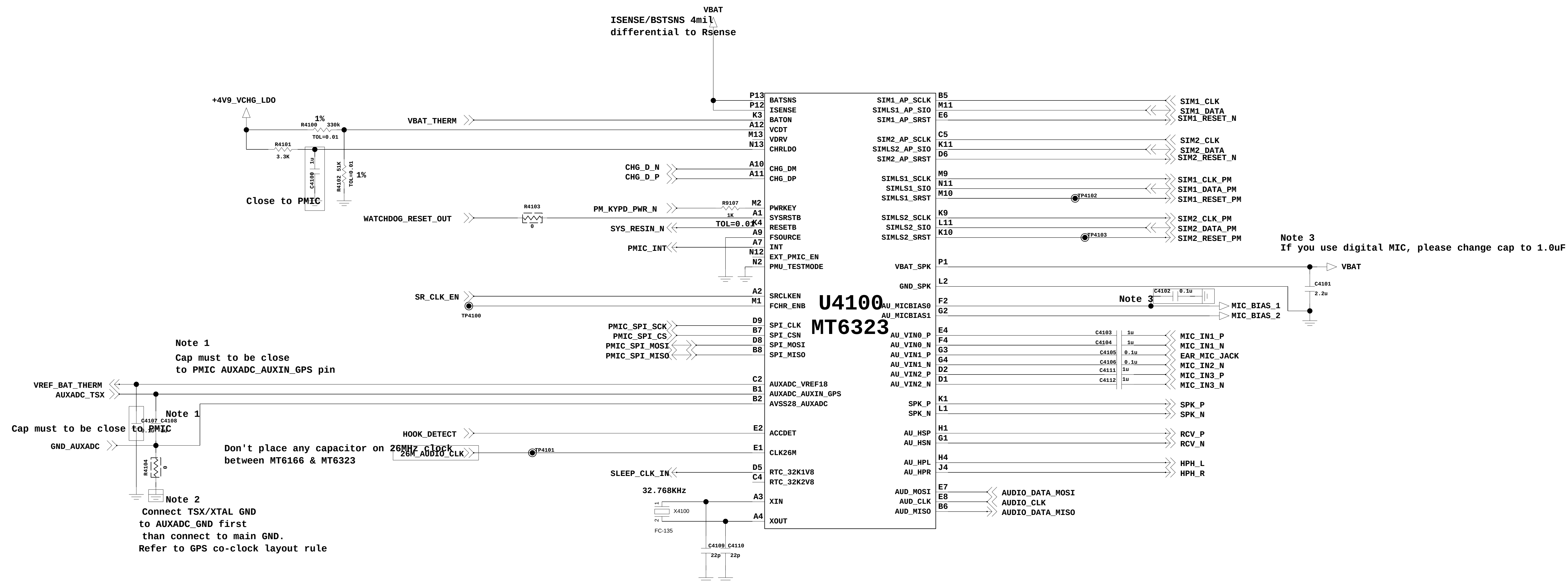
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<4-1-7-1_MT6323_DATA>

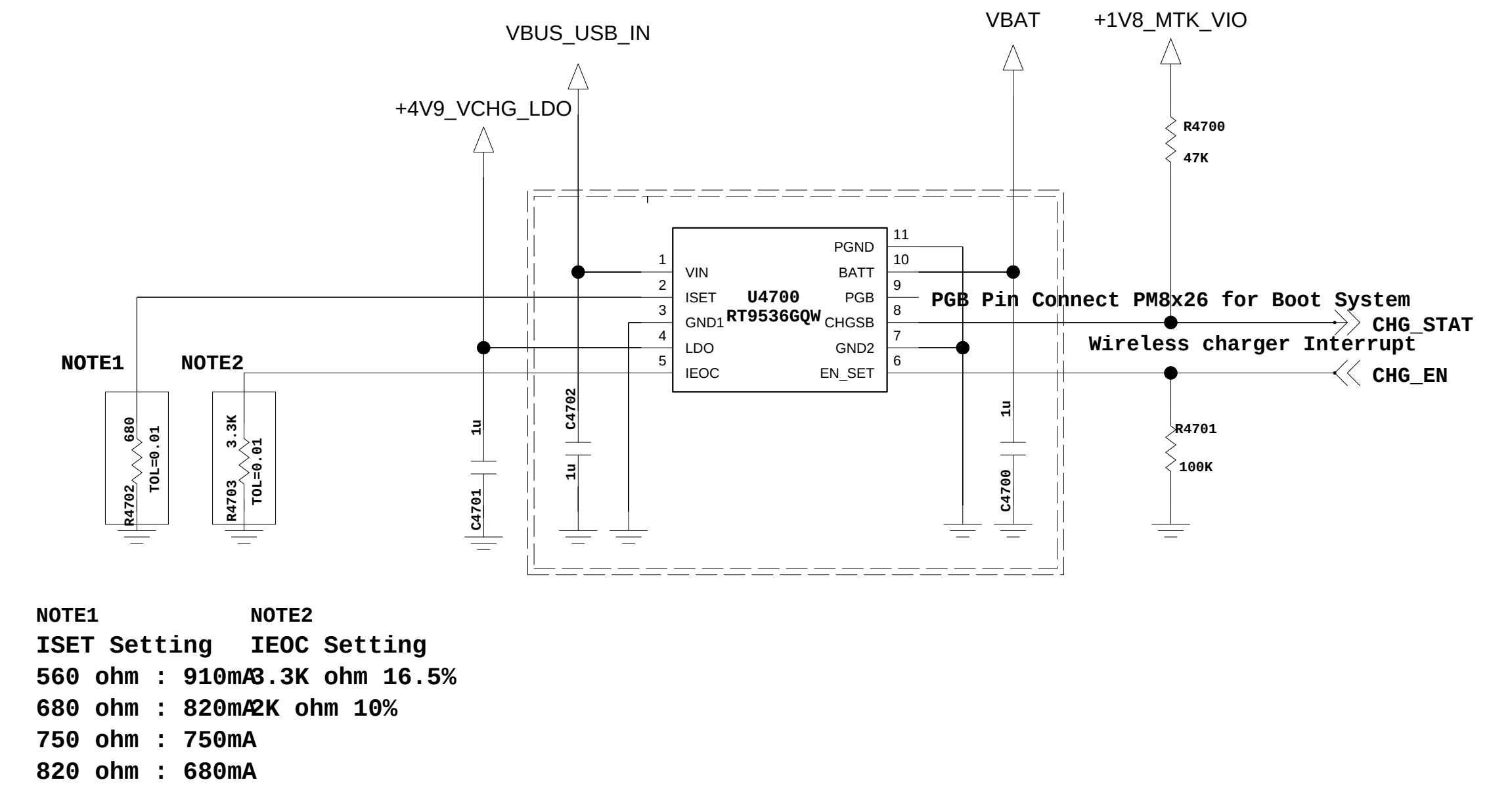
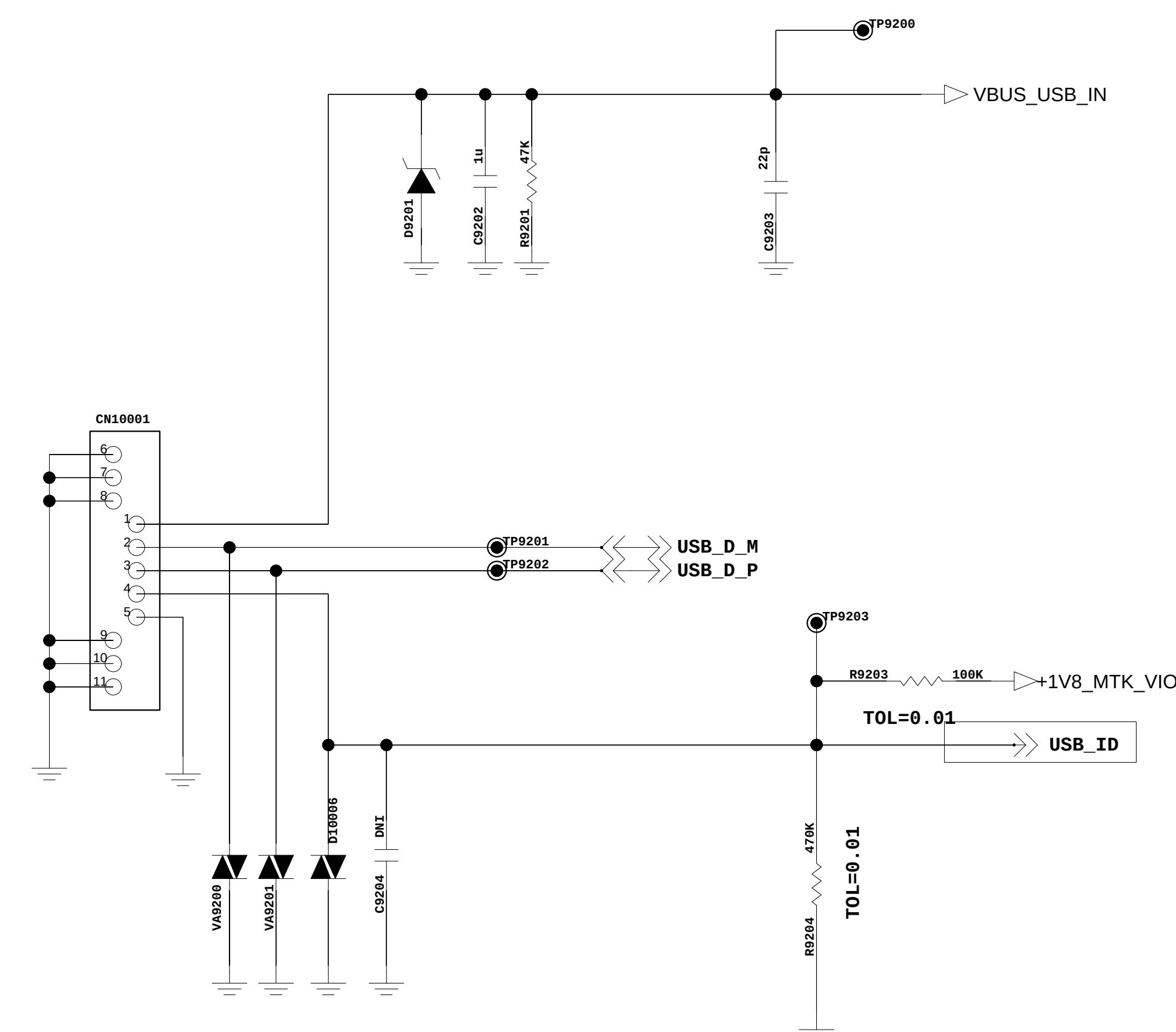
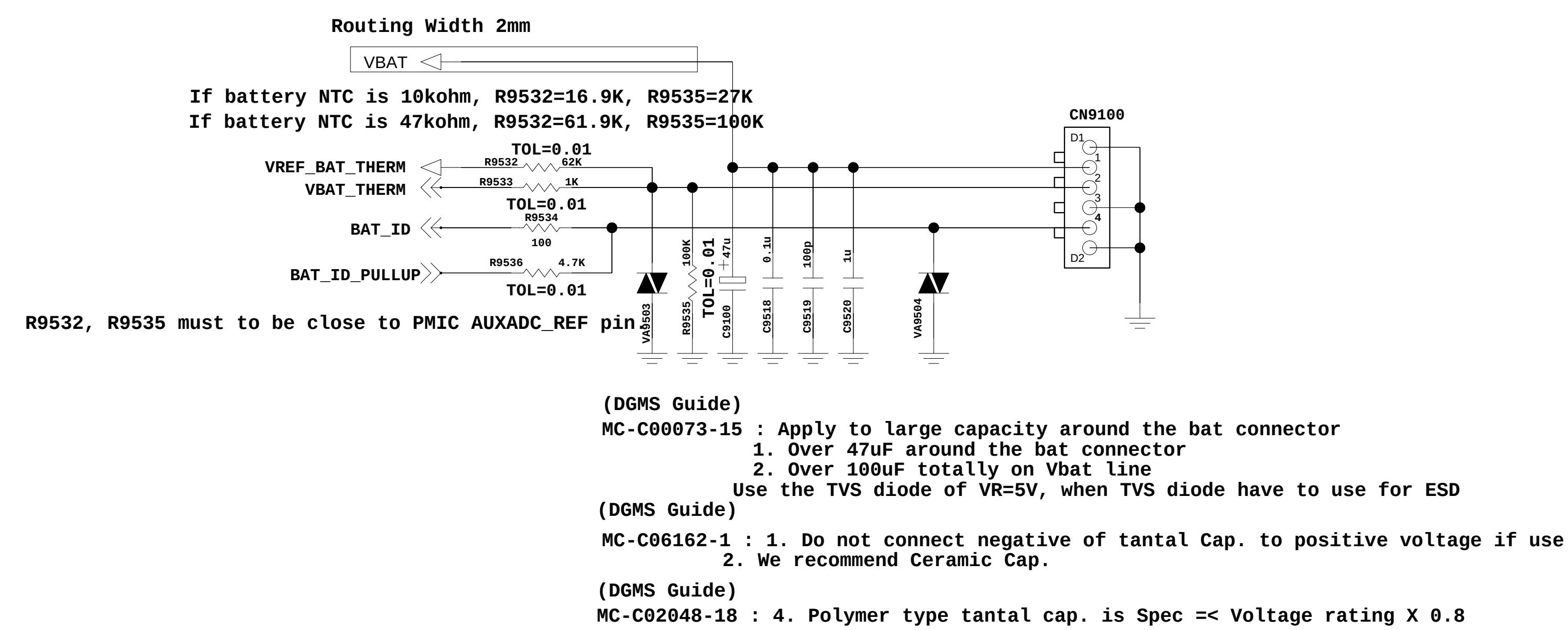
Rev_0.4



<9-1_Battery_CNT_4Pin> Rev_1.1

<9-2-1_USB_IO>
Rev_1.1

<4-7-1-1_Linear Charger> Rev_0.3



<4-1-7-2_MT6323_POWER>

REV_0.4

BACKUP BATT

Note 6
Refer to MT6323 design notice for Zener selection
500mW Between IC and IO port
Add Zener Diode
Place on the path from VBAT to IC (Battery connector or test point or IO connector)
--> But LG Using 150mW Zener Diode
--> There is no issue by L30 & L50

Note 1
Refer to MT6323 design notice for Buck GND layout rule.
Choose input cap MT6572 between MT6582

Ref.	MT6572	MT6582
C4136	DNI	1uF
C4137	10uF	4.7uF
C4138	4.7uF	4.7uF
C4139	10uF	4.7uF
C4150	DNI	4.7uF

Place input cap. as close MT6323 as possible and priority as VPROC>VPA>VSY

Cap closed to MT6323.

U4100
MT6323

Dedicate VSS ball, must return to cap then to main GND:
1. GND_VREF(N14) => Cap

Buck converter brief specifications

VSYS	0.7-1.3V/1.35V (optional) (6.25mV/step)	2,800	Processor / Digital Core
VSYS	2.2	1,400	System LDO input
VPA	0.5V - 3.4V (0.1V/step)	600	3G power amplifier

LDO types and brief specifications

Type	LDOname	Vout (Volt)	Imax (mA)	Application
ALDO	VCN28	2.8	30	RF chip
ALDO	VTCXO	2.8	40	13/26MHz reference clock
ALDO	VA	2.8	150	Analog baseband
ALDO	VCAMA	2.8	150	Analog power for camera module
DLDO	VCN_3V3	3.3/3.4/3.5/3.6	240	WiFi
DLDO	VIO28	2.8	200	Digital IO
DLDO	VSIM1	1.8/3.0	50	1st Sim card
DLDO	VSIM2	1.8/3.0	50	2st SIM card
DLDO	VSUB	3.3	20	USB
DLDO	VGPI	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	General purpose LDO
DLDO	VGPI	1.2/1.3/1.5/1.8 2.0/2.5/2.8/3.0	100	General purpose LDO
DLDO	VEMC_3V3	3.0/3.3	400	3.3V EMMC
DLDO	VCAM_AF	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	AF application
ALDO	VMC	1.8/3.3	100	SD 2.0/3.0 memory card
ALDO	VMCH	3.0/3.3	400	SD 3.0 memory card
DLDO	VIBR	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	Vibrator
PTCLDO	VRTC	2.8	2	Real-time clock
VSYS LDO	VM	1.24/1.39/1.54/1.84	700	Memory power
VSYS LDO	VRF18	1.825	200	RF application
VSYS LDO	VIO18	1.8	300	IO pad power
VSYS LDO	VCAMD	1.2/1.3/1.5/1.8	150	Camera application
VSYS LDO	VCAM_IO	1.8	100	Camera IO application
VSYS LDO	VGPI	1.2/1.3/1.5/1.8	200	General purpose LDO
VSYS LDO	VCN18	1.8	120	General purpose LDO

DATE

13

13

REV

1

DATE

13

13

REV

1

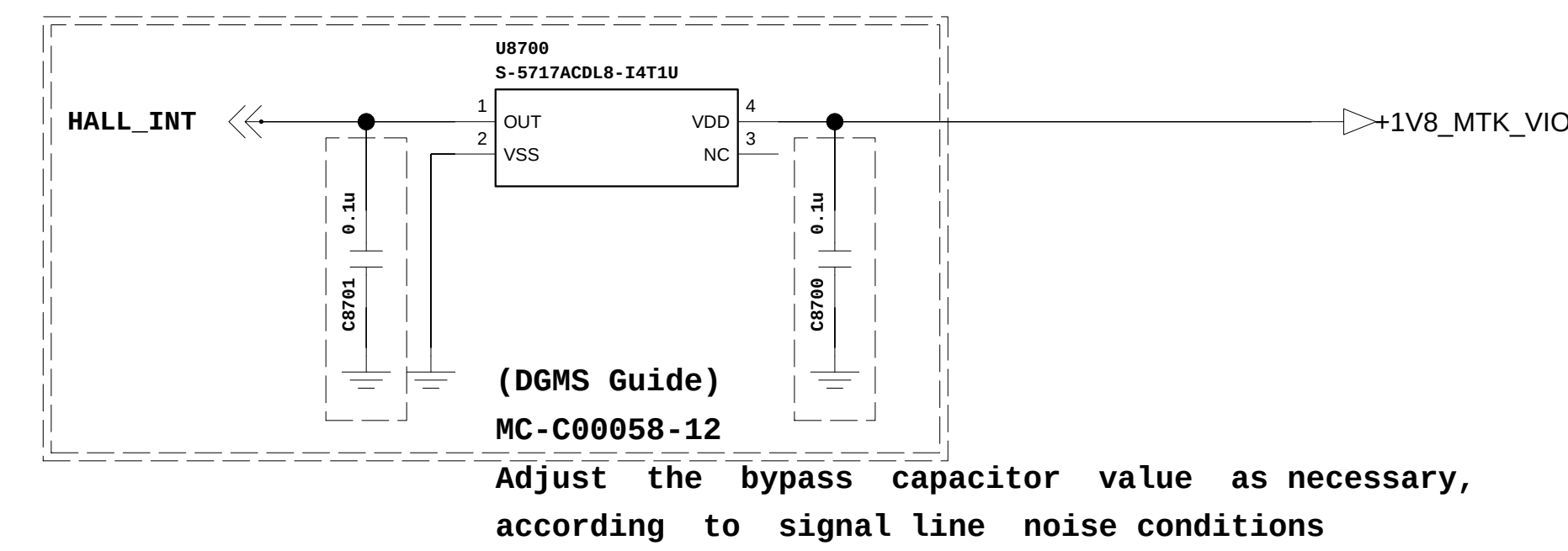
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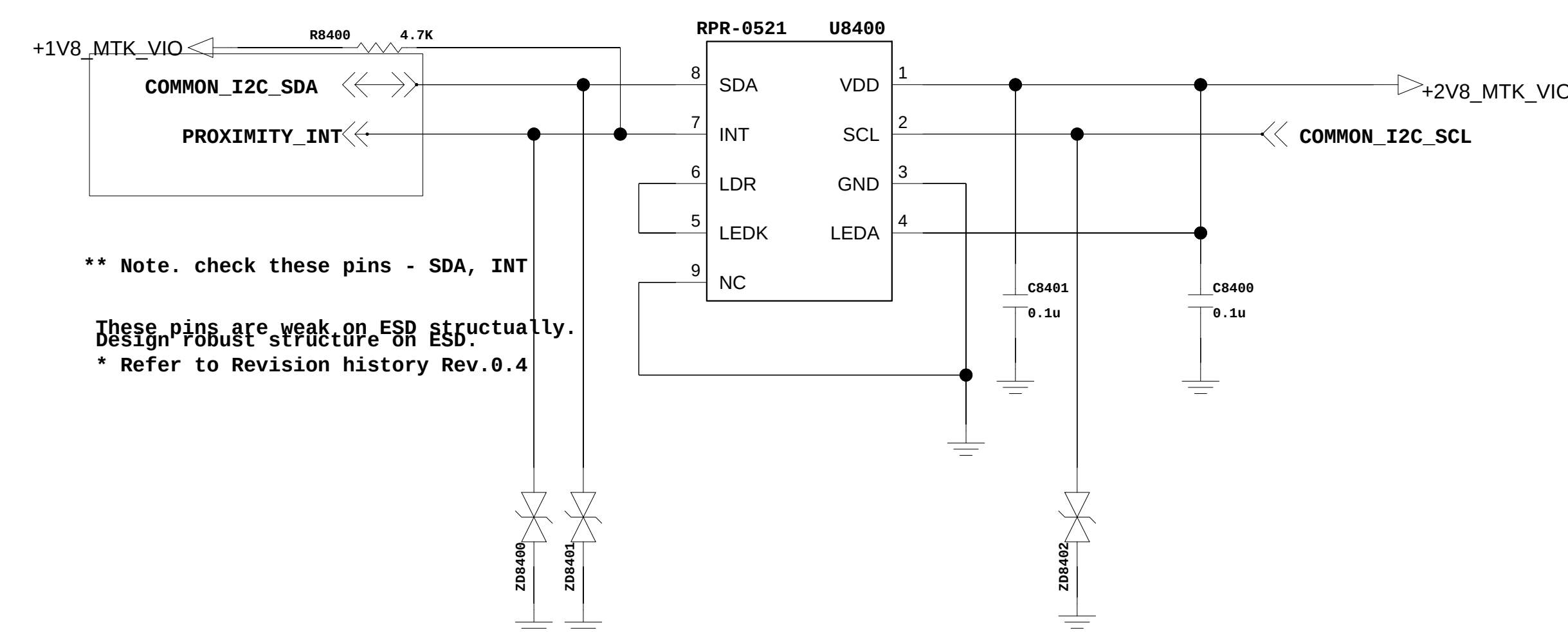
30pin 8M AF



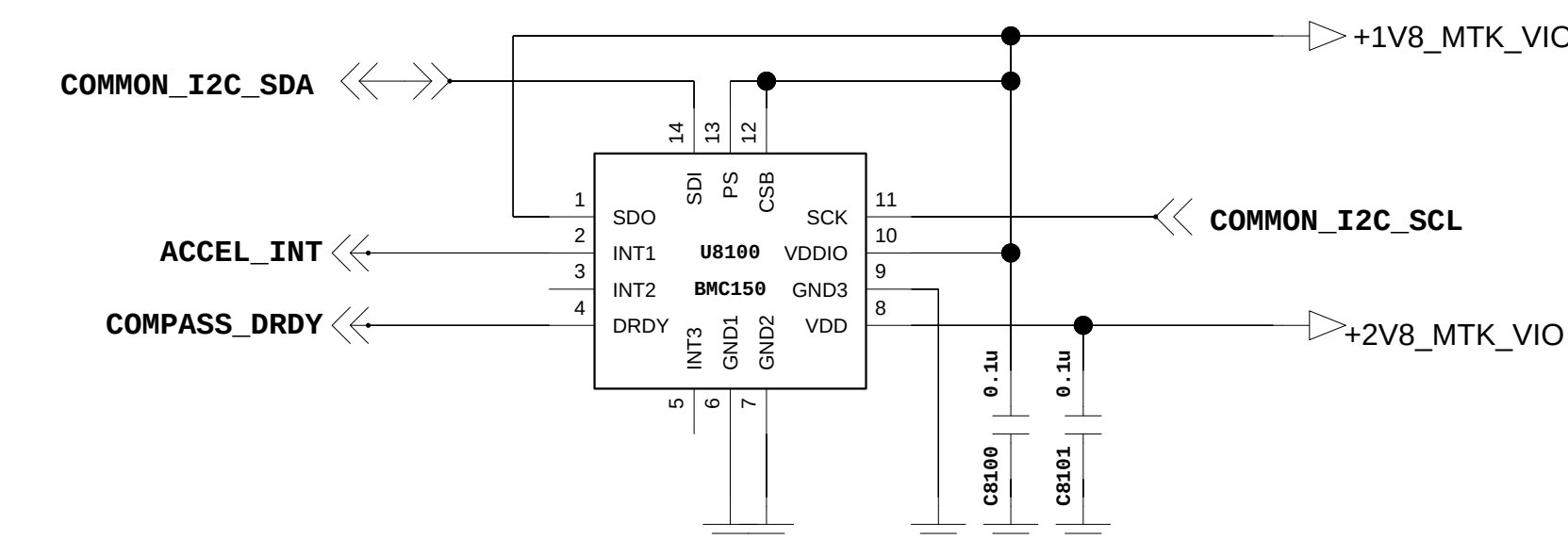
<8-7-1-5_Hall_IC_S57Rev>_1.1



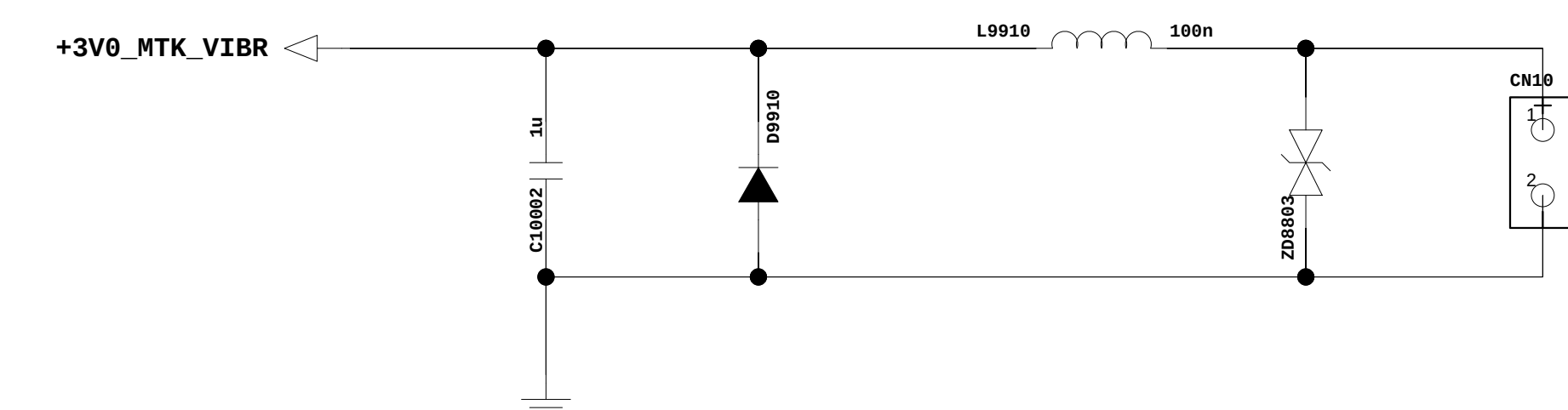
**<8-4-3-5_Proximity_RPR-0521>
Rev_0.3**



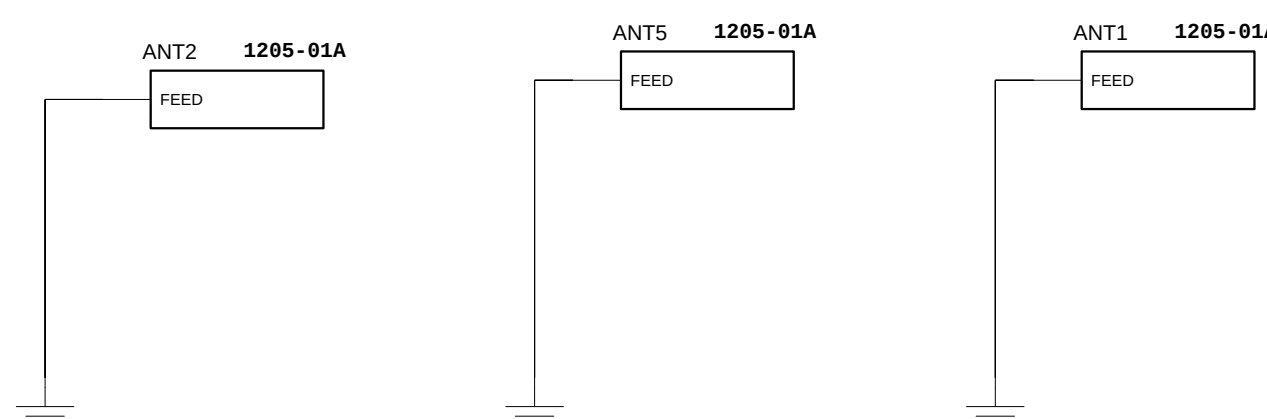
**<8-1-2-1_Accel_Compass_BMC150>
REV_1.0**



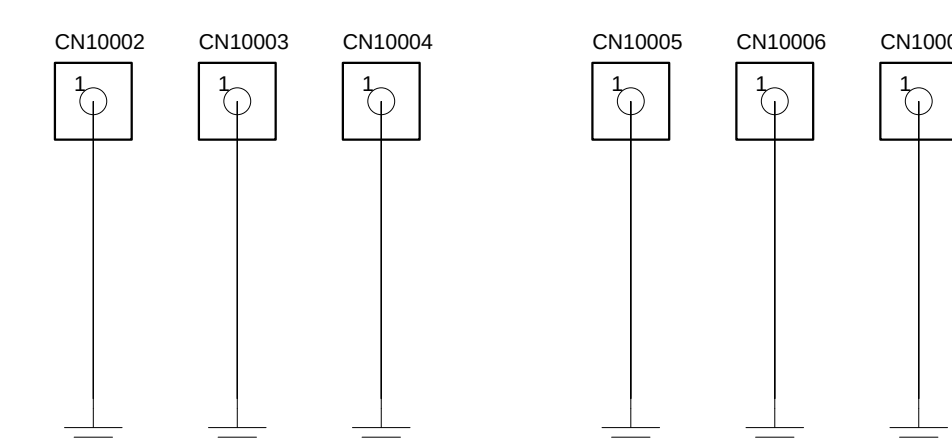
< 9-9-1-1_Motor >
Rev_0.3



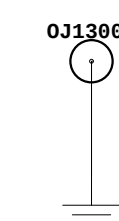
PCB Contact



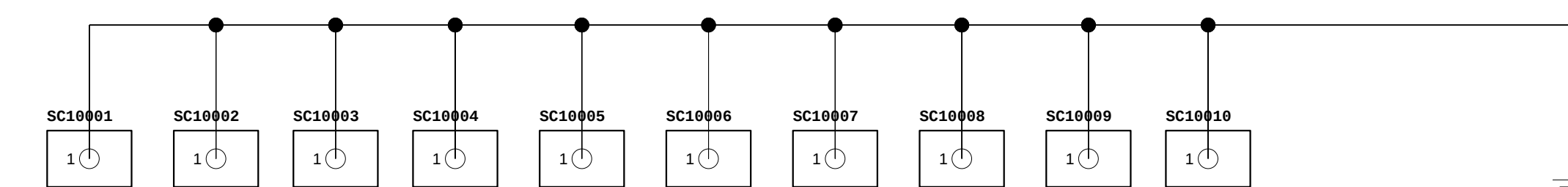
SMT GASKET



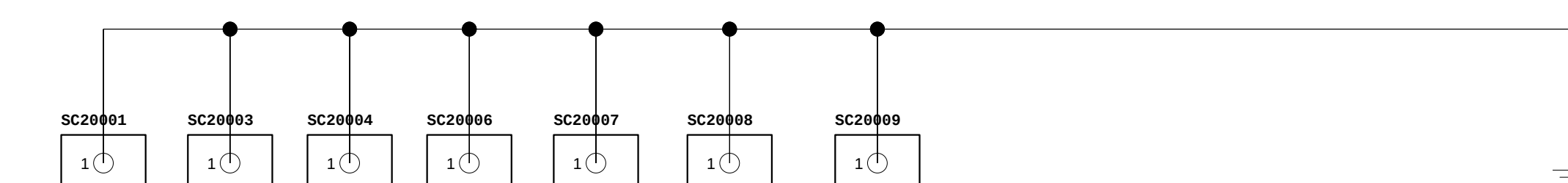
SCREW_HOLE



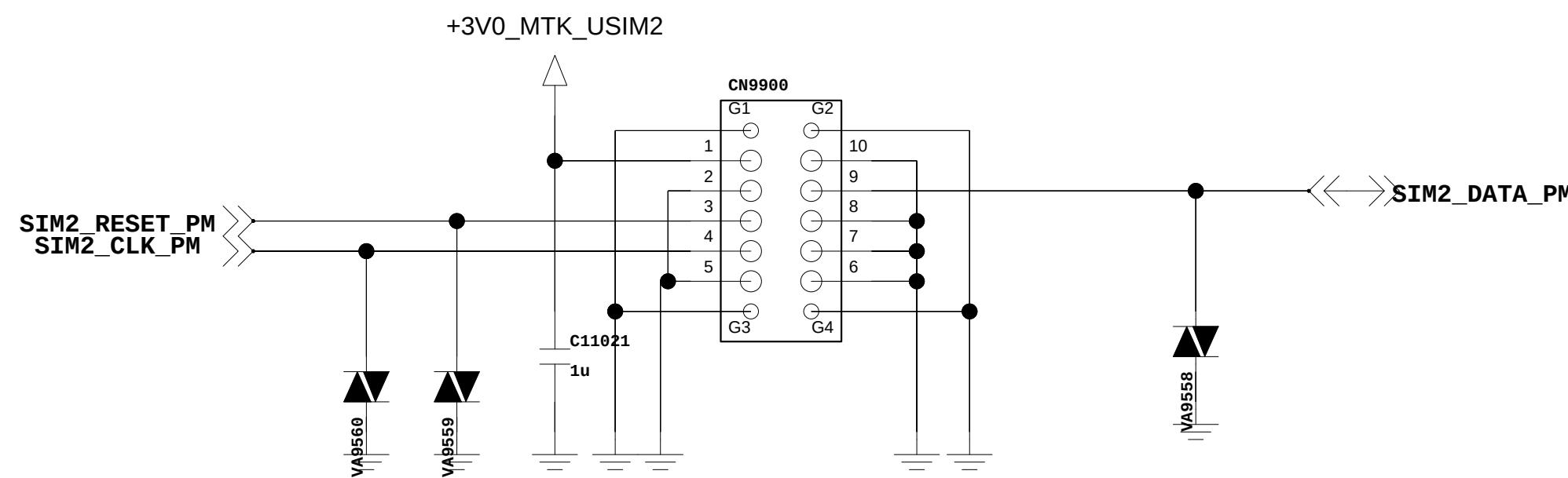
TOP SHIELD CAN CLIP



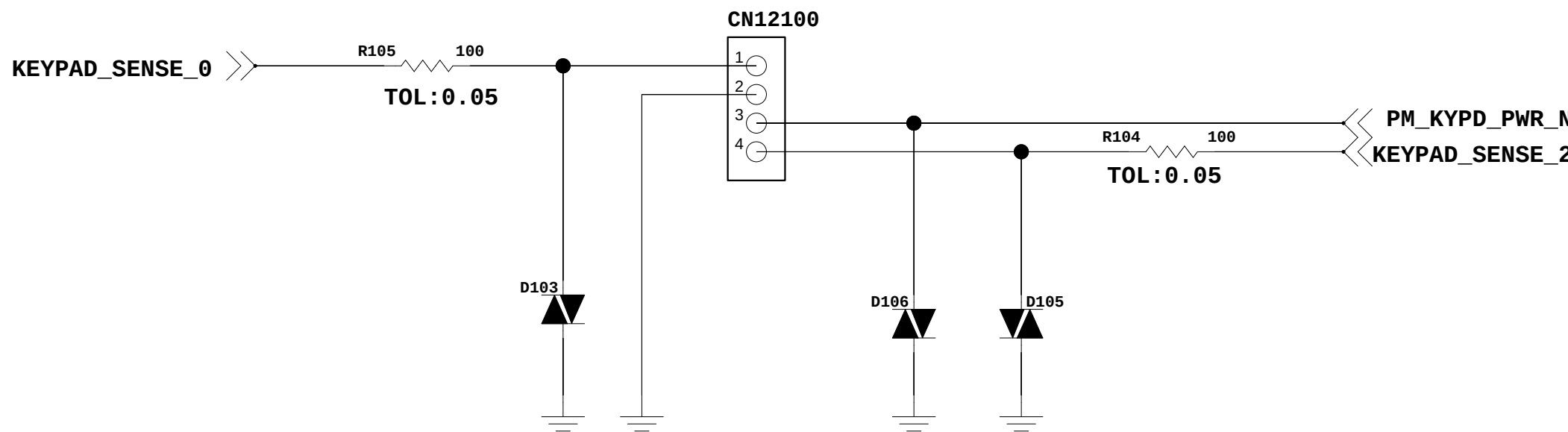
BOTTOM SHIELD CAN CLIP



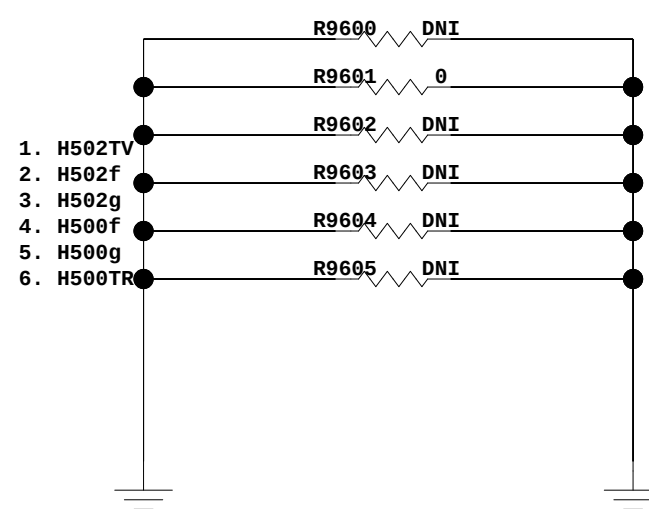
Sub SIM Connector



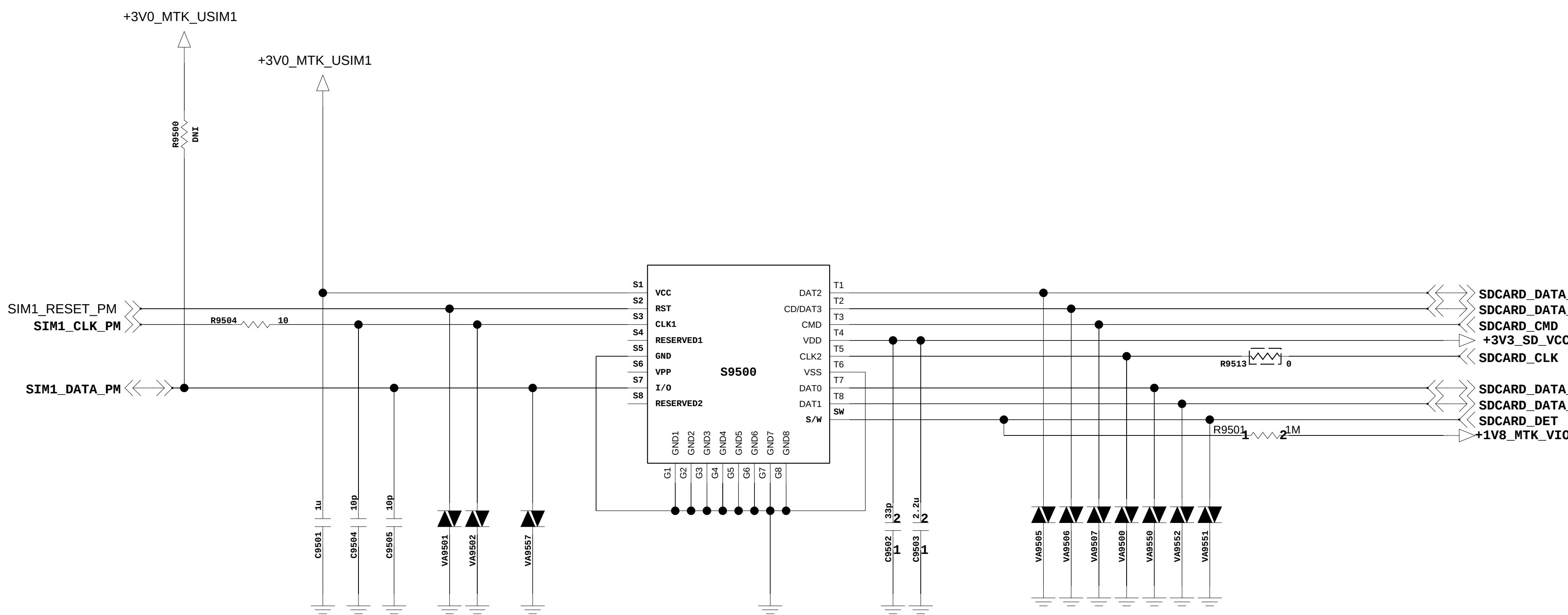
Back Volume Key



Model Name

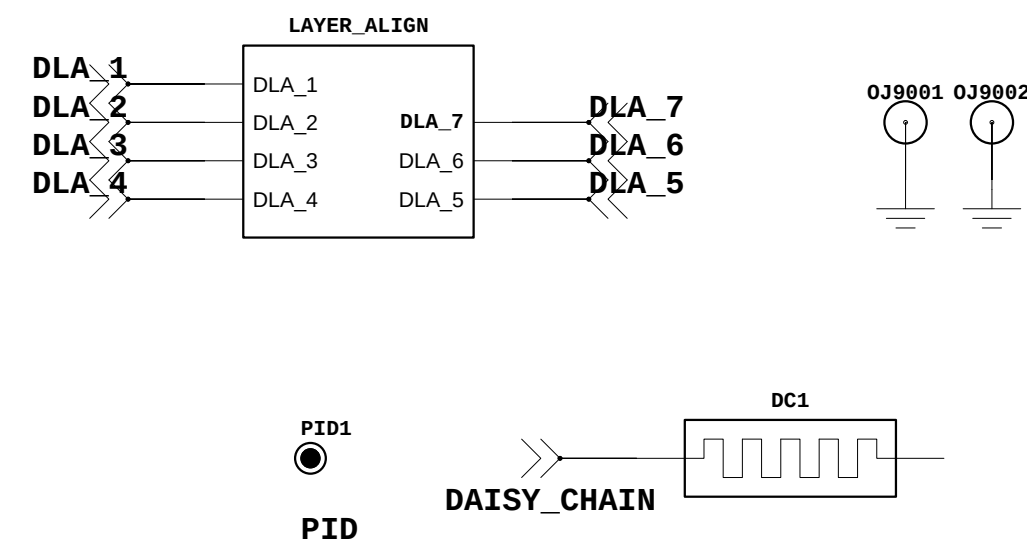


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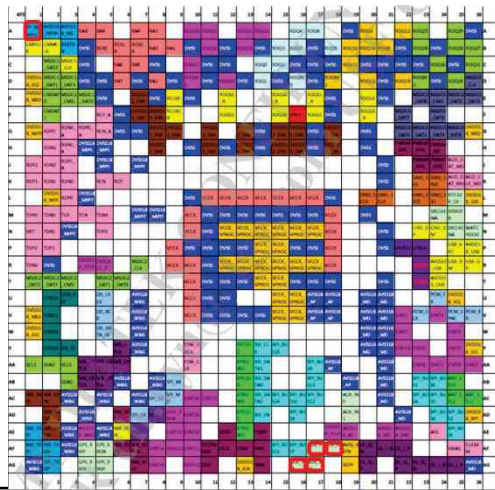
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Item for Manufacturing



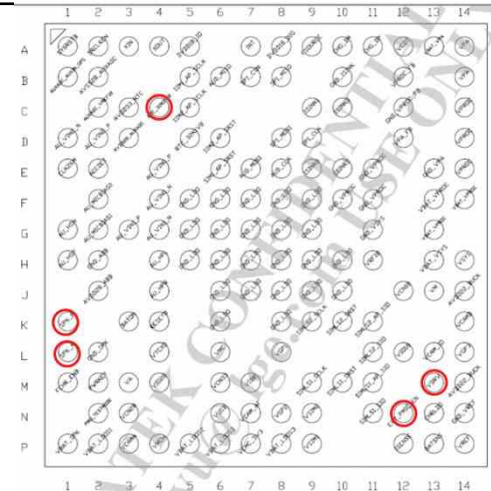
6. BGA PIN MAP

U2100_MT6582_LPDDR2



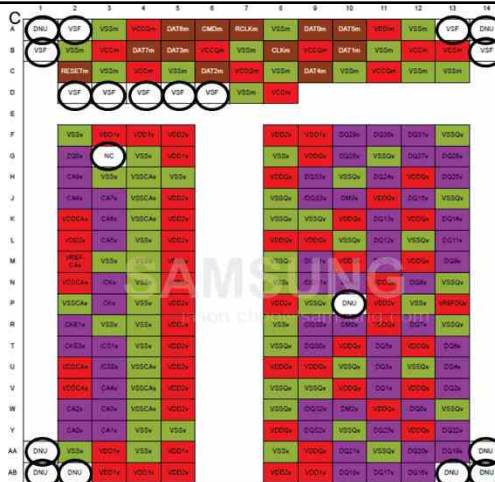
O Not used

U4100_MT6323_PMIC

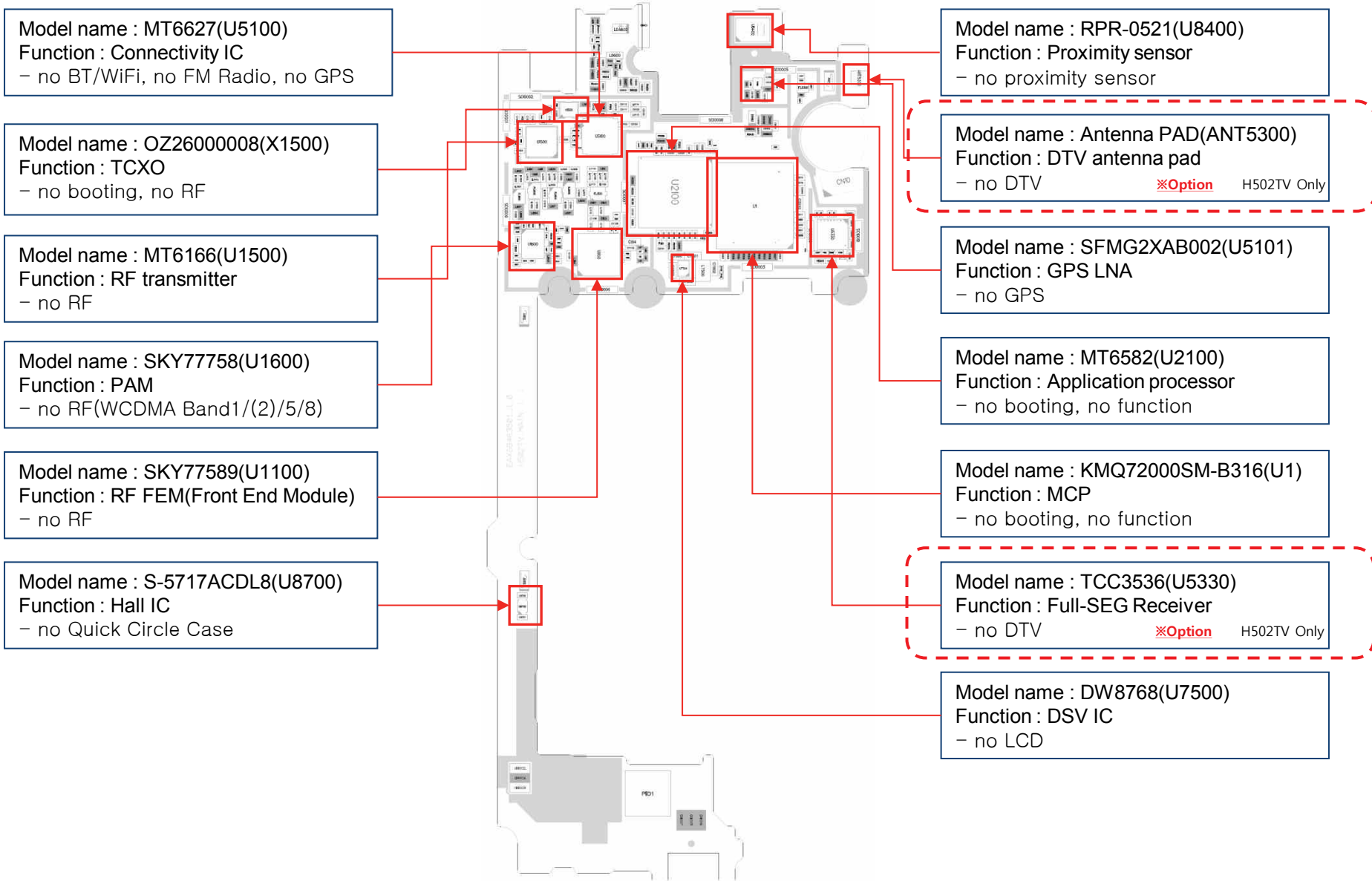


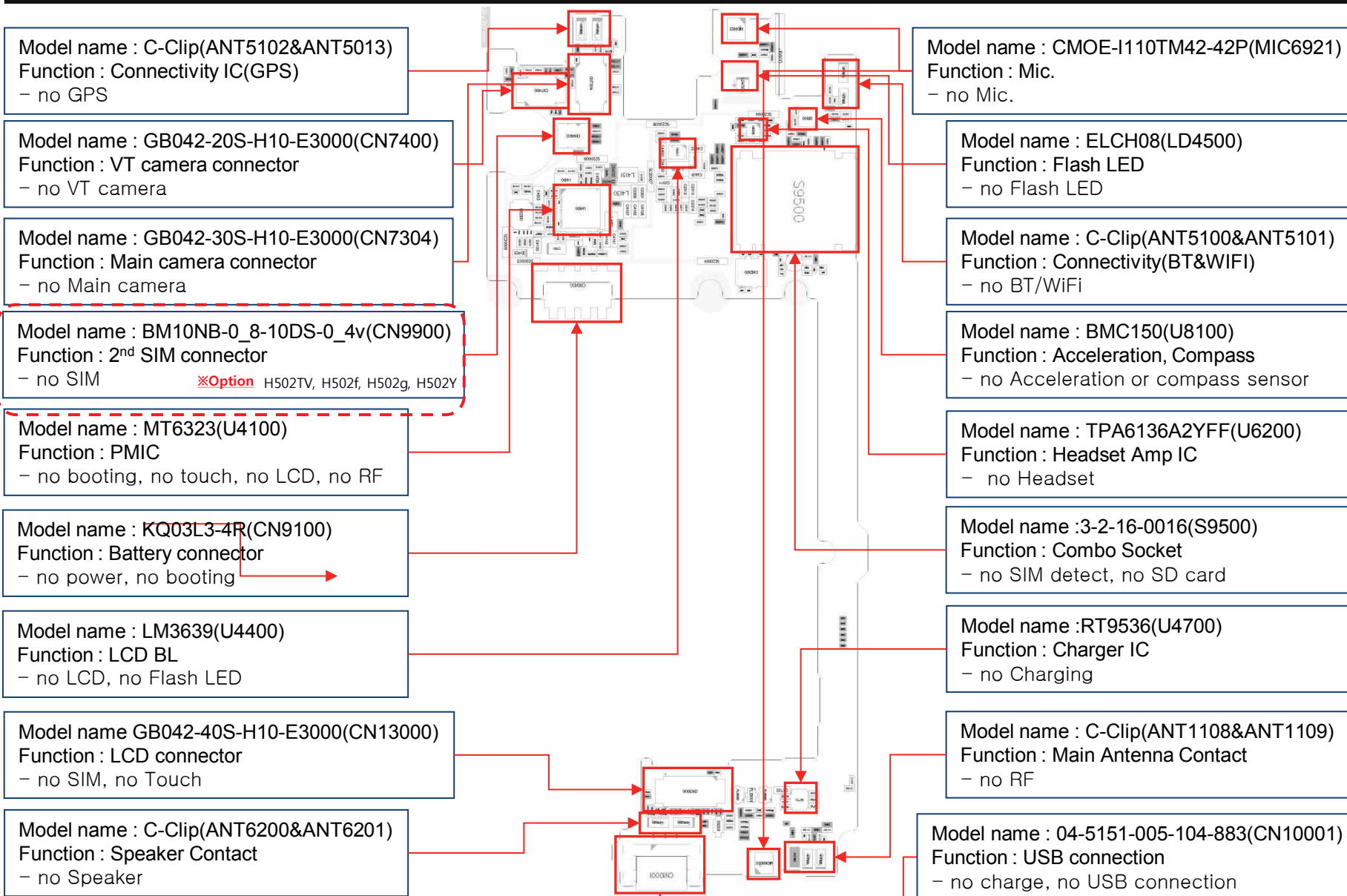
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U1_eMMC

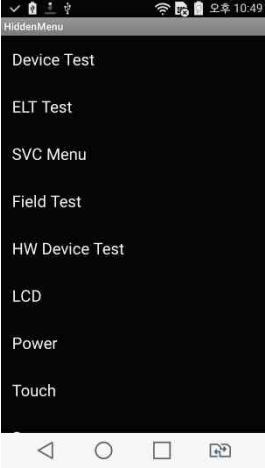
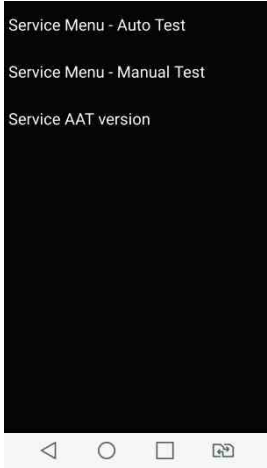
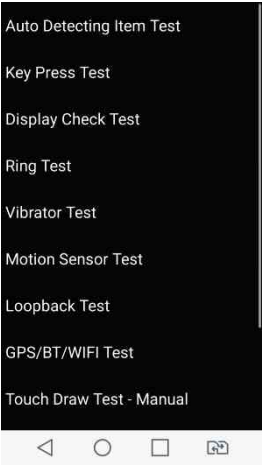


O Not used







1. Hidden Menu Start  - Start shortcut key: *#546368##502# - Hidden Menu List : Start the desired menu, click	2. Device Test  - Service Menu - Auto test : All auto test
2. Device Test List  - Service Menu - Manual test : Start the desired item, click	

8. HIDDEN MENU

1. Auto Detection Item Test Auto Detecting Item Pass <table><thead><tr><th>Item</th><th>Result</th></tr></thead><tbody><tr><td>USIM Card</td><td>Pass</td></tr><tr><td>USIM Card2</td><td>Pass</td></tr><tr><td>SD Card</td><td>Pass</td></tr><tr><td>Ear Phone wrong</td><td>Pass</td></tr><tr><td>TA wrong</td><td>Fail</td></tr><tr><td>Battery</td><td>59%</td></tr></tbody></table> Don't touch Proximity Sensor Test 후 근접센서 가리지 말 것 PASS FAIL	Item	Result	USIM Card	Pass	USIM Card2	Pass	SD Card	Pass	Ear Phone wrong	Pass	TA wrong	Fail	Battery	59%	2. Key Press Test EXIT the 'EXIT' button to exit END UP DOWN RIGHT LEFT Proximity Cal Pass:16 Block the sensor with your hand Proximity : FAR Block the light with your hand Light:246.00 PASS FAIL	3. Display check test PASS FAIL	4. Ring test EXIT the 'EXIT' button to exit Ring Test ON SOUND PASS FAIL
Item	Result																
USIM Card	Pass																
USIM Card2	Pass																
SD Card	Pass																
Ear Phone wrong	Pass																
TA wrong	Fail																
Battery	59%																
5. Vibrator test EXIT the 'EXIT' button to exit Vibrator Test ON VIBRATION PASS FAIL	6. Motion Sensor test Calibration LV : 0 PASS <table><thead><tr><th></th><th>X</th><th>Y</th><th>Z</th></tr></thead><tbody><tr><td>Magnetic</td><td>39.50</td><td>-1.50</td><td>143.50</td></tr><tr><td>Accelerometer</td><td>-0.02</td><td>9.19</td><td>2.89</td></tr></tbody></table> Motion Accel(N/A) Gyroscope Sensor(N/A) Stand the phone vertically. EXIT PASS FAIL		X	Y	Z	Magnetic	39.50	-1.50	143.50	Accelerometer	-0.02	9.19	2.89	7. Loopback test EXIT the 'EXIT' button to exit LOOPBACK TEST AUDIO ON PASS FAIL	8. GPS/BT/WIFI test GPS(RF) Test 12 Sec GPS Test GLO Test CNo : 38.6 dB N/A Bluetooth Scanning Bluetooth Scanning... 6E:B9:46:E5:82:8D Pass WiFi Scanning WiFi Scan Complete. BSSID, 00:00:00:00:00:00 Pass PASS Retry FAIL		
	X	Y	Z														
Magnetic	39.50	-1.50	143.50														
Accelerometer	-0.02	9.19	2.89														

9. Touch Draw test-Manual	10. Camera(Main) test	11. Camcorder test	12. Camera(VT) test
			

1. Summary

Tool Version	DLL name	USB Driver	
LGFLAS Hv190	LGH502f_20150326_LGFLASHv190_Download	LGUnitedMobileDriver_S50MAN311AP22_ML_WHQL_Ver_3.11.3	
Please Check the Version to “B2B”			
H/W			
	Name	Part No.	SW
D/L Cable	Micro 5P (56-open-910K) USB DLC	RAD321 67835	TOT

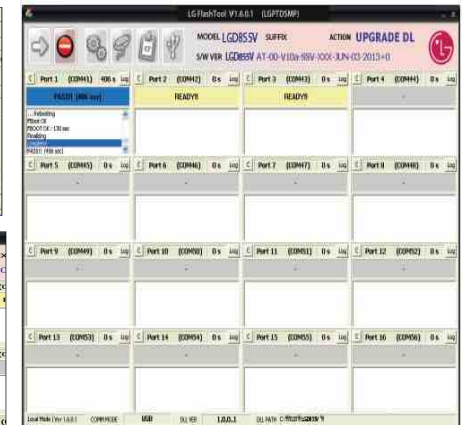
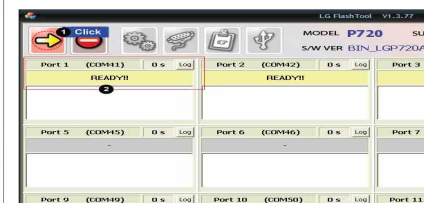
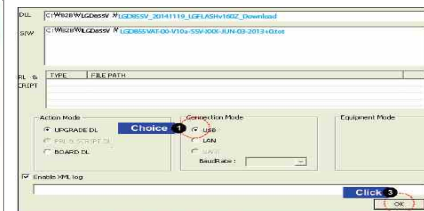
2. USB COM port Setting



3. USB D/L Cable setting



4. Flash tool D/L setting



※ If you want more information, please refer B2B's Notification "Download User Guide".

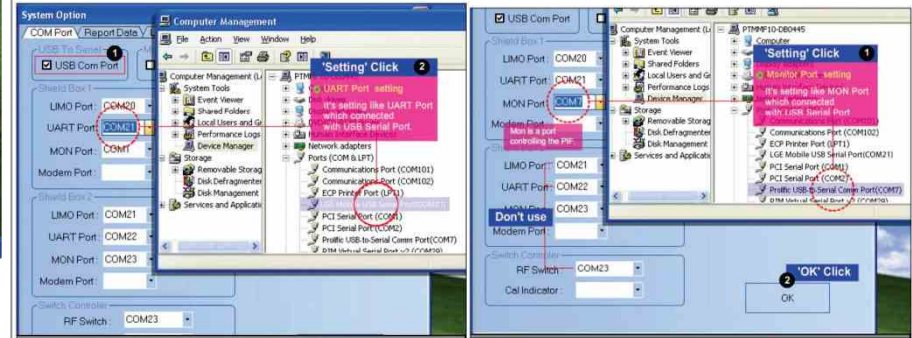
1. Summary

CAL INFORMATION		
S/W VERSION		
[TachyonV2013] LGH502F 20150325 ALL File Calibration		
Please Check the Version to "B2B"		
H/W		
	Name	Part No.
PIF	PIF200	BJAY0024021
USB Cable	USB Cable 2.0	RAD32247898
Power Cable	DC Power Cable	RAD32247878
I/O Cable	5P E-SATA DC Plug	RAD32167861
RF Cable Main	MXFJM3WX6000	RAD32827895
Power Supply PIF	Power supply 5.2V	
Power Supply Phone	Power supply 5.0V	
PF Test Equipment	Agilent/CMW500	
Notice	1. Use the Battery (Refer to Attached ppt) 1) Phone states: Power off 2) If do not use the battery, TX fails. 2. Port Setting (Refer to Attached ppt) 1) Uart Port1 : Use the "LGE Mobile USB Serial Port"	
CMW 500 RF Cable connection		

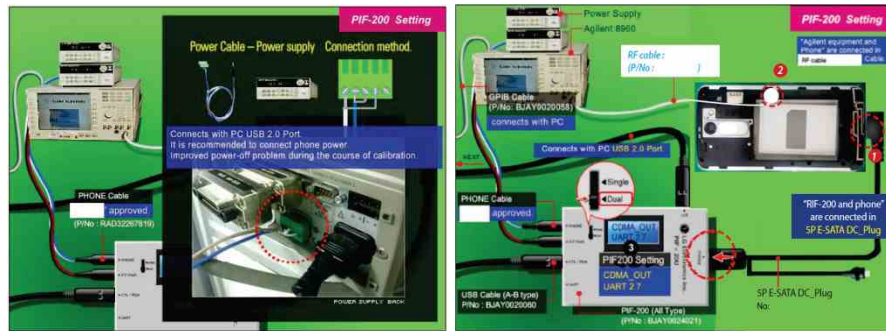


LGH502TV

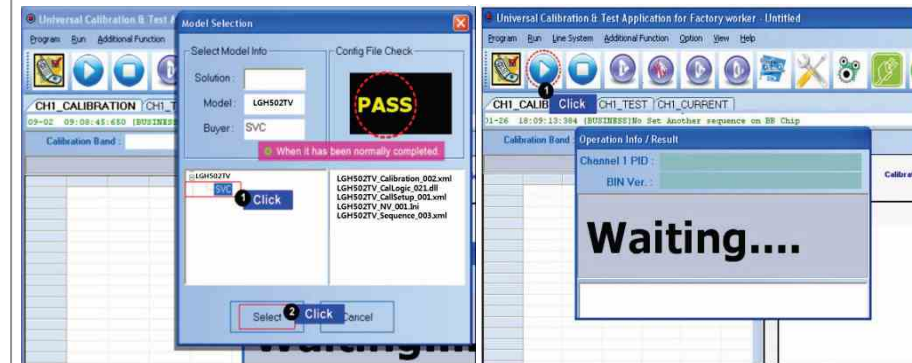
2. . USB COM port Setting



3. Calibration Cable setting



4. Flash tool D/L setting



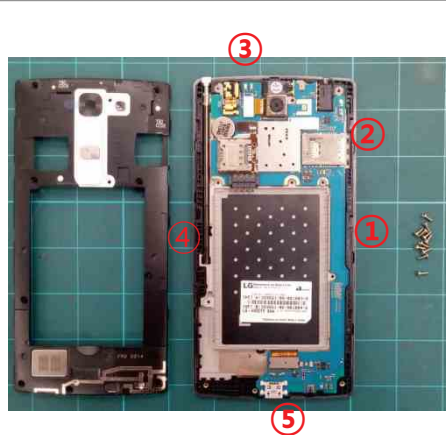
※ If you want more information, please refer B2B's Notification "RF Calibration User Guide".

11. DISASSEMBLE GUIDE

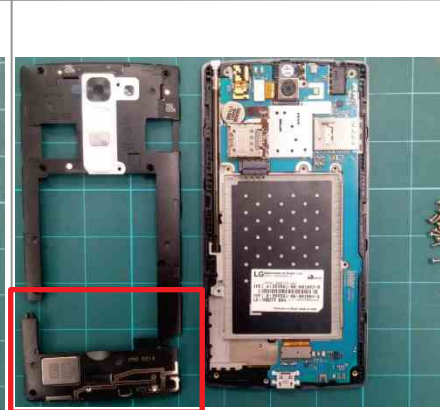
1. Disassemble Screw (11ea)



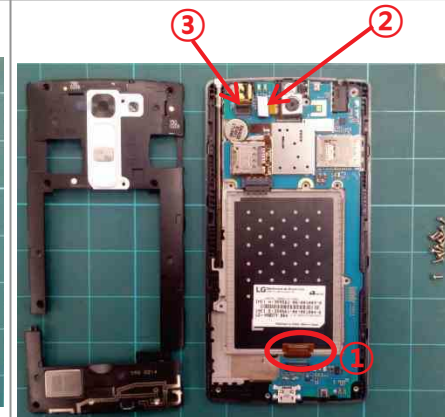
2. Disassemble Rear cover



3. Disassemble Main Antenna



4. Disassemble Connector



5. Disassemble Main PCB



Hook Disassemble

6. Disassemble HW Parts (5ea)



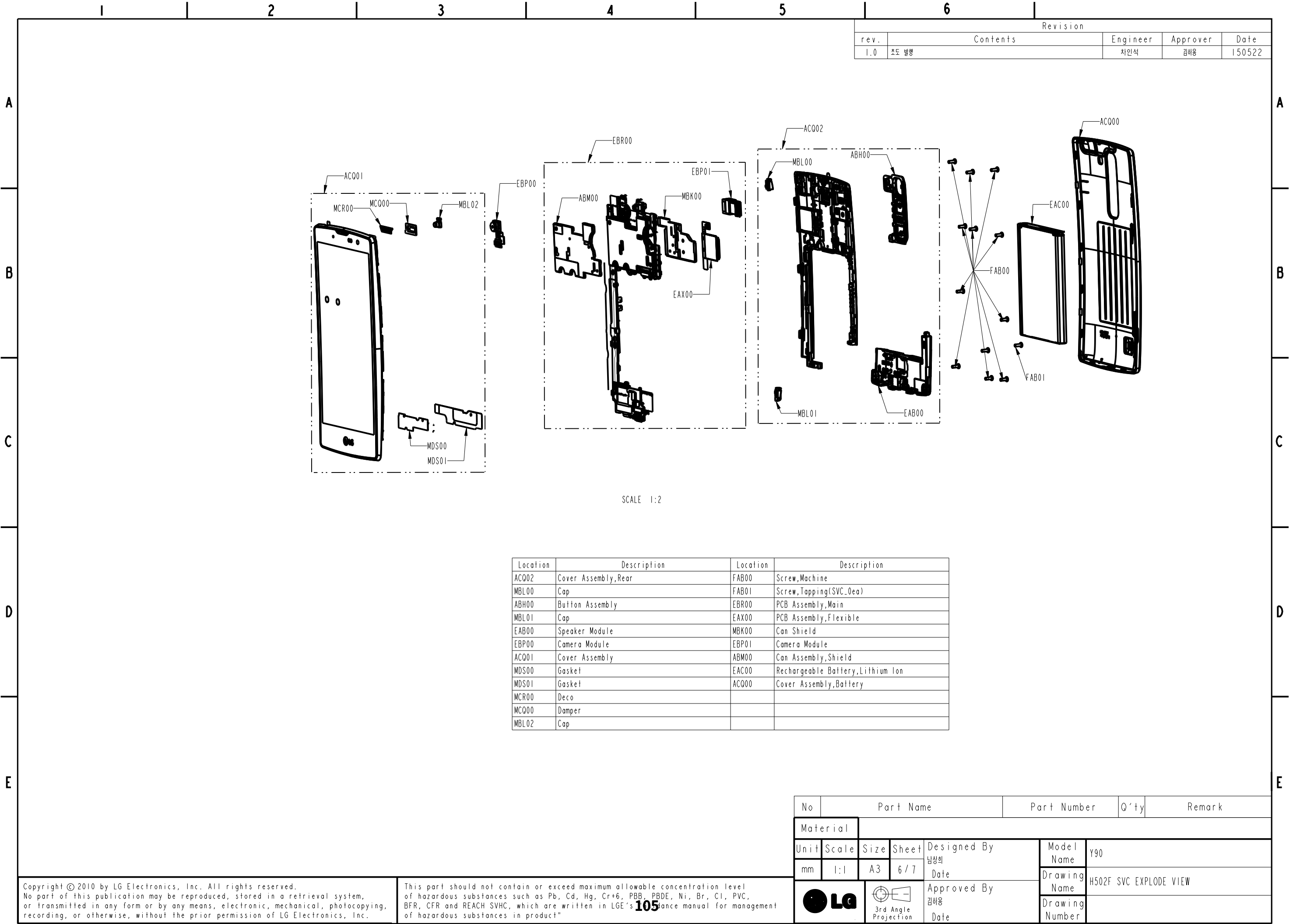
7. LCD Disassemble
(by Heating machine)



8. Complete Disassembling



12. EXPLODED VIEW



13. REPLACEMENT PART LIST

No	P/N	Description	Qty	Location no
1	AGQ88647608	Phone Assembly	1	AGQ000000
2	MLAZ0054701	Label	1	MEZ000000
3	MEZ66387402	Label,Approval	1	MEZ002100
4	ACQ88372308	Cover Assembly,EMS	1	ACQ100400
5	ACQ87831906	Cover Assembly,Rear	1	ACQ02
6	ACQ87852406	Cover Assembly,Rear(SVC)	1	ACQ105800
7	EAA63805201	PIFA Antenna,Bluetooth	1	EAA030101
8	MJN69568101	Tape,Protect	1	MJN061100
9	MJN69388601	Tape	1	MJN000000
10	MFB63813101	Lens,Flash	1	MFB029600
11	MEZ64319901	Label,After Service	1	MEZ000900
12	MCQ68545101	Damper	1	MCQ000000
13	MCQ68166901	Damper,Motor	1	MCQ049800
14	MCQ68166801	Damper,Camera	1	MCQ009400
15	MCK68589401	Cover,Rear	1	MCK063300
16	MBL66357701	Cap	1	MBL00
17	ABH75459902	Button Assembly	1	ABH00
18	MBG65483302	Button	1	MBG000000
19	MBG65562902	Button	1	MBG000001
20	MCR65887702	Decor	1	MCR000000
21	MJN69388701	Tape,Protect	1	MJN061100
22	EAX66264501	PCB,Flexible	1	EAX010700
23	EAA63865501	PIFA Antenna,GPS	1	EAA030100
24	MKC65379201	Window,Camera	1	MKC009400
25	MBL66318301	Cap	1	MBL01
26	EAB63668401	Speaker Module	1	EAB00
27	ACQ87832032	Cover Assembly,Bar	1	ACQ003400
28	EAU62283801	Motor,DC	1	EAU010000
29	EAG63930601	Jack,Phone	1	EAG140300
30	EAB63288401	Receiver	1	EAB010400
31	EBP62481901	Camera Module	1	EBP00
32	ACQ88378002	Cover Assembly	1	ACQ01
33	MDS65190701	Gasket	1	MDS00
34	MCQ68166601	Damper,Camera	1	MCQ009400
35	ACQ87791602	Cover Assembly,Front	1	ACQ032700
36	MDS65013401	Gasket	1	MDS01
37	MHK64907101	Sheet	1	MHK000002
38	MHK64945601	Sheet	1	MHK000000
39	MJN69408301	Tape	1	MJN000001

13. REPLACEMENT PART LIST

40	MJN69428101	Tape	1	MJN000000
41	MCR65966502	Decor	1	MCR00
42	MCQ68264701	Damper	1	MCQ00
43	MCQ68166701	Damper	1	MCQ000001
44	ACQ87852302	Cover Assembly,Front(Sub)	1	ACQ033200
45	MICE0016905	Insert,Nut	4	MET099500
46	MCK68588902	Cover,Front	1	MCK032700
47	MDQ64336301	Frame	1	MDQ000000
48	MBL66397101	Cap	1	MBL02
49	EAT62613701	Module,Hybrid Touch LCD	1	EAT130000
50	MJN69527606	Tape,USP Film	1	MJN107400
51	MJN69527701	Tape,Protect	1	MJN061100
52	MLAZ0038301	Label	1	MEZ000000
53	FAB31918701	Screw,Machine	11	FAB00
54	FAB32218901	Screw,Tapping	0	FAB01
55	EBR81145706	PCB Assembly,Main	1	EBR00
56	EBR81162601	PCB Assembly,Main,Insert	1	EBR071500
57	EBR76354001	PCB Assembly,Flexible	1	EAX00
58	EBR81021701	PCB Assembly,Flexible,Insert	1	EBR070100
59	MDS65330901	Gasket	1	MDS000000
60	EBR76354101	PCB Assembly,Flexible,SMT	1	EBR070400
61	EBR76328601	PCB Assembly,Flexible,SMT Top	1	EBR070300
62	EAX66307501	PCB,Flexible	1	EAX010700
63	EAG63393301	Connector,BtoB	1	CN9501
64	EBR76354201	PCB Assembly,Flexible,SMT Bottom	1	EBR070200
65	EAG64229801	Socket,DIMM/SIMM	1	J9500
66	MBK64173001	Can,Shield	1	MBK00
67	EBP62522102	Camera Module	1	EBP01
68	ABM74716301	Can Assembly,Shield	1	ABM00
69	MDS65310901	Gasket	1	MDS000000
70	MDS65331001	Gasket	1	MDS000002
71	MEV65471401	Insulator	1	MEV000002
72	MEV65590801	Insulator	1	MEV000001
73	MBK64153601	Can,Shield	1	MBK070300
74	MDS65290801	Gasket	1	MDS000001
75	BRAH0001305	Resin,PC	0	RAA050100
76	EBR81086205	PCB Assembly,Main,SMT	1	EBR071800
77	EBR81162501	PCB Assembly,Main,SMT Top	1	EBR071701
78	EAX66463501	PCB,Main	1	EAX010000

13. REPLACEMENT PART LIST

79	SAFP0000401	Wire Pad,Short	10	R1505,R1506,R1509,R5301,R6603,R6609,R6611,R7500,R7501,R9513
80	ECZH0025916	Capacitor,Ceramic,Chip	4	C1111,C1117,C1605,C1606
81	EAG64389801	C-Clip	3	ANT1,ANT2,ANT5
82	ERHY0009303	Resistor,Chip	1	R1100
83	ERHZ0000404	Resistor,Chip	3	R1101,R1102,R6928
84	ECZH0004402	Capacitor,Ceramic,Chip	10	C2321,C2323,C2325,C2326,C2327,C2329,C2330,C2333,C2334,C2336
85	ERHY0009536	Resistor,Chip	1	R1510
86	ECCH0009504	Capacitor,Ceramic,Chip	1	C1635
87	EAE62843001	Capacitor,Ceramic,Chip	4	C7500,C7501,C7502,C7506
88	EAE62945301	Capacitor(High Frequency),Ceramic,Chip	2	C5108,C5122
89	MBV62321701	Clip	10	SC10001,SC10002,SC10003,SC10004,SC10005,SC10006,SC10007,SC10008,SC10009,SC10010
90	EBC62456701	Resistor,Chip	2	R1601,R1602
91	ERHZ0000286	Resistor,Chip	1	R9536
92	EAP62447001	Inductor,Multilayer,Chip	1	C1658
93	EAN62969801	IC,Hall Effect Switch	1	U8700
94	ECCH0009216	Capacitor,Ceramic,Chip	5	C1109,C1110,C1115,C1116,C1646
95	ECZH0001217	Capacitor,Ceramic,Chip	6	C1501,C1502,C1503,C1504,C1505,C1506
96	EAN63569001	IC,Proximity	1	U8400
97	ERHY0009504	Resistor,Chip	2	R1104,R1105
98	ECZH0025920	Capacitor,Ceramic,Chip	2	C1604,C1628
99	ELCH0004703	Inductor,Multilayer,Chip	7	C1101,C1102,C1634,C1650,L1625,L1636,L1641
100	ECCH0000182	Capacitor,Ceramic,Chip	1	C5102
101	EAM62710701	Filter,Bead	3	FB6601,FB6602,FB6603
102	ERHZ0000250	Resistor,Chip	1	R3309
103	EAE62762301	Capacitor,Ceramic,Chip	2	C5116,C5119
104	EAM62510901	Filter,Duplexer	1	FL1600
105	ELCH0004721	Inductor,Multilayer,Chip	3	C1602,L1100,L1107
106	ERHZ0000236	Resistor,Chip	1	R1514
107	ERHY0000254	Resistor,Chip	4	R2204,R2205,R3110,R8400
108	ECZH0025917	Capacitor,Ceramic,Chip	1	C1112
109	EAE62883601	Capacitor(High Frequency),Ceramic,Chip	2	C1105,C5128
110	ECCH0009103	Capacitor,Ceramic,Chip	7	C1607,C1640,C1641,C1656,C1660,C5114,C5120
111	EAE62962301	Capacitor,Ceramic,Chip	2	C5112,C5113
112	ECCH0002001	Capacitor,Ceramic,Chip	11	C3304,C3307,C3311,C3314,C3316,C3340,C3341,C8400,C8401,C8700,C8701
113	ELCH0004720	Inductor,Multilayer,Chip	1	L5113
114	EAE62685301	Capacitor,Ceramic,Chip	1	C10002
115	EAP62107801	Inductor,Wire Wound,Chip	1	L7500
116	ECCH0000151	Capacitor,Ceramic,Chip	2	C6601,C6602
117	ERHZ0000406	Resistor,Chip	4	R2105,R2219,R2222,R2226
118	SFBH0008105	Filter,Bead	1	FB6600

13. REPLACEMENT PART LIST

119	EAG64451501	Connector,Terminal	2	CN10002,CN10003
120	ECCH0017301	Capacitor,Ceramic,Chip	1	C1500
121	EAN62887801	IC,RF Transceiver,3G	1	U1500
122	ECCH0000143	Capacitor,Ceramic,Chip	1	C6603
123	ELCH0004717	Inductor,Multilayer,Chip	1	L5108
124	EAE62945901	Capacitor(High Frequency),Ceramic,Chip	1	L1645
125	ERHZ0000401	Resistor,Chip	3	R5101,R5105,R5106
126	EAE62945401	Capacitor(High Frequency),Ceramic,Chip	1	L1102
127	ERHZ0000294	Resistor,Chip	1	R2101
128	EAH62033201	Diode,TVS	2	D6600,D6601
129	EAM62072101	Filter,Bead	2	FB6922,FB6923
130	EAE62883501	Capacitor(High Frequency),Ceramic,Chip	3	C1632,C1642,C1649
131	ELCH0004726	Inductor,Multilayer,Chip	1	L5114
132	EBC62463701	Resistor,Chip	2	R3310,R3311
133	ECCH0004904	Capacitor,Ceramic,Chip	5	C1507,C2320,C2335,C3306,C3342
134	EAN63366602	IC,MCP,eMMC	1	U1
135	EAW62283501	Crystal	1	X1500
136	ERHZ0000487	Resistor,Chip	1	R6600
137	ECZH0000813	Capacitor,Ceramic,Chip	6	C1113,C2101,C2102,C5103,L1112,L1500
138	EDLH0015104	LED,Chip	1	LD4602
139	EAT61914601	Module,Tx Module	1	U1100
140	EAP62526201	Inductor,Multilayer,Chip	1	L5116
141	ECZH0000816	Capacitor,Ceramic,Chip	1	C1600
142	EAN62909001	IC,WiFi	1	U5100
143	ECCH0009104	Capacitor,Ceramic,Chip	3	C6810,C6811,C6812
144	EAN63106301	IC,RF Amplifier	1	U5101
145	ELCH0004701	Inductor,Multilayer,Chip	2	L1111,L1601
146	EAM63790301	Filter,Separator,Switch	1	U1601
147	ERHY0009526	Resistor,Chip	2	R2200,R2201
148	EAE62506501	Capacitor,Ceramic,Chip	2	C1609,C1629
149	ECZH0000801	Capacitor,Ceramic,Chip	1	C1108
150	ELCH0004712	Inductor,Multilayer,Chip	2	L1600,L1603
151	ELCH0004716	Inductor,Multilayer,Chip	1	L1110
152	ELCH0004704	Inductor,Multilayer,Chip	4	L1628,L1629,L1638,L1644
153	EAN63388401	IC,Digital Baseband Processor,3G	1	U2100
154	ELCH0003826	Inductor,Multilayer,Chip	1	L1504
155	ELCH0004706	Inductor,Multilayer,Chip	2	C1106,L1633
156	ECZH0000830	Capacitor,Ceramic,Chip	4	C5107,C6928,C6929,L1104
157	ECCH0009106	Capacitor,Ceramic,Chip	1	C5115

13. REPLACEMENT PART LIST

158	MEZ65049701	Label	1	PID1
159	ELCH0001035	Inductor,Multilayer,Chip	1	C5109
160	EAE63067401	Capacitor,Ceramic,Chip	1	C7503
161	ERHZ0000443	Resistor,Chip	1	R6927
162	ERHY0009519	Resistor,Chip	1	R1103
163	ECCH0000198	Capacitor,Ceramic,Chip	1	C6930
164	EAM62531301	Filter,Saw,Dual	1	FL1100
165	EAN62796101	IC,Power Amplifier	1	U1600
166	ERHZ0000486	Resistor,Chip	1	R6601
167	EAE62969601	Capacitor,Ceramic,Chip	1	C1114
168	EAE62946601	Capacitor(High Frequency),Ceramic,Chip	1	C1630
169	EAP62245901	Inductor,Multilayer,Chip	1	C1657
170	EAP62109801	Inductor,Wire Wound,Chip	1	L6600
171	ECZH0001216	Capacitor,Ceramic,Chip	1	C5111
172	ERHY0009506	Resistor,Chip	1	R2224
173	EAN63688501	IC,DC,DC Converter	1	U7500
174	ECCH0009228	Capacitor,Ceramic,Chip	1	C5117
175	ECZH0003103	Capacitor,Ceramic,Chip	1	C5105
176	ERHY0024501	Resistor,Chip	1	R1603
177	EAM62890301	Filter,Duplexer	1	FL1604
178	ERHZ0000451	Resistor,Chip	1	R3109
179	ERHY0003301	Resistor,Chip	1	R9534
180	ERHZ0000519	Resistor,Chip	1	R2106
181	ELCH0004718	Inductor,Multilayer,Chip	3	L1109,L1113,L1605
182	EAP62226501	Inductor,Multilayer,Chip	1	L1630
183	EAM62930901	Filter,Duplexer	1	FL1606
184	ERHZ0000220	Resistor,Chip	1	R2102
185	ERHZ0000529	Resistor,Chip	1	R6926
186	ERHY0042409	Resistor,Chip	1	R1607
187	EAE62963601	Capacitor(High Frequency),Ceramic,Chip	1	C5123
188	ECCH0006201	Capacitor,Ceramic,Chip	1	C3310
189	EAH61532901	Diode,Switching	1	D9910
190	ELCH0001430	Inductor,Multilayer,Chip	1	L9910
191	EBR81218801	PCB Assembly,Main,SMT Bottom	1	EBR071601
192	ELCH0004703	Inductor,Multilayer,Chip	1	C1101,C1102,C1634,C1650,L1625,L1636,L1641
193	ERHY0009501	Resistor,Chip	1	R9601
194	ECCH0007804	Capacitor,Ceramic,Chip	1	C4101
195	ECCH0007802	Capacitor,Ceramic,Chip	4	C4137,C4138,C4139,C4150
196	SEVY0007902	Varistor	2	VA13002,VA13008
197	ECCH0009103	Capacitor,Ceramic,Chip	4	C1607,C1640,C1641,C1656,C1660,C5114,C5120

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198	ECZH0004402	Capacitor,Ceramic,Chip	15	C2321,C2323,C2325,C2326,C2327,C2329,C2330,C2333,C2334,C2336
199	ECCH0004904	Capacitor,Ceramic,Chip	22	C1507,C2320,C2335,C3306,C3342
200	ECCH0000198	Capacitor,Ceramic,Chip	13	C6930
201	ECCH0009230	Capacitor,Ceramic,Chip	2	C6700,C6701
202	MBV62321701	Clip	7	SC10001,SC10002,SC10003,SC10004,SC10005,SC10006,SC10007,SC10008,SC10009,SC10010
203	EAN62888101	IC,PMIC	1	U4100
204	ECCH0000110	Capacitor,Ceramic,Chip	2	C9504,C9505
205	EAE62487001	Capacitor,Low ESL	1	C3301
206	ECZH0003103	Capacitor,Ceramic,Chip	3	C5105
207	ENBY0040701	Connector,BtoB	1	CN7304
208	ERHZ0000406	Resistor,Chip	3	R2105,R2219,R2222,R2226
209	EAG64409801	C-Clip	2	ANT6200,ANT6201
210	ECCH0005604	Capacitor,Ceramic,Chip	7	C2300,C2301,C2312,C2313,C2314,C4141,C4142
211	ERHZ0000288	Resistor,Chip	1	R9204
212	EDNY0013602	Diode,Zener	1	ZD4131
213	ERHZ0000402	Resistor,Chip	1	R9504
214	ECZH0000813	Capacitor,Ceramic,Chip	9	C1113,C2101,C2102,C5103,L1112,L1500
215	EAM62072101	Filter,Bead	2	FB6922,FB6923
216	ECCH0006201	Capacitor,Ceramic,Chip	2	C3310
217	ERHY0003201	Resistor,Chip	2	R9107,R9533
218	EAN62572401	IC,DC,DC Converter	1	U4400
219	EAP62106301	Inductor,Wire Wound,Chip	1	L4400
220	EAE63284101	Capacitor,TA,Polymer	1	C9100
221	EAN62677301	IC,Charger	1	U4700
222	ERHZ0000465	Resistor,Chip	1	R4101
223	ECZH0001215	Capacitor,Ceramic,Chip	4	C4103,C4104,C4108,C4136
224	EAP62725901	Inductor,Wire Wound,Chip	1	L4130
225	EXXY0018701	Crystal	1	X4100
226	SEVY0005402	Varistor	1	VA9504
227	ERHZ0000299	Resistor,Chip	1	R4132
228	EAG63772101	Connector,RF	1	SW1101
229	SEVY0008901	Varistor	5	D103,D105,D106,VA6923,VA6924
230	ERHZ0000295	Resistor,Chip	1	R4102
231	EAH61992701	Diode,Schottky	1	D4400
232	ECZH0000830	Capacitor,Ceramic,Chip	3	C5107,C6928,C6929,L1104
233	ERHY0009507	Resistor,Chip	4	R1115,R5103,R5104,R9501
234	ENBY0034201	Connector,BtoB	1	CN7400
235	EAG64389801	C-Clip	4	ANT1,ANT2,ANT5
236	EAM62710701	Filter,Bead	2	FB6601,FB6602,FB6603

13. REPLACEMENT PART LIST

237	EAG63530401	Connector,Terminal Block	1	CN9100
238	EAE62685301	Capacitor,Ceramic,Chip	4	C10002
239	SAFP0000401	Wire Pad,Short	10	R1505,R1506,R1509,R5301,R6603,R6609,R6611,R7500,R7501,R9513
240	EAG64370001	Connector,Terminal Block	1	CN12100
241	ERHY0000298	Resistor,Chip	1	R1113
242	ERHZ0000204	Resistor,Chip	2	R9203,R9535
243	EAP61866701	Inductor,Multilayer,Chip	1	L1108
244	ECCH0017301	Capacitor,Ceramic,Chip	4	C1500
245	ELCH0003841	Inductor,Multilayer,Chip	2	L5102,L5107
246	ECCH0002001	Capacitor,Ceramic,Chip	7	C3304,C3307,C3311,C3314,C3316,C3340,C3341,C8400,C8401,C8700,C8701
247	ERHZ0000486	Resistor,Chip	2	R6601
248	EAE62882601	Capacitor(High Frequency),Ceramic,Chip	1	L5106
249	ECCH0000182	Capacitor,Ceramic,Chip	5	C5102
250	ECCH0000113	Capacitor,Ceramic,Chip	1	C5100
251	EUSY0412501	IC,Headphone Amplifier	1	U6200
252	ECZH0025916	Capacitor,Ceramic,Chip	1	C1111,C1117,C1605,C1606
253	EAN62637201	IC,Acceleration Sensor	1	U8100
254	EAB62830401	Microphone,Condenser	2	MIC69020,MIC6921
255	EAM62770201	Filter,EMI/Power	1	FL13001
256	EAP63106901	Inductor,Wire Wound,Chip	1	L4132
257	ERHZ0000443	Resistor,Chip	2	R6927
258	ERHZ0000310	Resistor,Chip	1	R4702
259	ECCH0000144	Capacitor,Ceramic,Chip	1	C4407
260	EAE62882501	Capacitor(High Frequency),Ceramic,Chip	1	C1160
261	ERHZ0000203	Resistor,Chip	1	R1111
262	ECCH0000115	Capacitor,Ceramic,Chip	3	C4109,C4110,C9203
263	EAG64091101	Socket,DIMM/SIMM	1	S9500
264	EAG63652201	C-Clip	2	ANT5102,ANT5103
265	ERHY0009506	Resistor,Chip	2	R2224
266	ERHY0009503	Resistor,Chip	2	R104,R105
267	EAG64149801	Connector,I/O	1	CN10001
268	EAE62767801	Capacitor,Ceramic,Chip	1	C4153
269	ERHY0009505	Resistor,Chip	2	R13000,R13001
270	ERHZ0000308	Resistor,Chip	1	R9532
271	EAP63185901	Inductor,Multilayer,Chip	1	L4401
272	EAM62770301	Filter,EMI/Power	2	FL13000,FL13002
273	EAE62969601	Capacitor,Ceramic,Chip	3	C1114
274	ERHZ0000538	Resistor,Chip	1	R4100
275	SFBH0008105	Filter,Bead	1	FB6600

13. REPLACEMENT PART LIST

276	EAE62883401	Capacitor(High Frequency),Ceramic,Chip	1	C5101
277	ELCH0004723	Inductor,Multilayer,Chip	1	C1100
278	EAG63393201	Connector,BtoB	1	CN9900
279	EAV63012101	LED,Flash	1	LD4500
280	ENBY0036001	Connector,BtoB	1	CN13000
281	ERHY0009316	Resistor,Chip	3	R1116,R5102,R5107
282	EAH61872601	Diode,TVS	1	D9201
283	ELCH0004709	Inductor,Multilayer,Chip	1	C1157
284	ERHZ0000267	Resistor,Chip	1	R4703
285	EAP62245901	Inductor,Multilayer,Chip	1	C1657
286	EAE62505701	Capacitor,Ceramic,Chip	2	C4400,C4702
287	ERHY0009303	Resistor,Chip	1	R1100
288	EAP62666001	Inductor,Wire Wound,Chip	1	L4131
289	EAE62685201	Capacitor,Ceramic,Chip	1	C9518
290	ERHZ0000316	Resistor,Chip	1	R1112
291	BRAH0001304	Resin,PC	0	RAA050100
292	SAD35467401	Software,Mobile	1	SAD010000
293	AGF78099905	Package Assembly	1	AGF000000
294	MEZ65773003	Label,Unit Box	1	MEZ084101
295	MLAJ0004402	Label,Master Box	0	MEZ047200
296	MAY67109603	Box,Unit	1	MAY084000
297	AGJ74178410	Pallet Assembly	1	AGJ000000
298	MAY66471606	Box,Carton	0	MAY010800
299	MAY66471717	Box,Pallet Sleeve	0	MAY115200
300	MPCY0021602	Pallet	0	MGA000000
301	MLAZ0050901	Label	0	MEZ000000
302	MBL65957705	Cap,Box	0	MBL007001
303	MLAJ0004402	Label,Master Box	0	MEZ047200
304	MAF63267402	Bag,Vinyl	1	MAF086500
305	MAY66470404	Box,Master	0	MAY047100
306	AAD87649415	Addition Assembly	1	AAD000000
307	EAC62018201	Rechargeable Battery,Lithium Ion	1	EAC00
308	AFN77214001	Manual Assembly,Operation	1	AFN053800
309	MBM65218601	Card,Quick Reference	1	MBM062600
310	MBM65036301	Card,Service Guide	1	MBM068900
311	ACQ88371001	Cover Assembly,Battery	1	ACQ00
312	MCK68549401	Cover,Battery	1	MCK004100
313	MLAZ0051104	Label	1	MEZ000000
314	MLAA0071101	Label,Approval	1	MEZ002100
315	MFL69182901	Manual,Operation	0	MFL053800
316	MBM64995933	Card,Warranty	1	MBM087200
317	EAY62709802	Adapters	1	EAY060000
318	EAD62377902	Cable,Assembly	1	EAD010000
319	EAB62808211	Earphone,Stereo	1	EAB010200

13. REPLACEMENT PART LIST

320	SAF31060801	Software Assembly,Common	1	SAF010100
321	SAD35455001	Software,USB Driver	1	SAD010500
322	SAD35454901	Software,OSP	1	SAD010600
323	SAD35434001	Software,Application	1	SAD010100

